Attention and Transformers

Basics and Applications of Deep Learning

Fall 2026

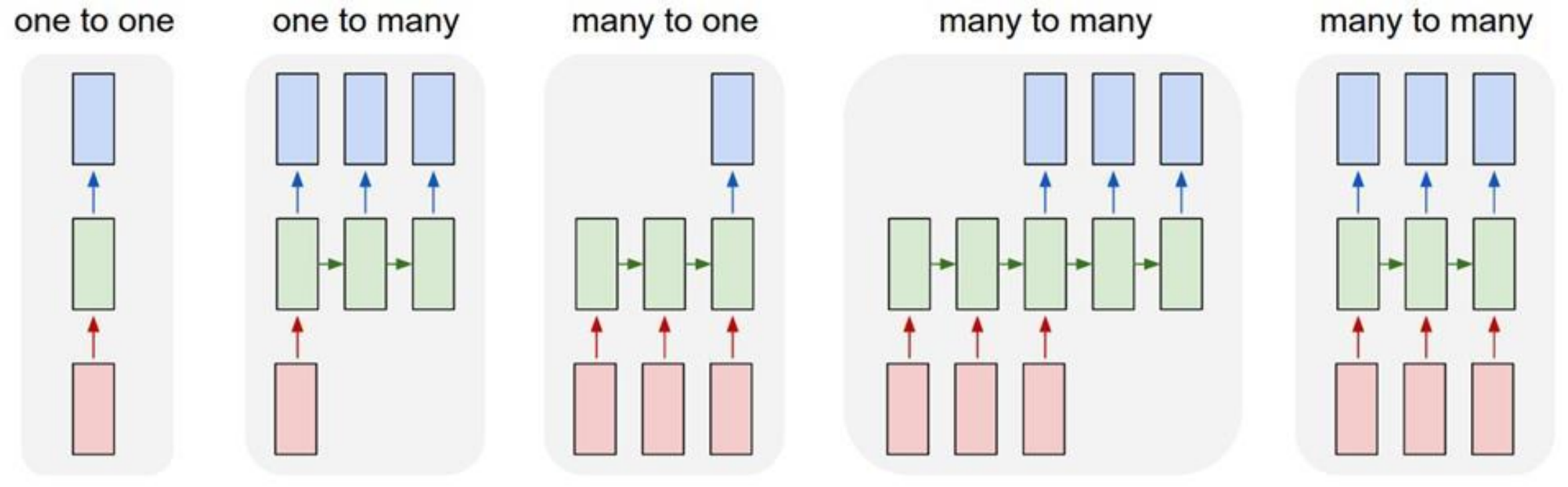
Department of Mathematical Data Science

Hanyang University ERICA

Lecturer: Sehyun Kwon

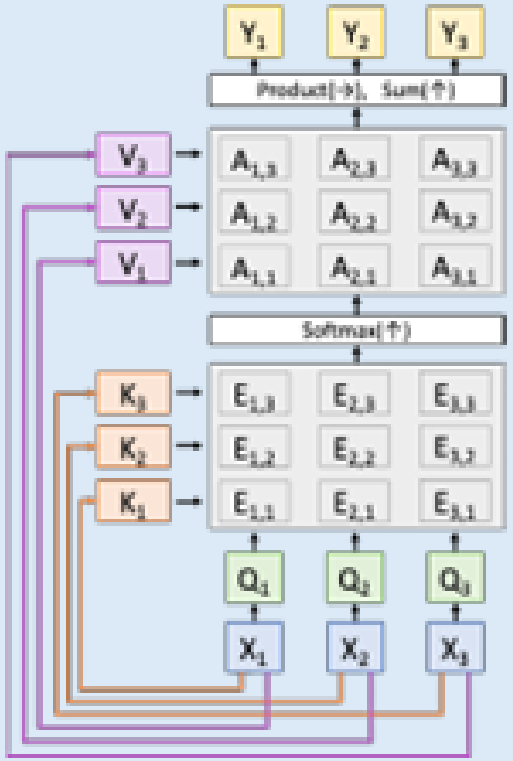


Last Time: Recurrent Neural Networks

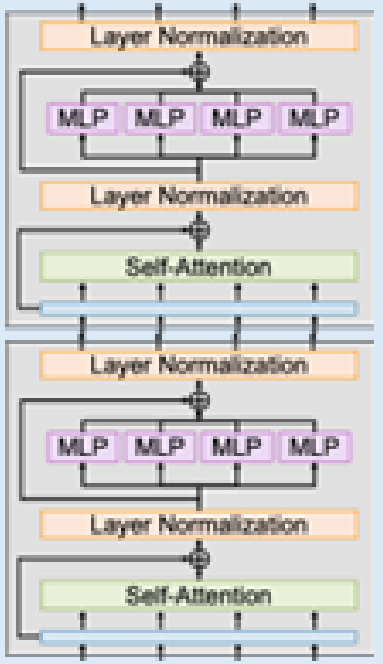


Today: Attention + Transformers

Attention : A new primitive that operates on sets of vectors

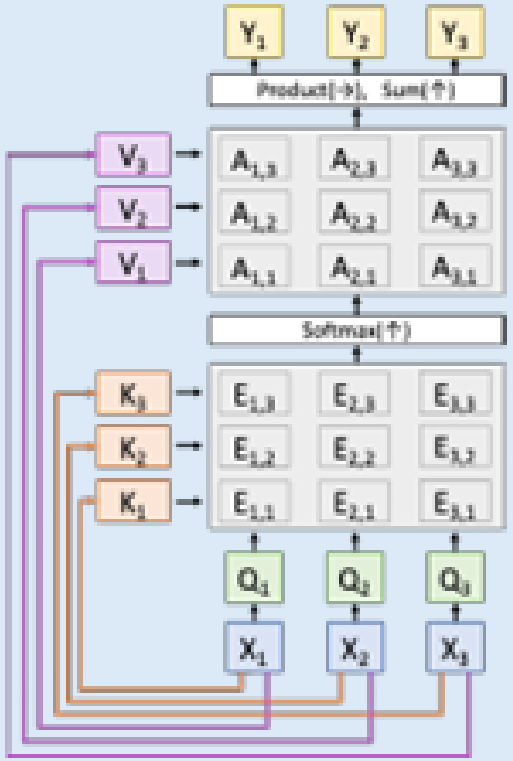


Transformer : A neural network architecture that uses attention everywhere



Today: Attention + Transformers

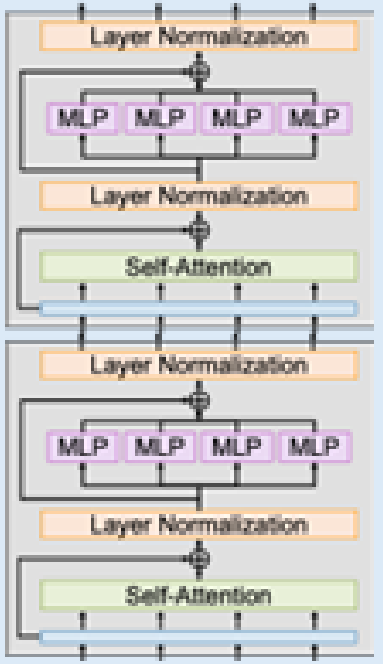
Attention : A new primitive that operates on sets of vectors



Transformers are used everywhere today!

But they developed as an offshoot of RNNs so let's start there

Transformer : A neural network architecture that uses attention everywhere

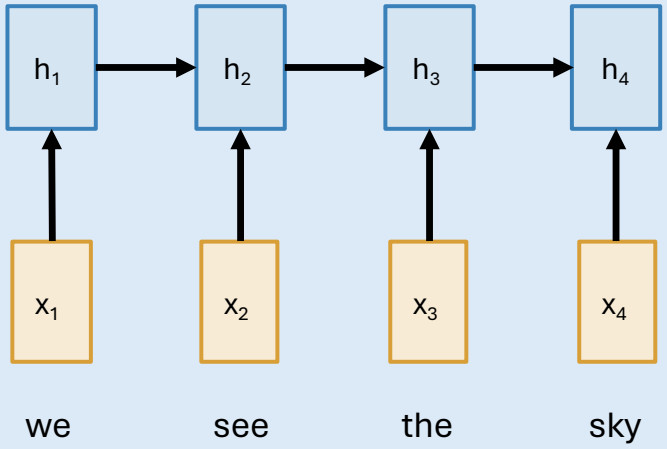


RNN Encoder–Decoder

Input: Sequence $x_1, \ldots x_T$
Output: Sequence $y_1, \ldots, y_{T'}$

A motivating example for today's discussion – machine translation! English → Italian

Encoder: $h_t = f_W(x_t, h_{t-1})$



Sutskever et al, "Sequence to sequence learning with neural networks", NeurIPS 2014

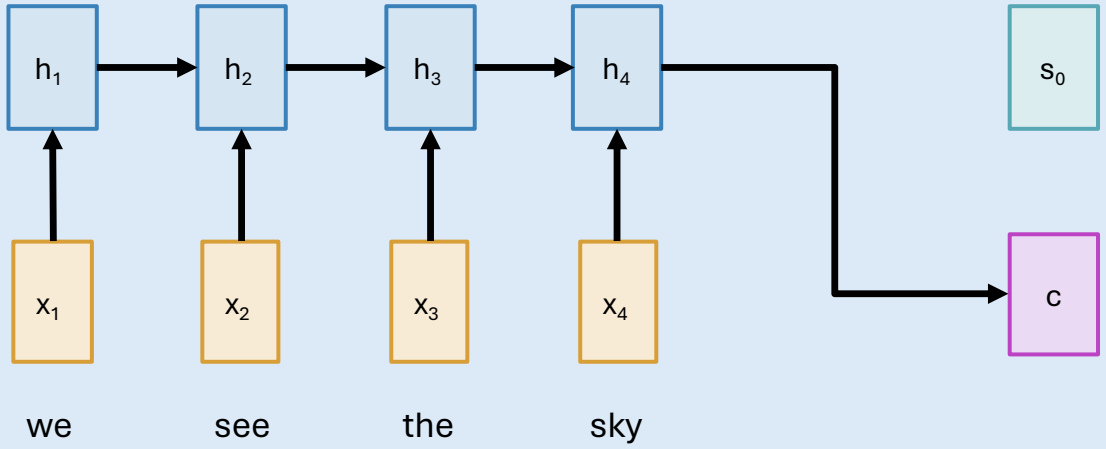
Sequence-to-Sequence with RNNs

Input: Sequence $x_1, \ldots x_T$

Output: Sequence $y_1, \ldots, y_{T'}$

Encoder: $h_t = f_W(x_t, h_{t-1})$

From final hidden state predict:
Initial decoder state s_0
Context vector c (often $c=h_T$)



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Sequence-to-Sequence with RNNs

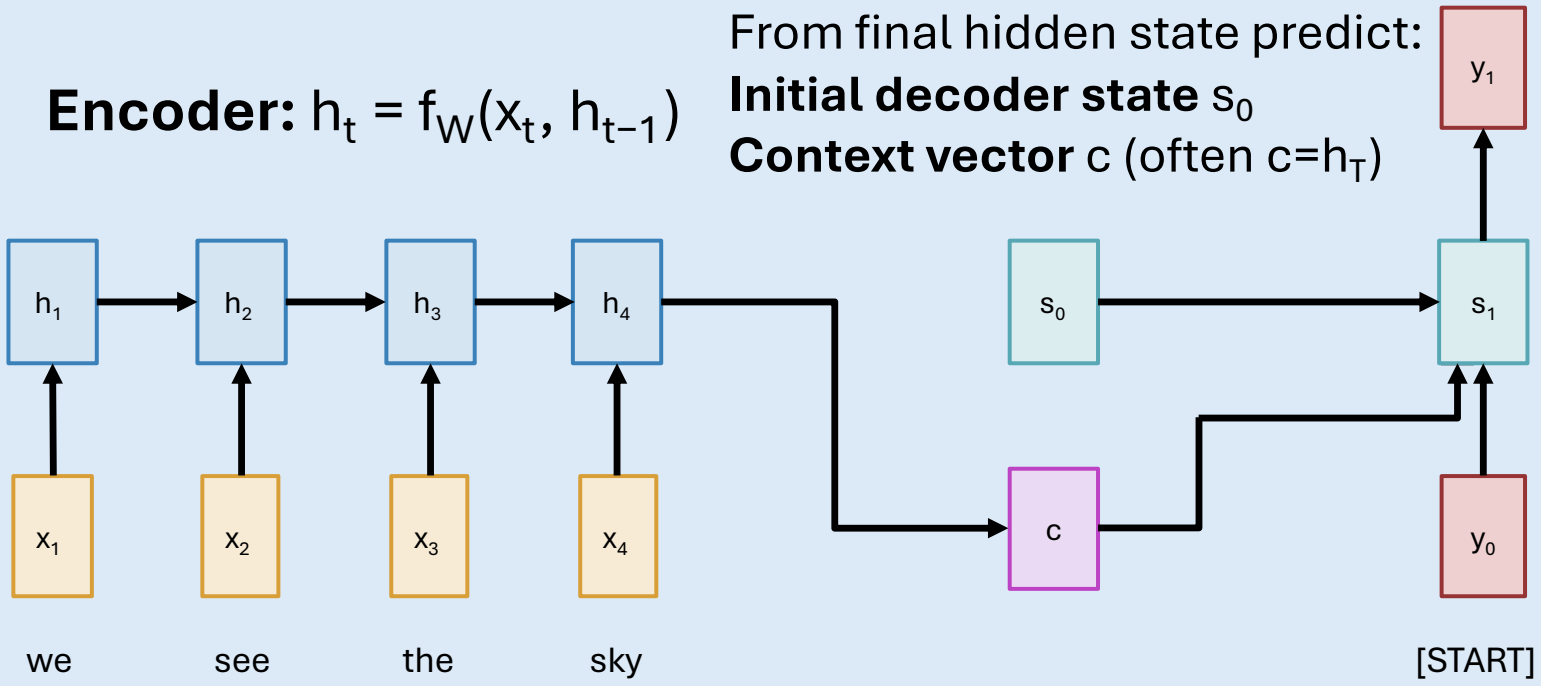
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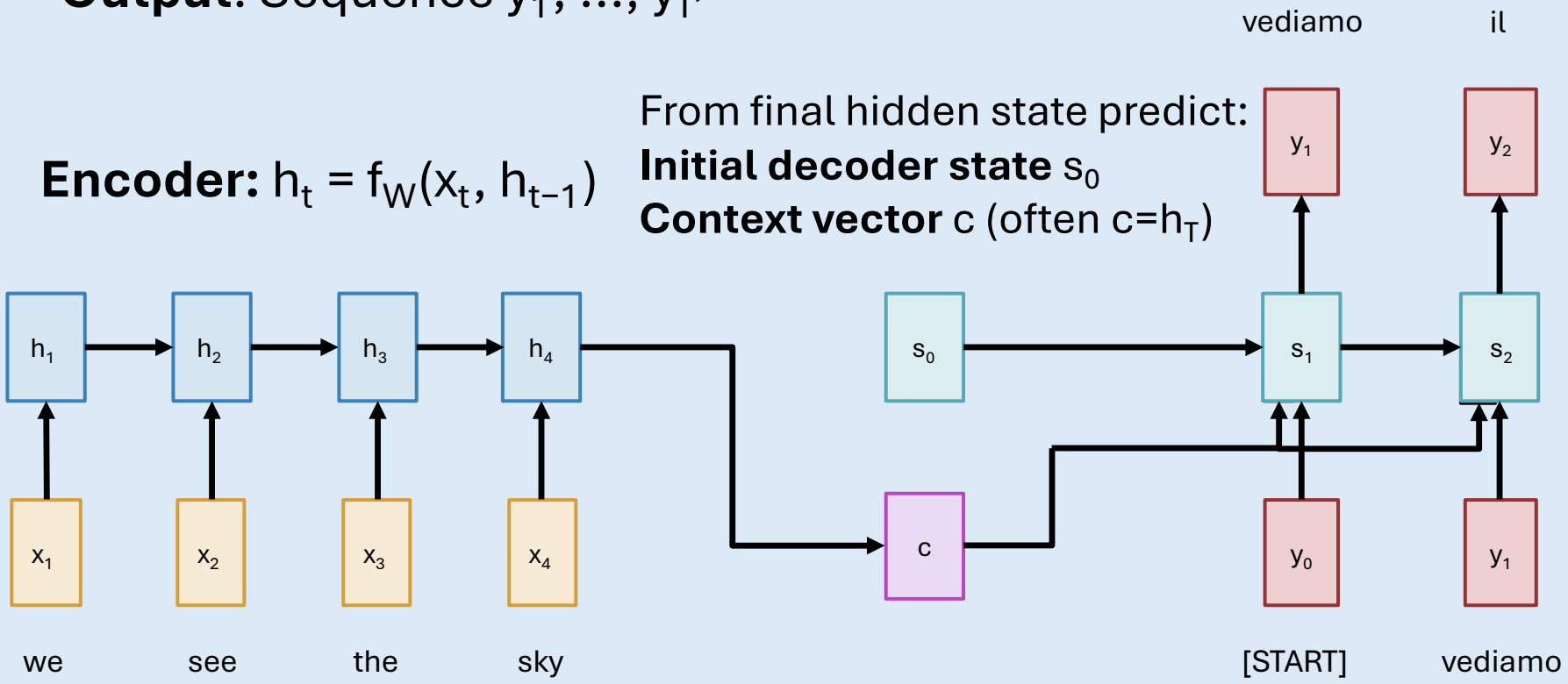
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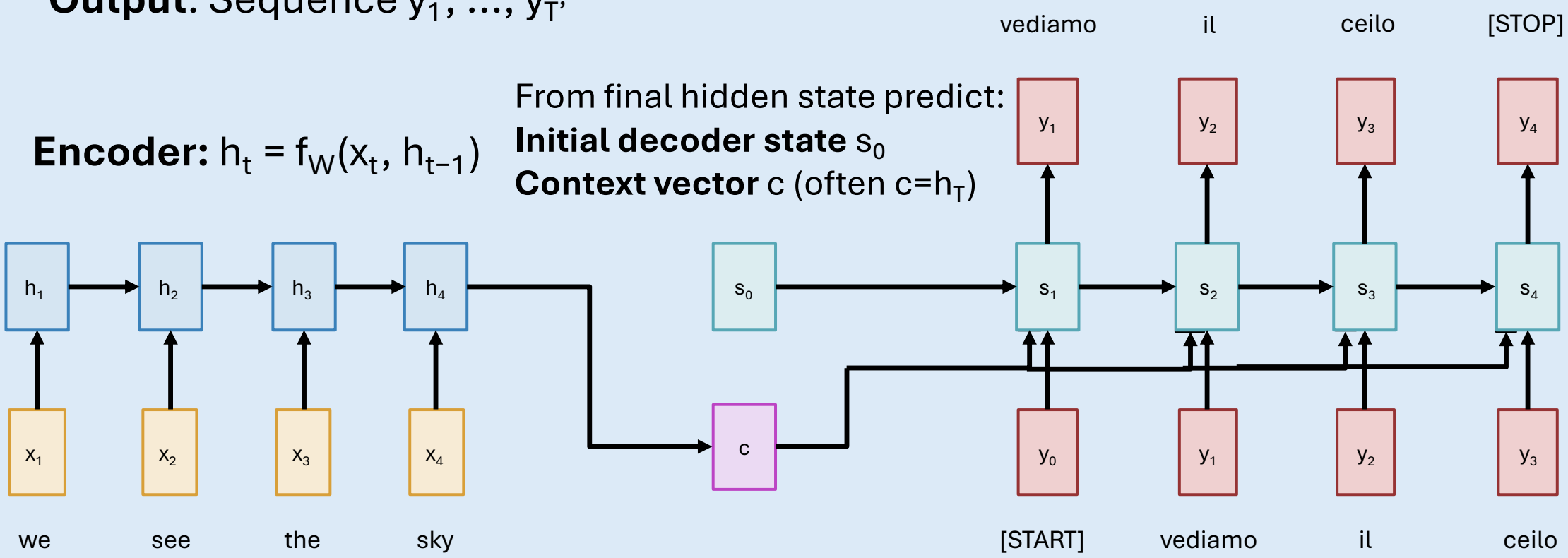
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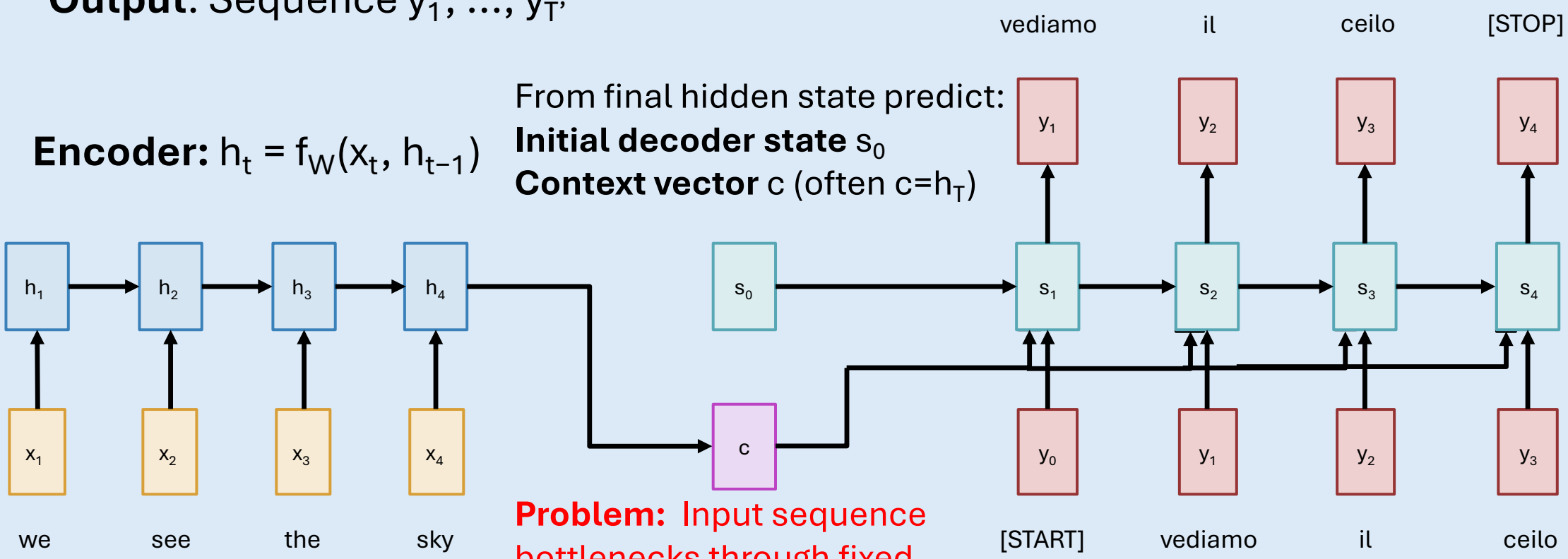
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Problem: Input sequence bottlenecks through fixed sized c. What if T=1000?

Sutskever et al, "Sequence to sequence learning with neural networks", NeurIPS 2014

Sequence-to-Sequence with RNNs

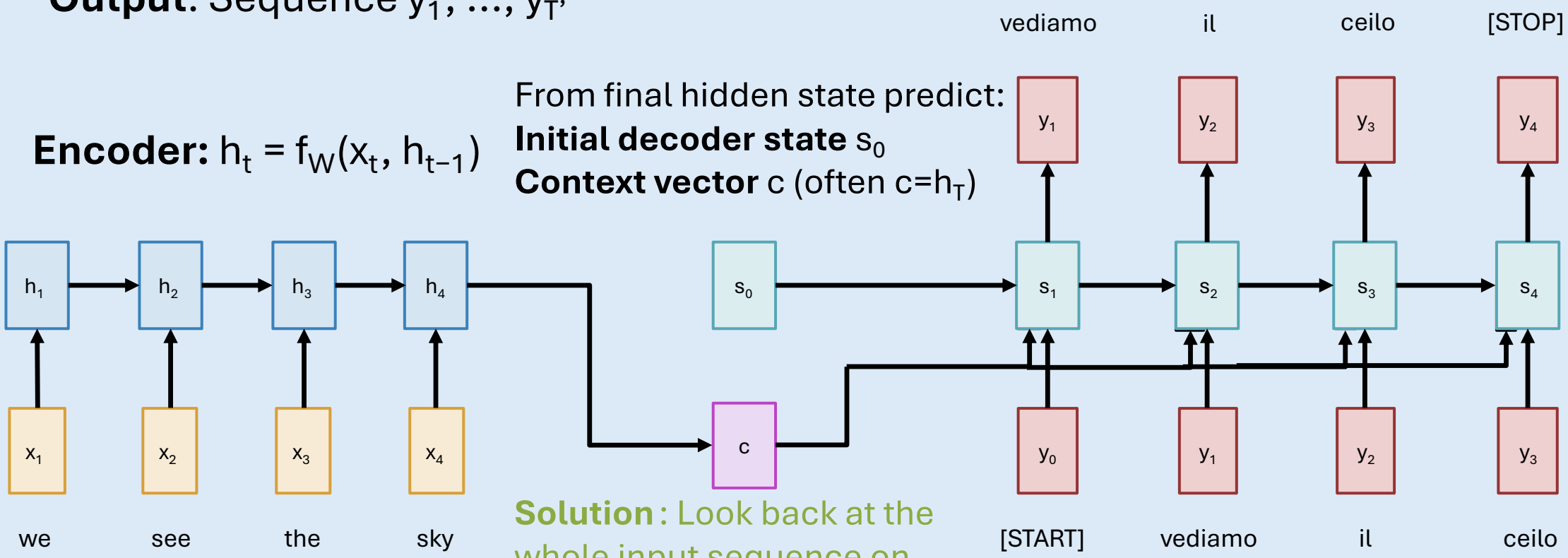
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Solution: Look back at the whole input sequence on each step of the output

Sutskever et al, "Sequence to sequence learning with neural networks", NeurIPS 2014

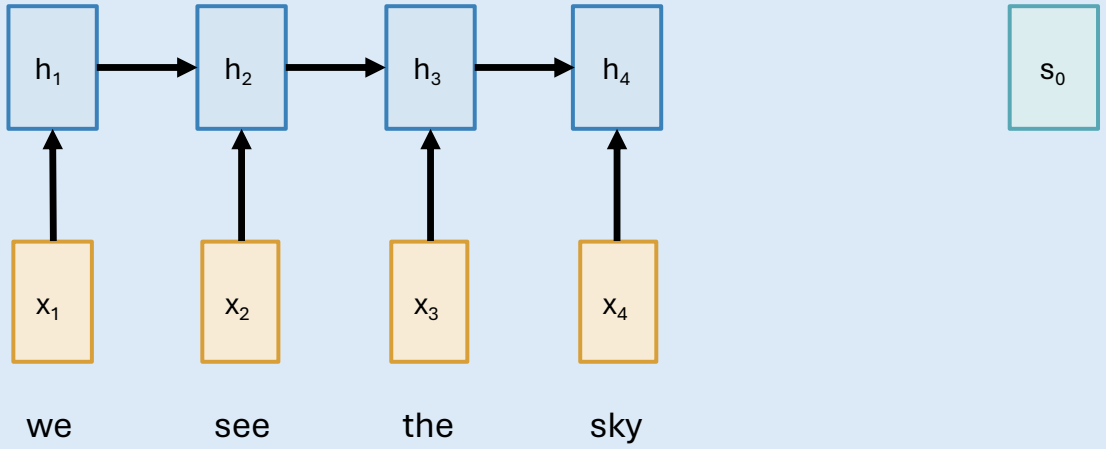
Sequence-to-Sequence with Attention

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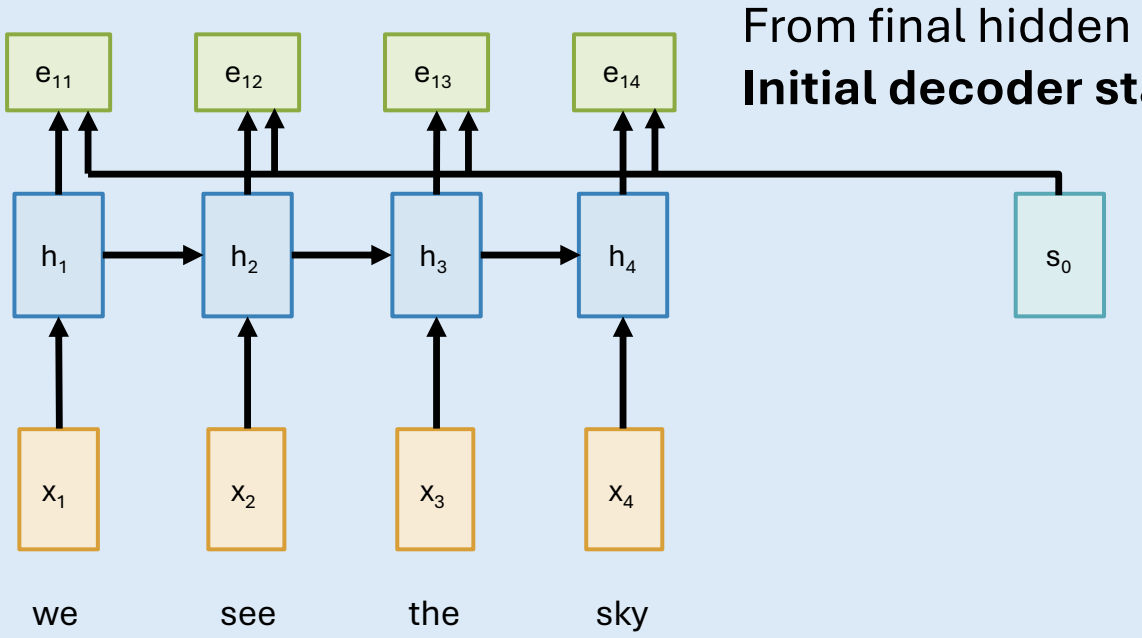


Bahdanau et al, "Neural machine translation by jointly learning to align and translate", ICLR 2015

Sequence-to-Sequence with Attention

Compute (scalar) **alignment scores**

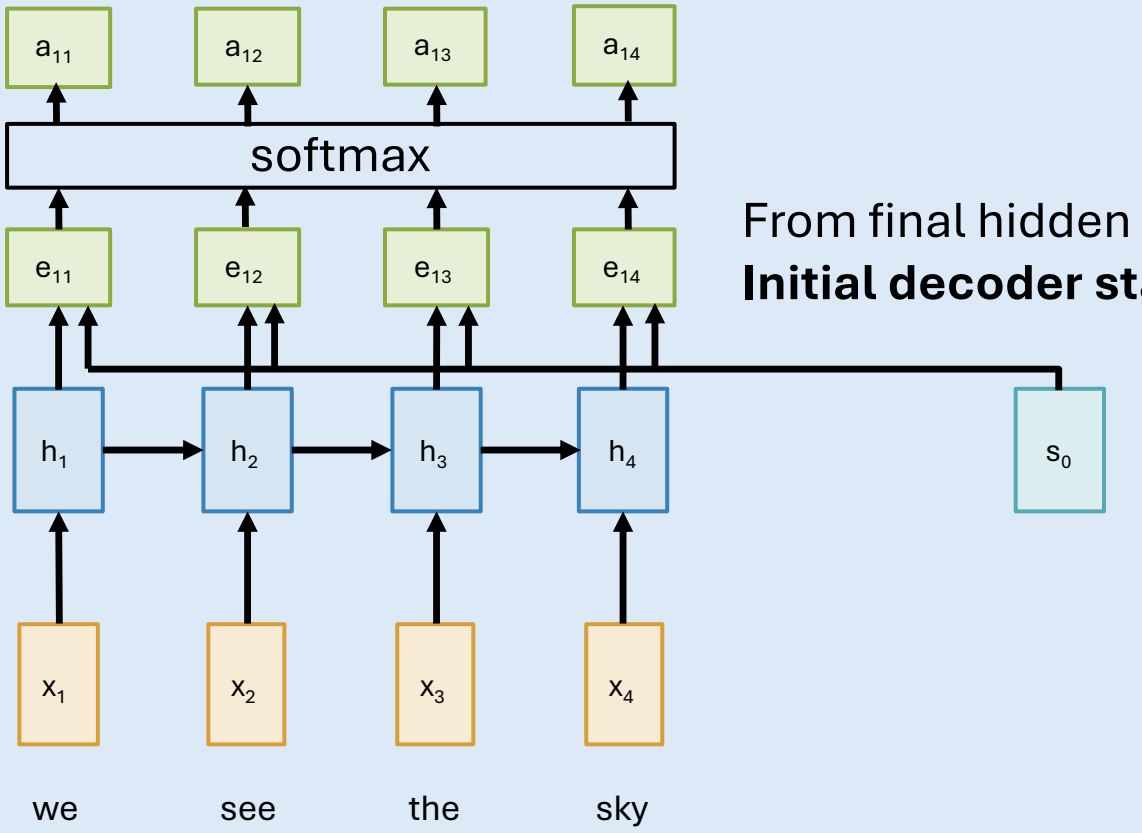
$e_{t,i} = f_{att}(s_{t-1}, h_i)$ (f_{att} is a Linear Layer)



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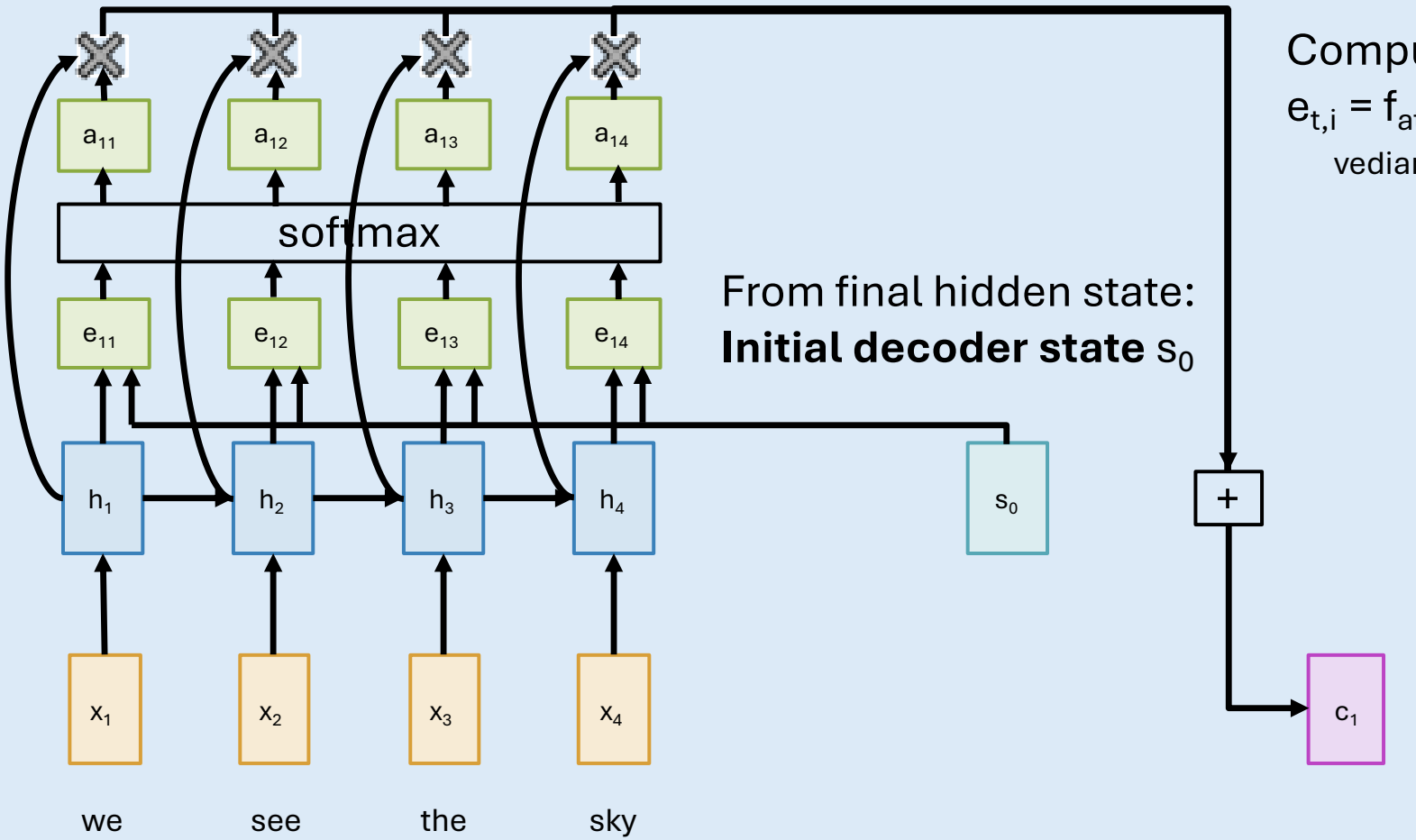
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Normalize alignment scores to get **attention weights**
$0 < a_{t,i} < 1 \quad \sum_i a_{t,i} = 1$

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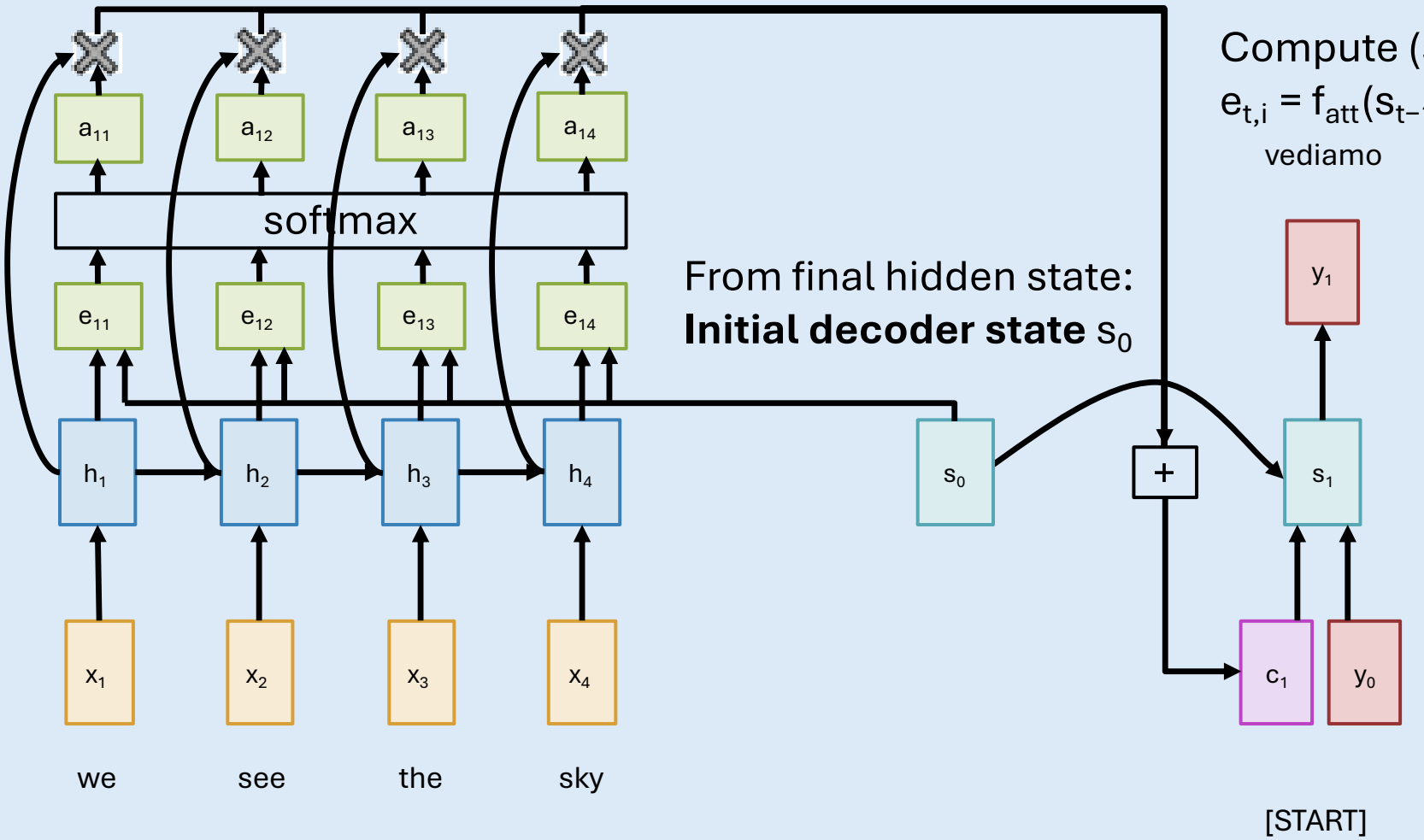
vediamo

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Compute context vector as **weighted sum of hidden states**
$c_t = \sum_i a_{t,i} h_i$

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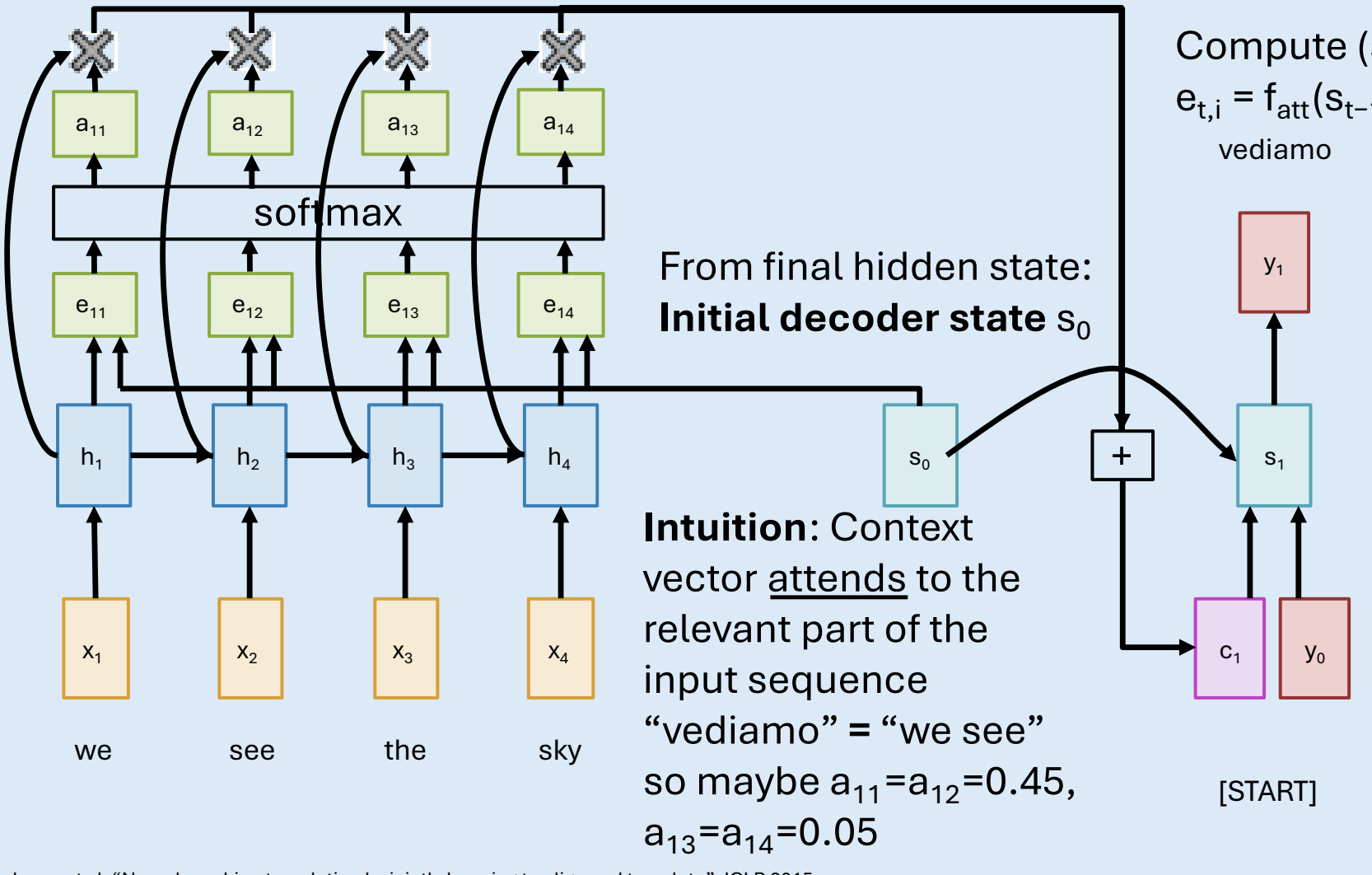
Compute **context vector** as weighted sum of hidden states
$c_t = \sum_i a_{t,i} h_i$

Use context vector in decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c_t)$

g_U is an RNN unit (e.g. LSTM, GRU)

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Sequence-to-Sequence with Attention



From final hidden state:
Initial decoder state s_0

Intuition: Context vector attends to the relevant part of the input sequence
"vediamo" = "we see"
so maybe $a_{11}=a_{12}=0.45$, $a_{13}=a_{14}=0.05$

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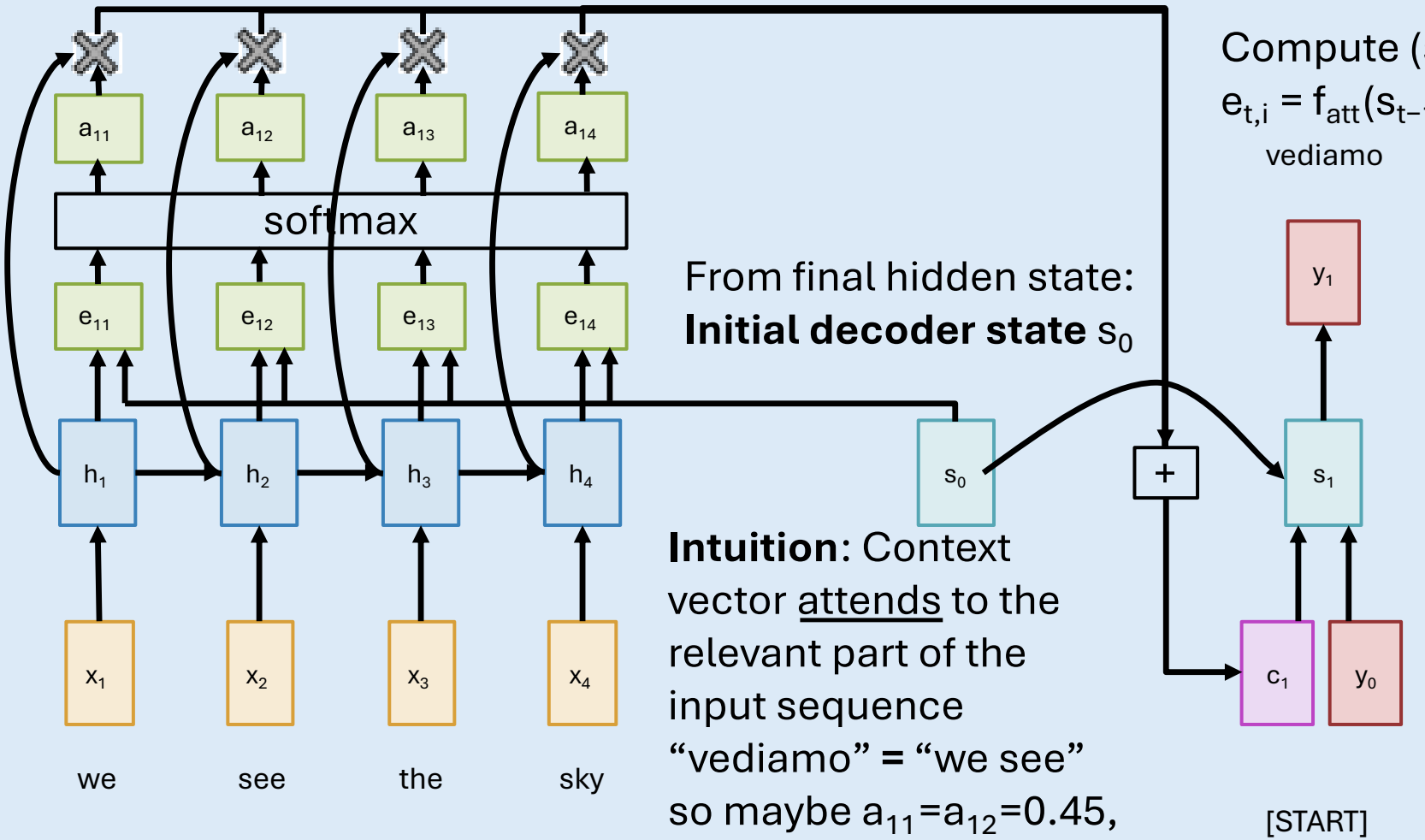
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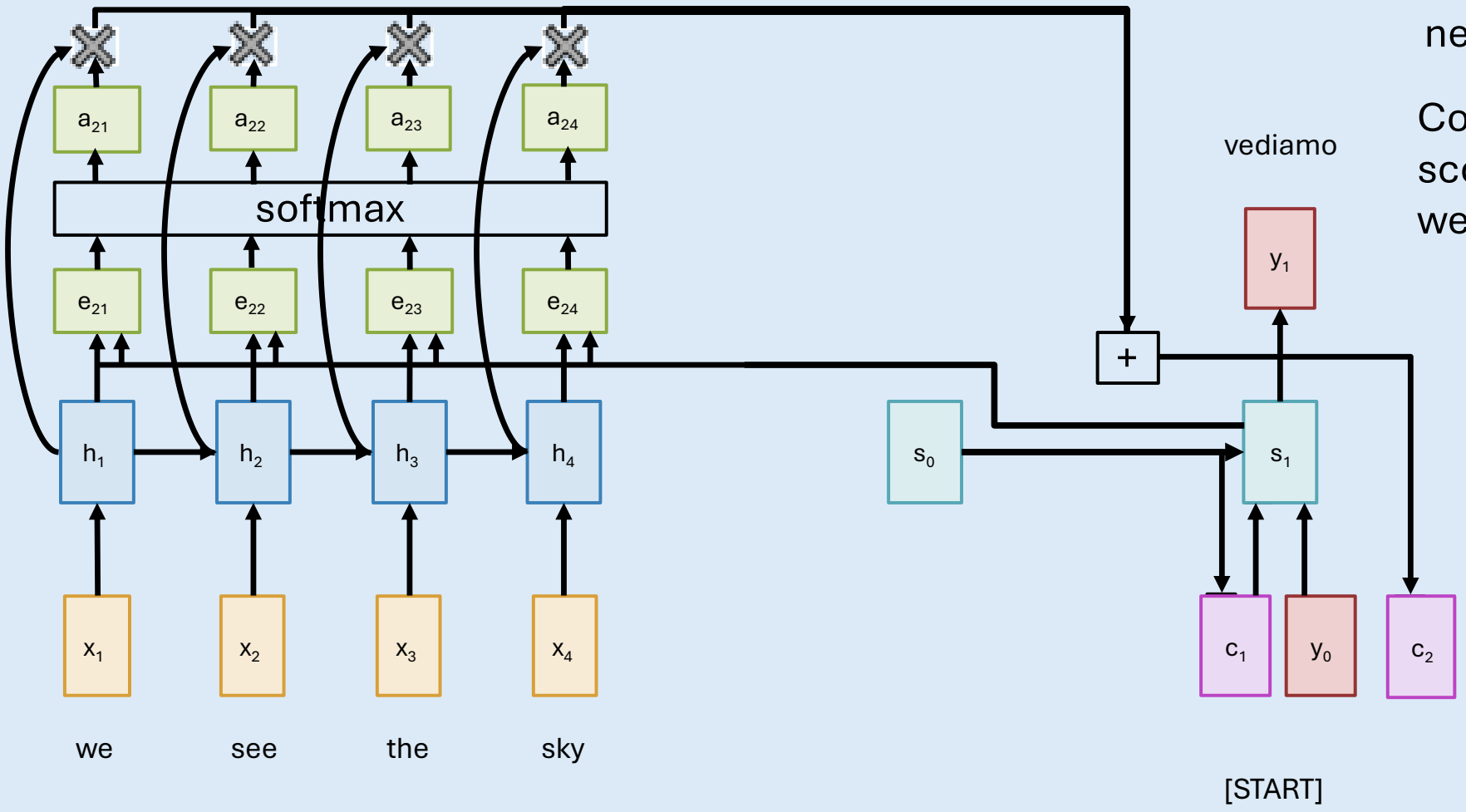
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All differentiable! No supervision on attention weights. Backprop through everything

Bahdanau et al, "Neural machine translation by jointly learning to align and translate", ICLR 2015

Sequence-to-Sequence with Attention



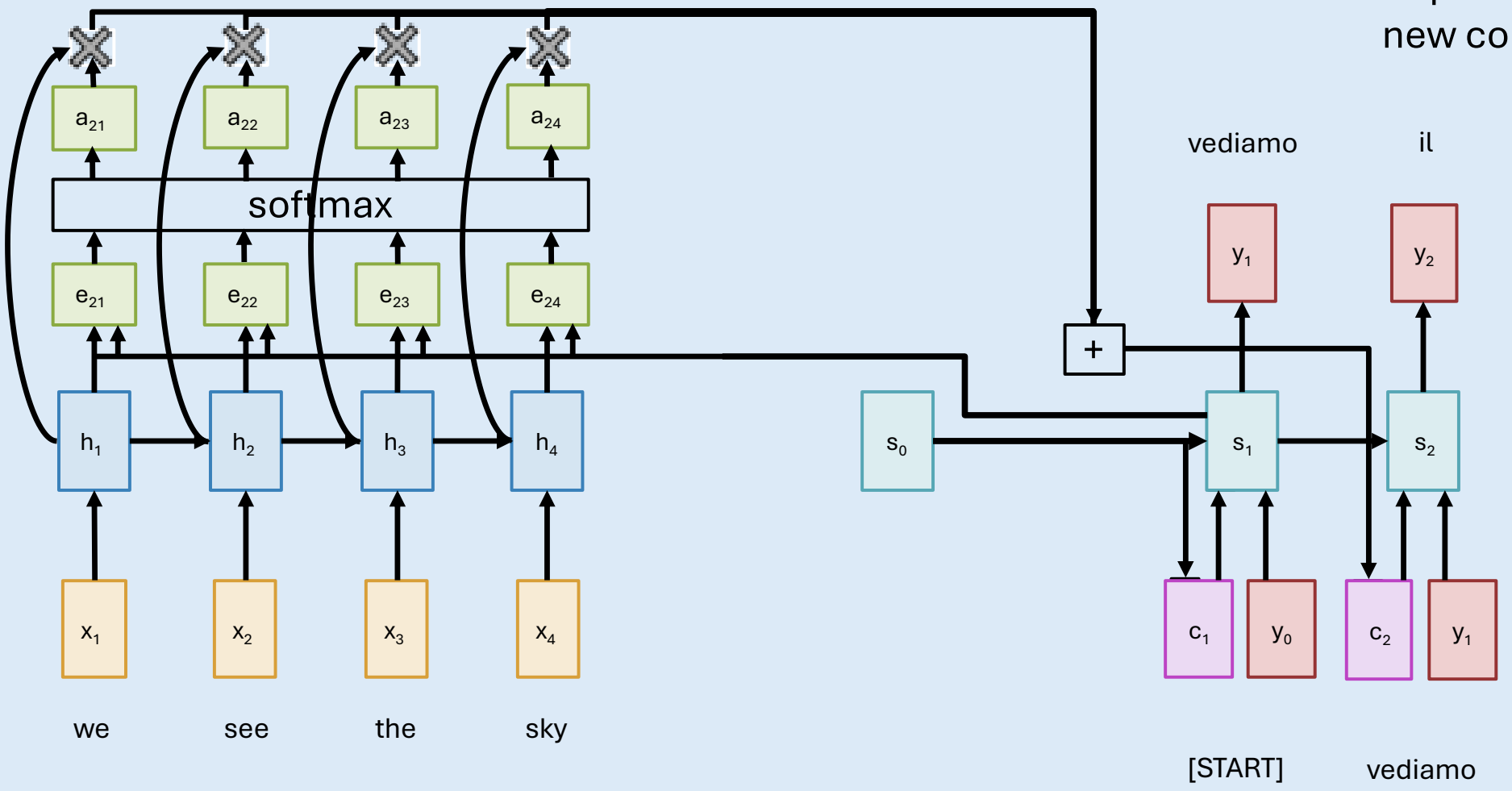
Repeat: Use s_1 to compute new context vector c_2

Compute new alignment scores $e_{2,i}$ and attention weights $a_{2,i}$

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Sequence-to-Sequence with Attention

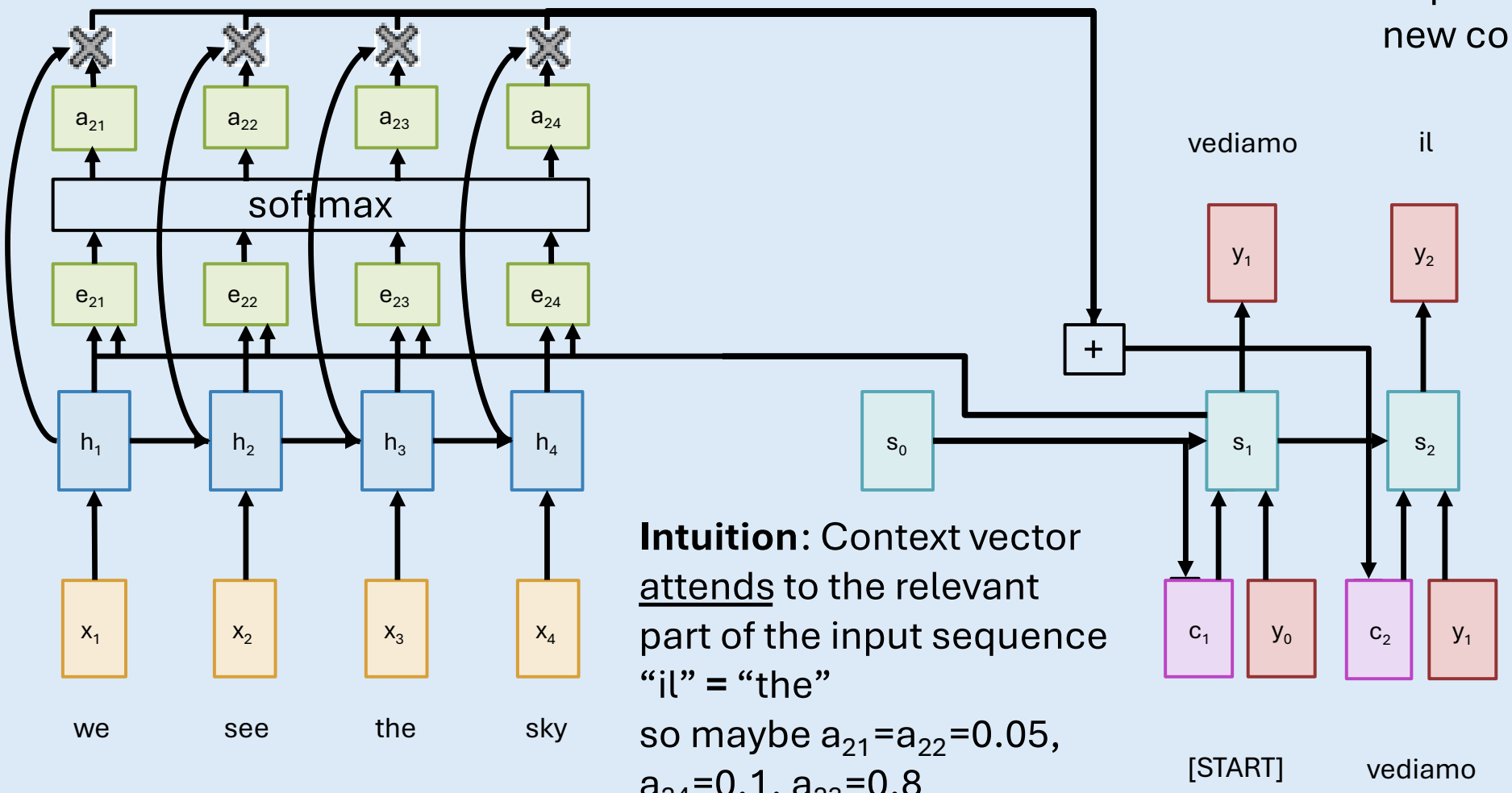
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Use context vector in decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c_t)$

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Sequence-to-Sequence with Attention

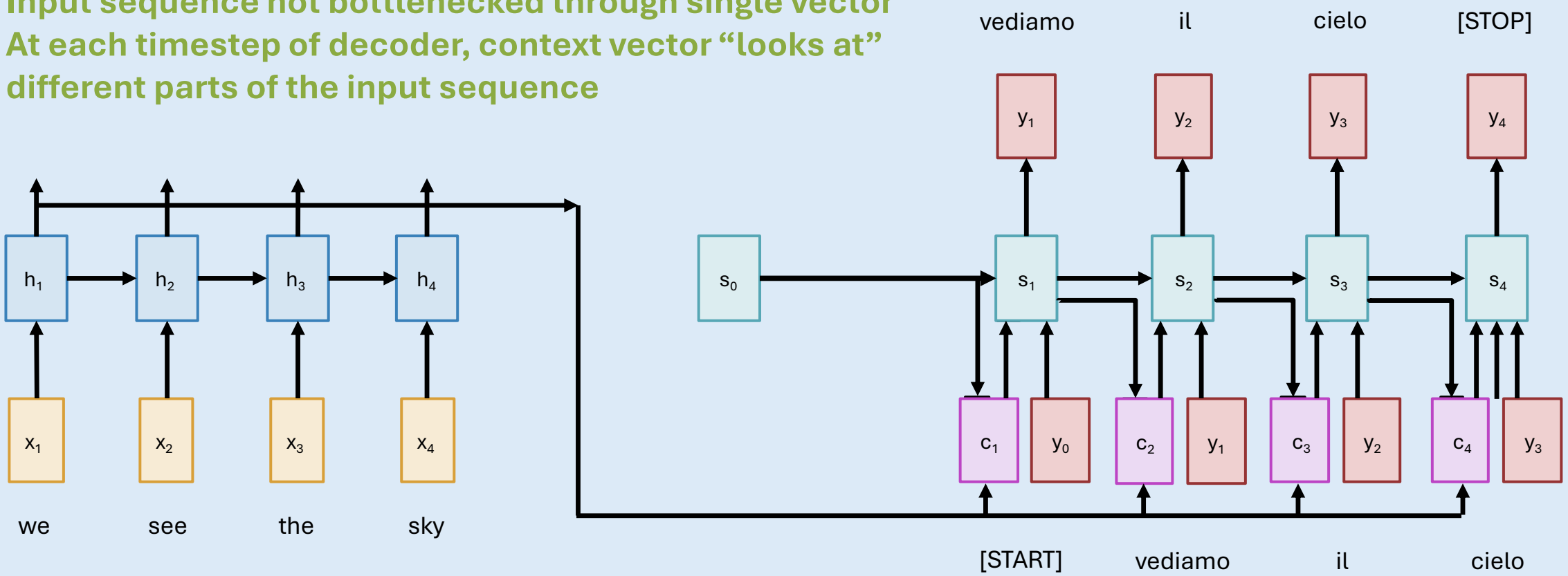


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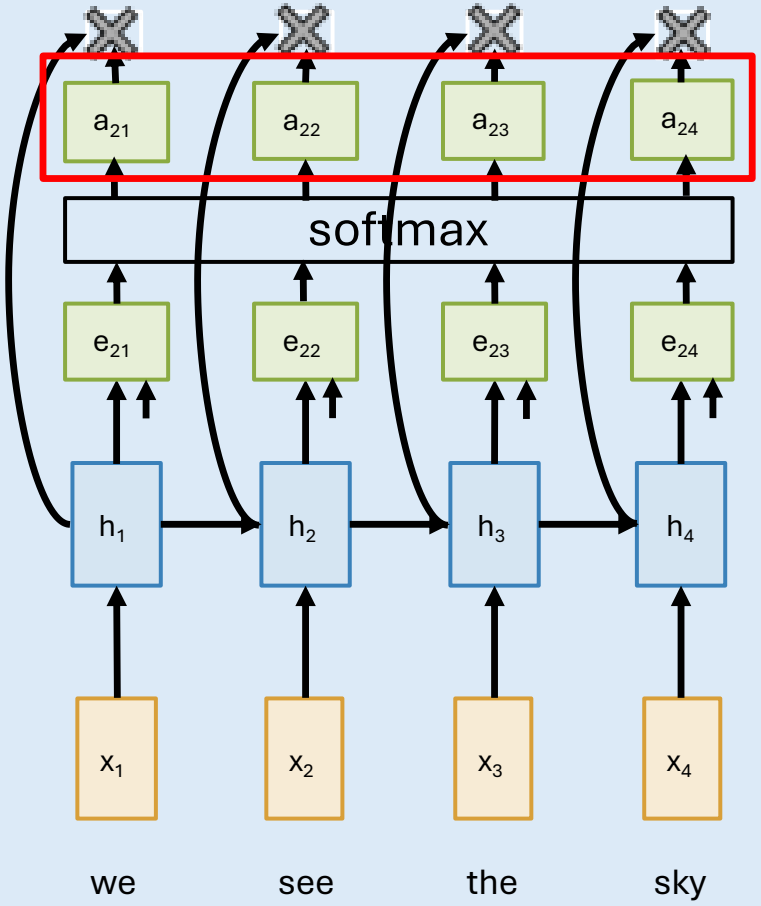
Use a different context vector in each timestep of decoder

- **Input sequence not bottlenecked through single vector**
- **At each timestep of decoder, context vector "looks at" different parts of the input sequence**



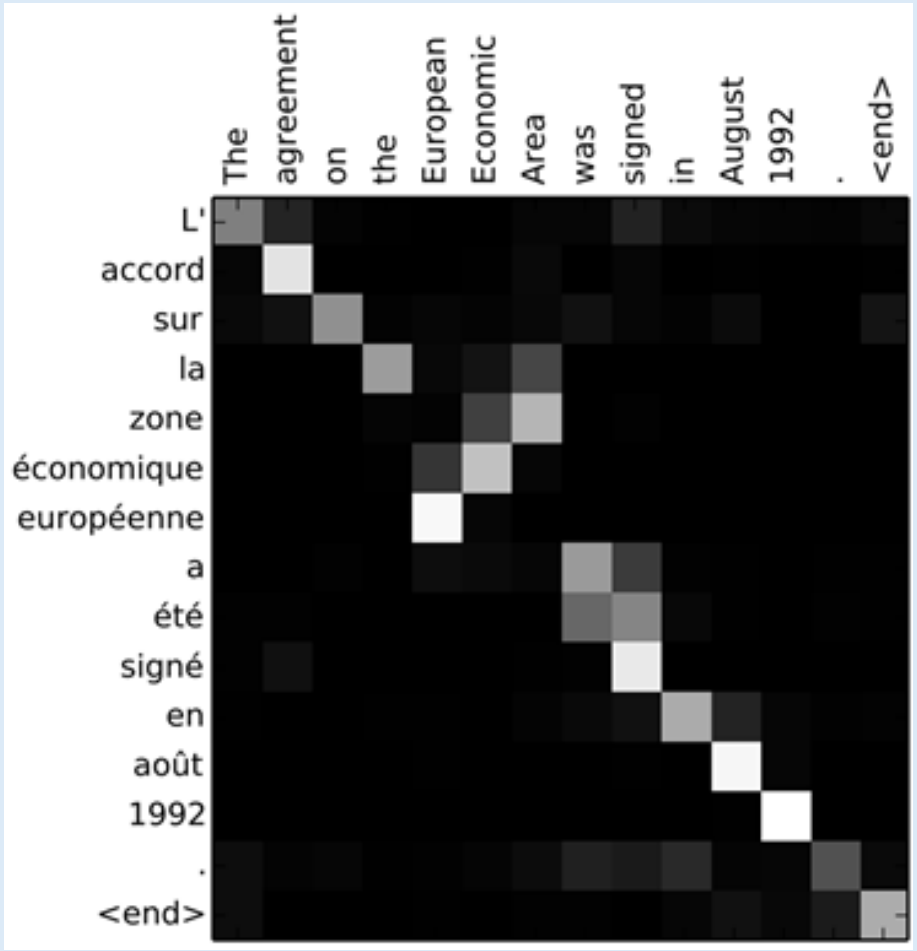
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Sequence-to-Sequence with Attention



Example : English to French translation

Visualize attention weights $a_{t,i}$

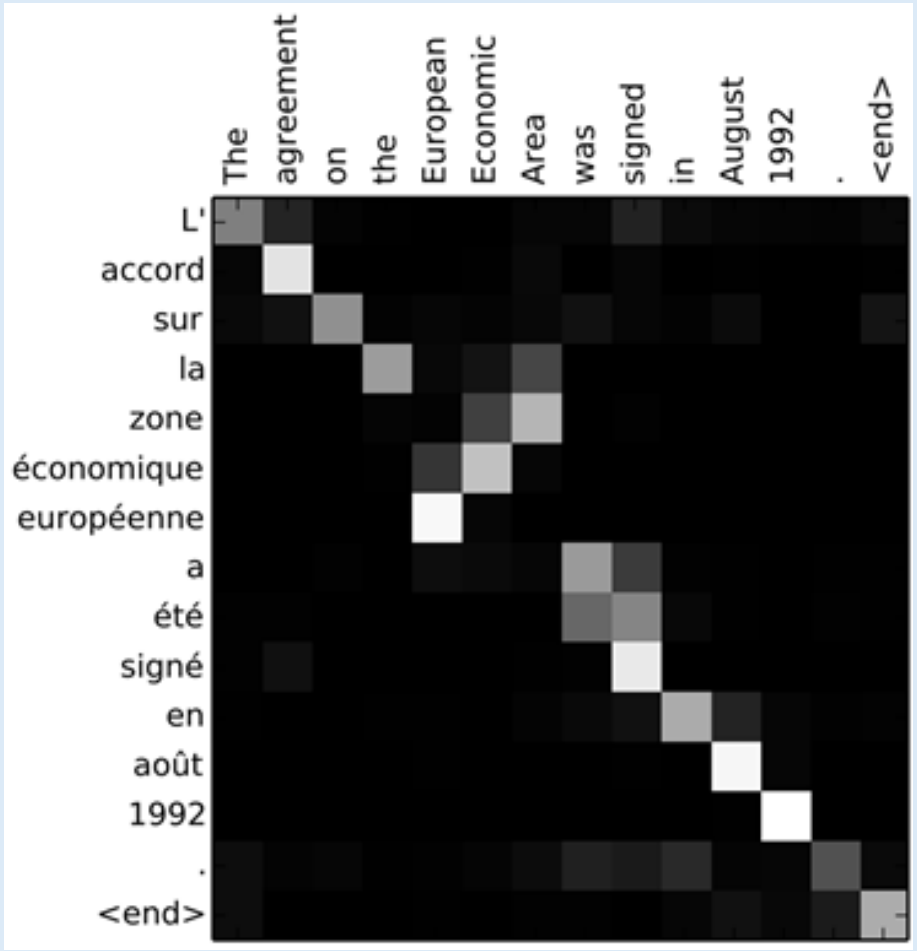


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Sequence-to-Sequence with Attention

Example : English to French translation

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Input: "The agreement on the European Economic Area was signed in August 1992."

Output: "L'accord sur la zone économique européenne a été signé en août 1992."

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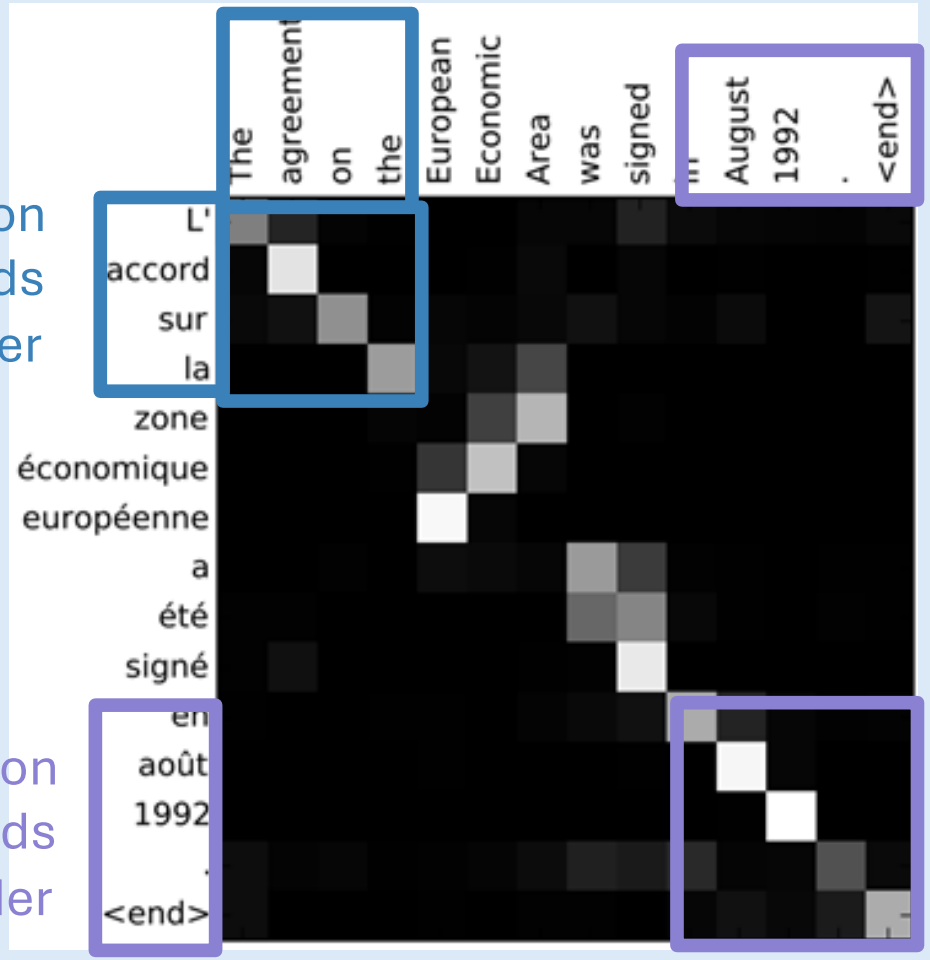
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Diagonal attention means words correspond in order

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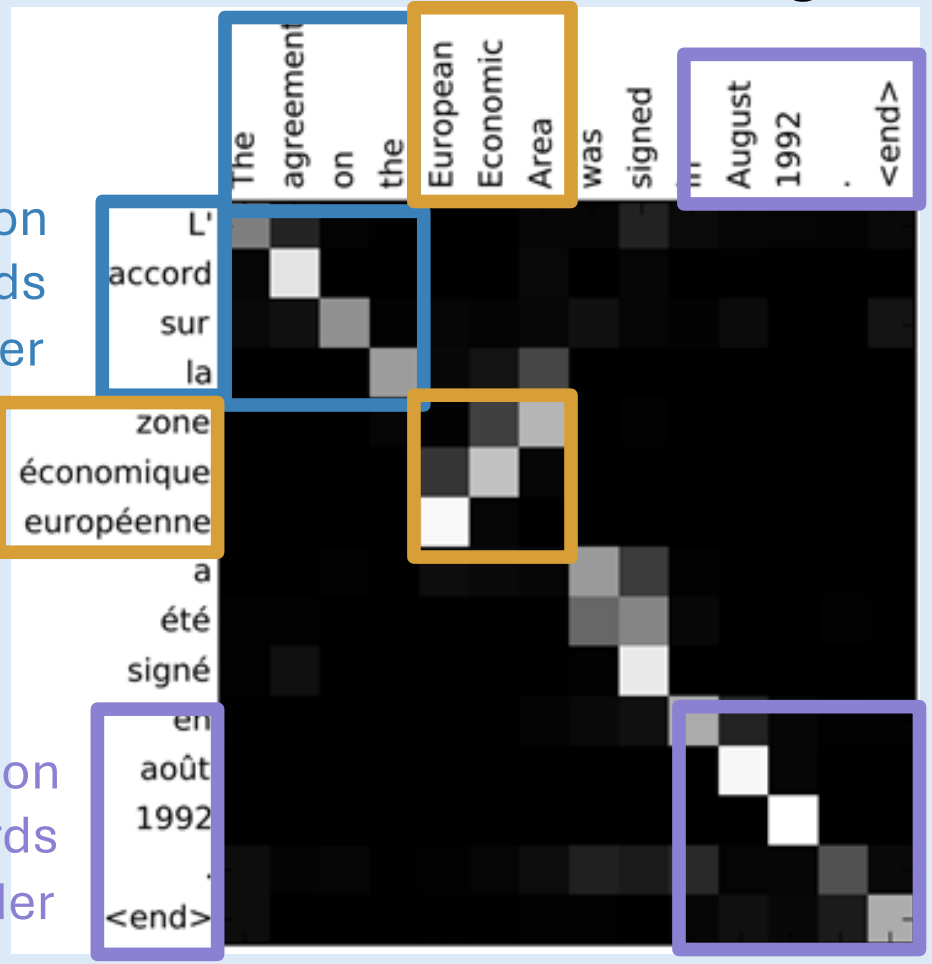
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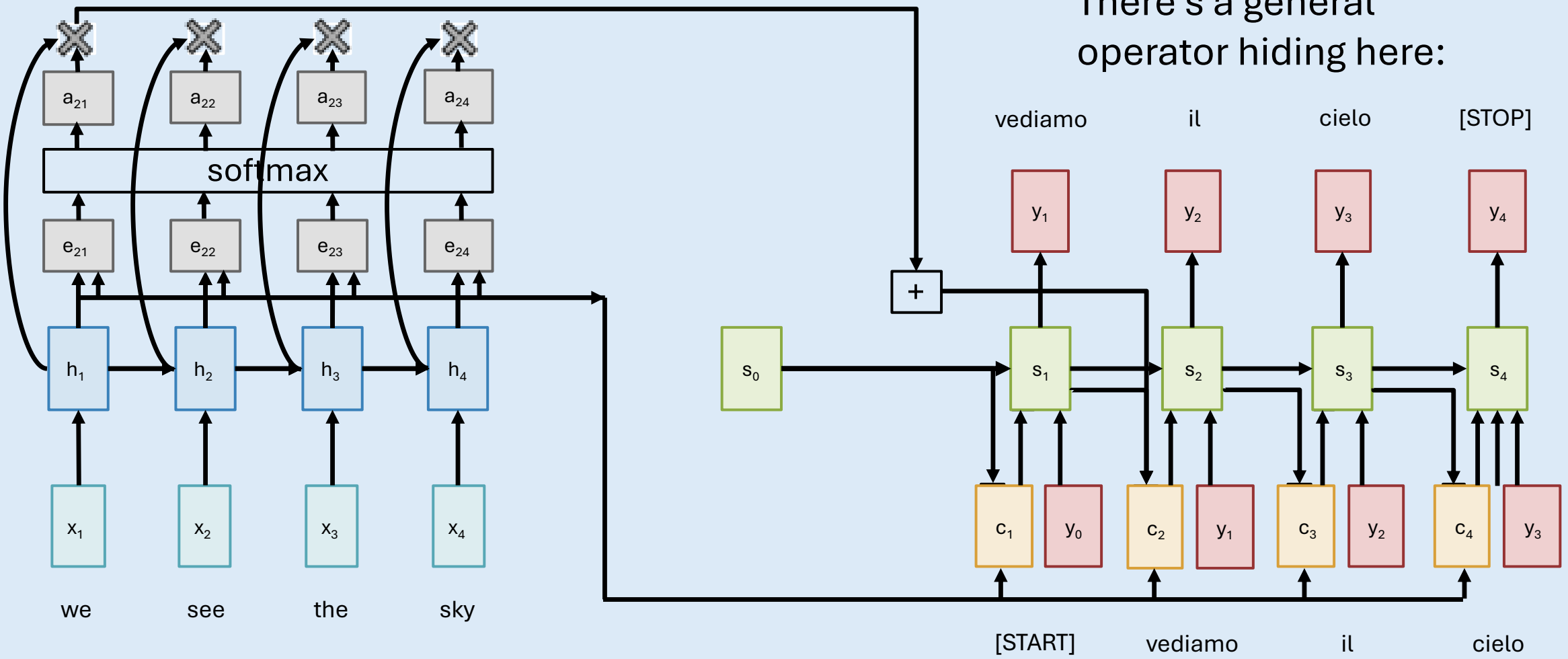
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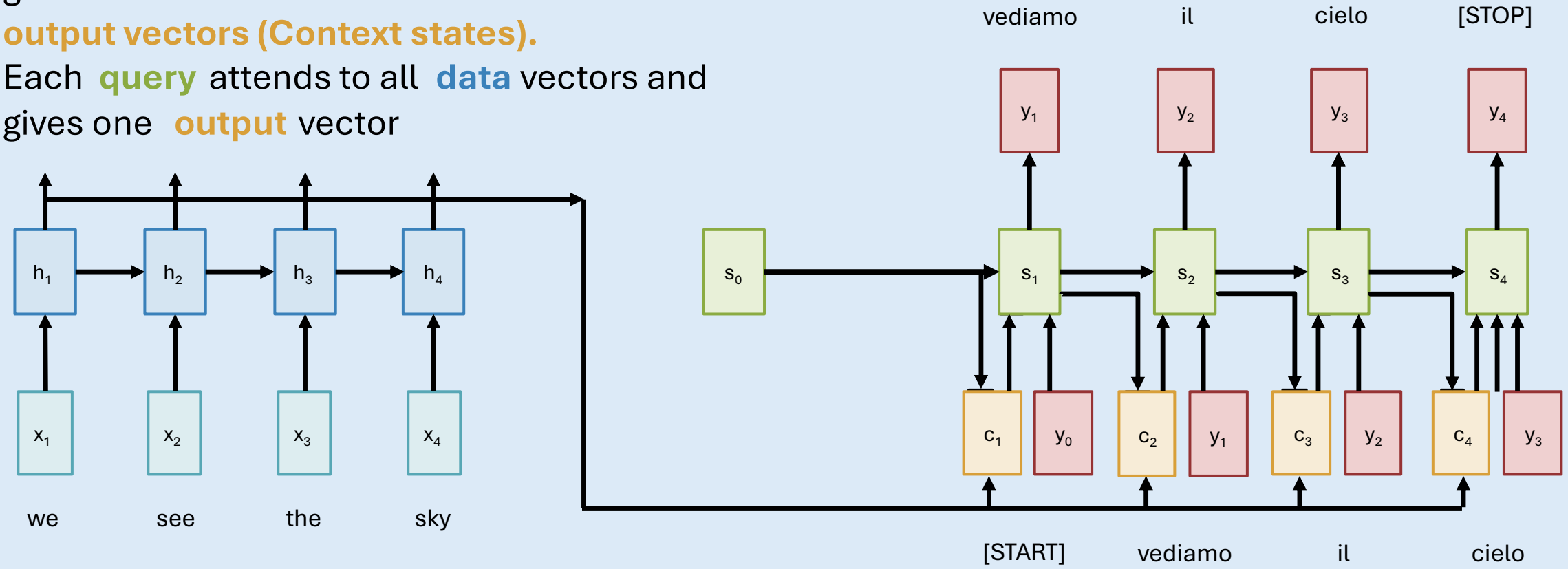
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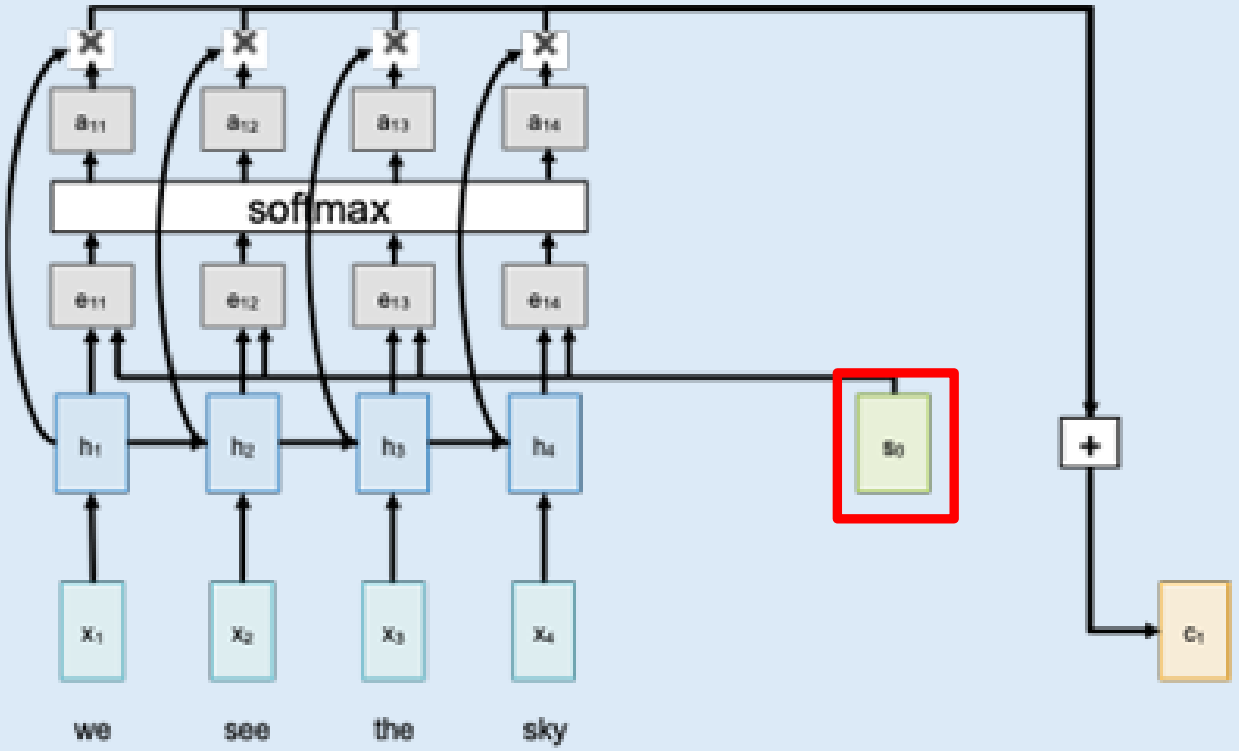
Query vectors (decoder RNN states) and
data vectors (encoder RNN states)
get transformed to
output vectors (Context states).
Each **query** attends to all **data** vectors and gives one **output** vector

There's a general operator hiding here:



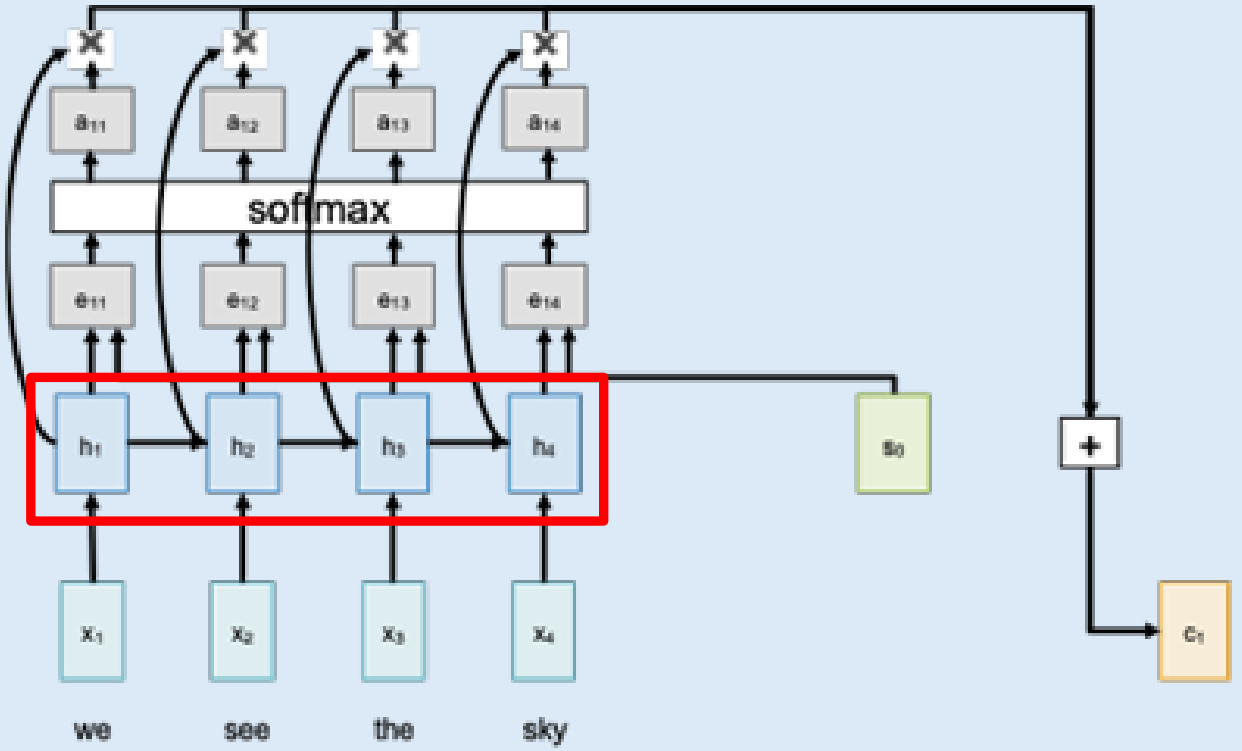
Attention Layer

Inputs:
Query vector: **q** $[D_Q]$



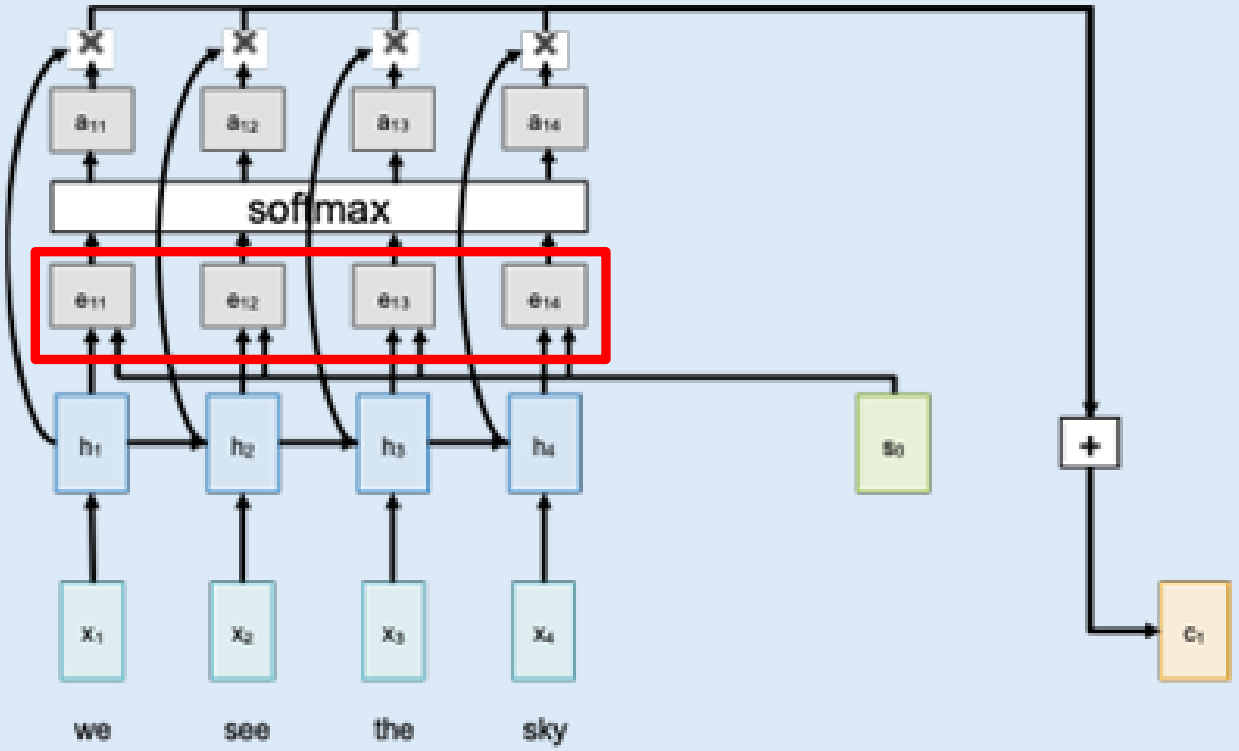
Attention Layer

Inputs:
Query vector: **q** $[D_Q]$
Data vectors: **X** $[N_X \times D_X]$



Attention Layer

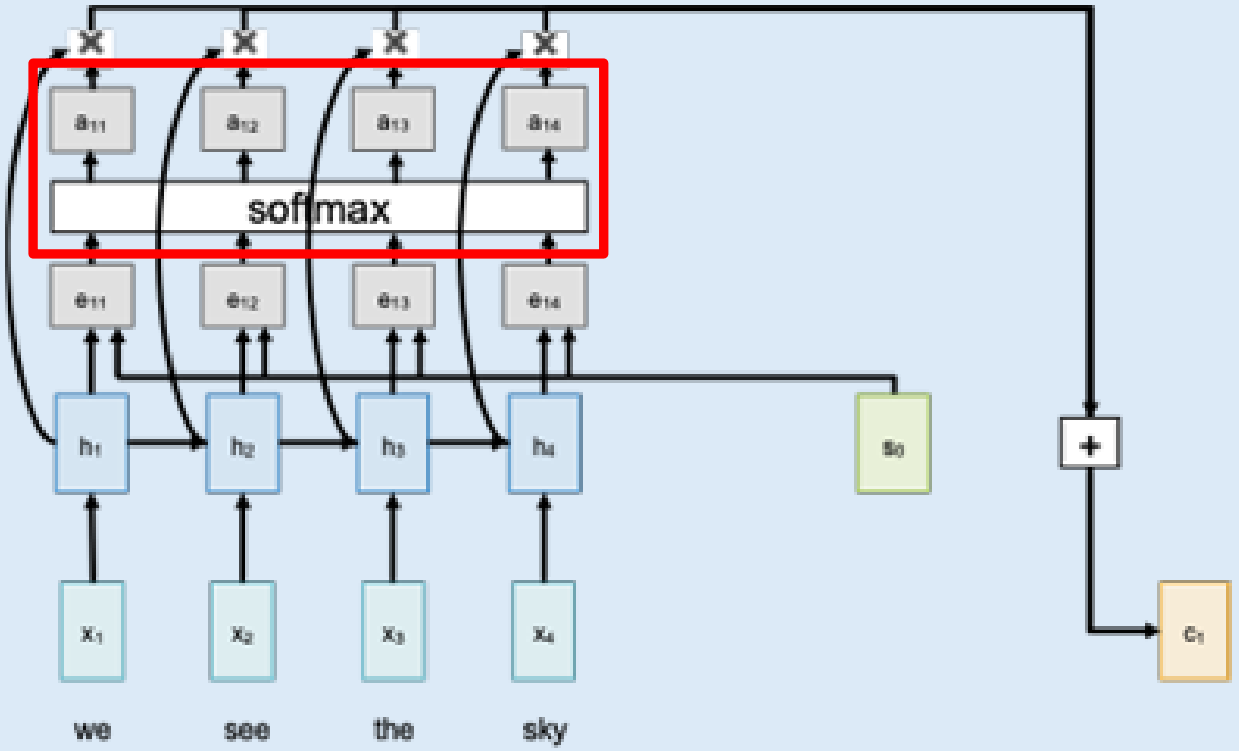
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Query vector: **q** $[D_Q]$
Data vectors: **X** $[N_X \times D_X]$



Computation:
Similarities: e $[N_X]$ $e_i = f_{att}(\mathbf{q}, \mathbf{X}_i)$

Attention Layer

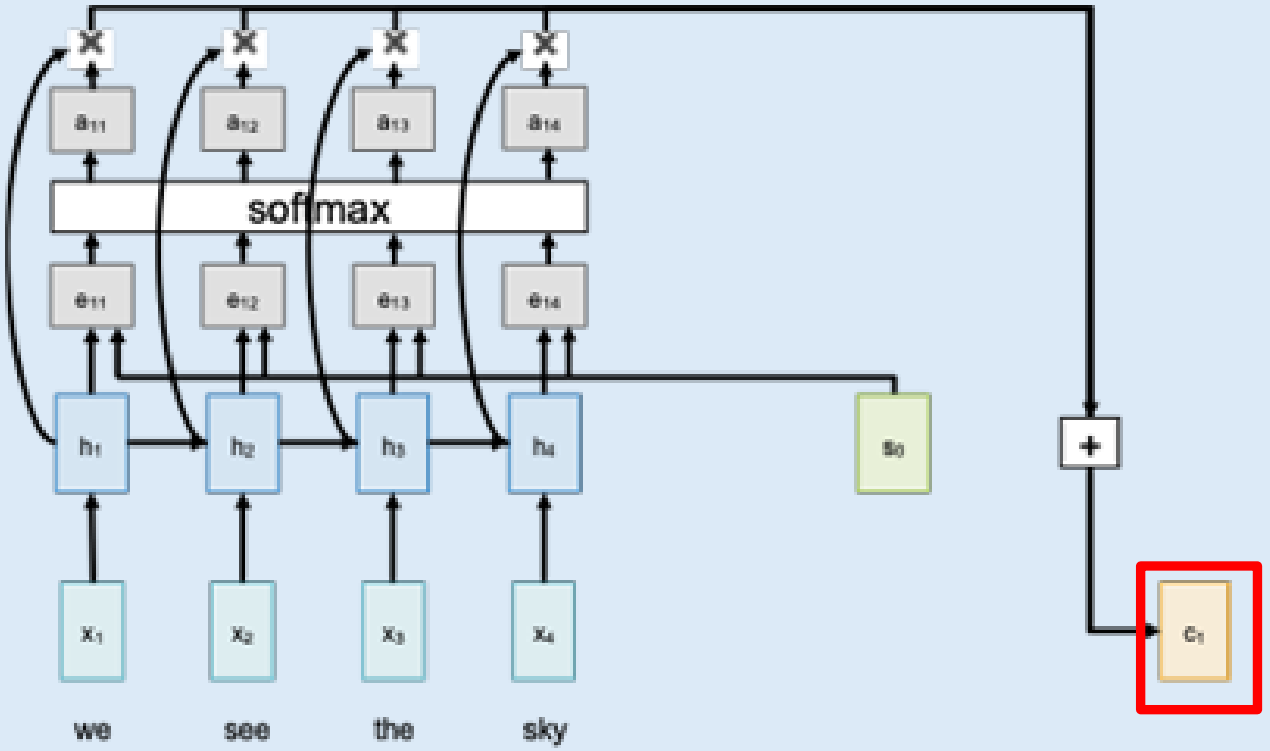
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Computation:
Similarities: e $[N_X]$ $e_i = f_{att}(\mathbf{q}, \mathbf{X}_i)$
Attention weights: a = softmax(e) $[N_X]$

Attention Layer

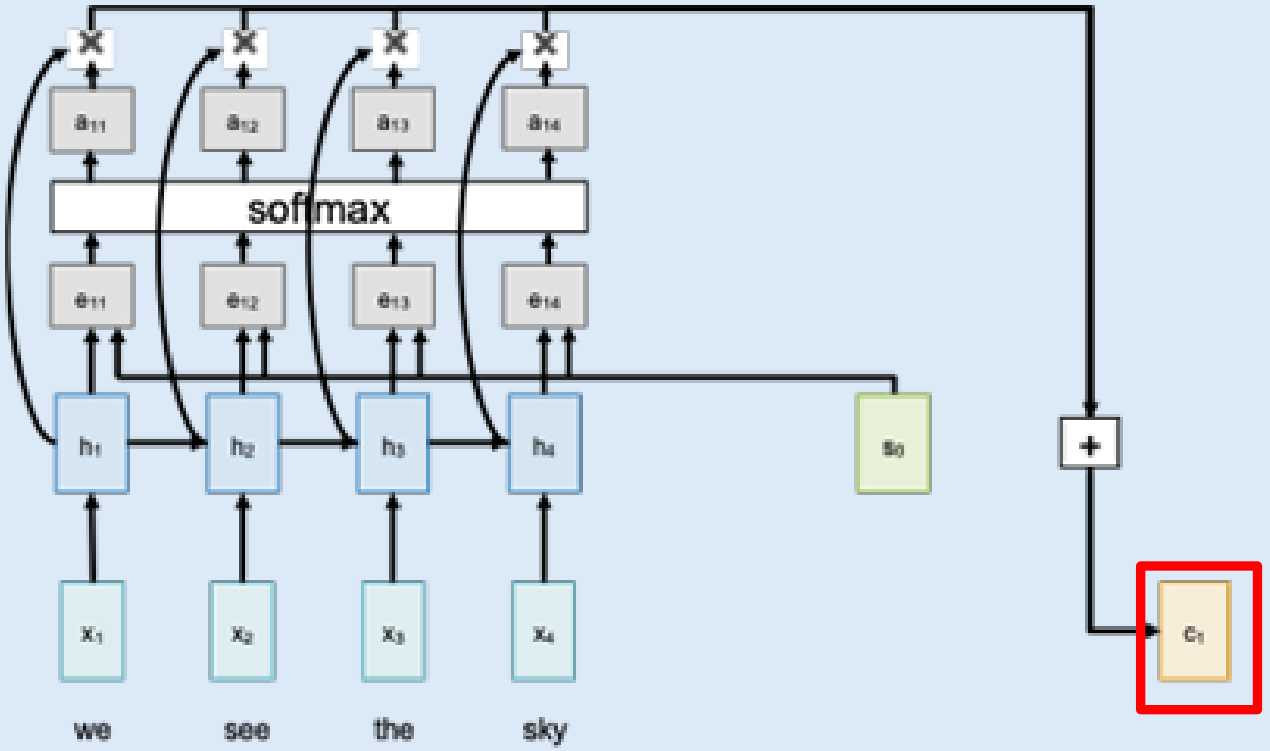
Inputs:
Query vector: **q** $[D_Q]$
Data vectors: **X** $[N_X \times D_X]$



Computation:
Similarities: e $[N_X]$ $e_i = f_{att}(\mathbf{q}, \mathbf{X}_i)$
Attention weights: a = softmax(e) $[N_X]$
Output vector: **y** $= \sum_i a_i \mathbf{X}_i$ $[D_X]$

Attention Layer

Inputs:
Query vector: **q** $[D_Q]$
Data vectors: **X** $[N_X \times D_X]$

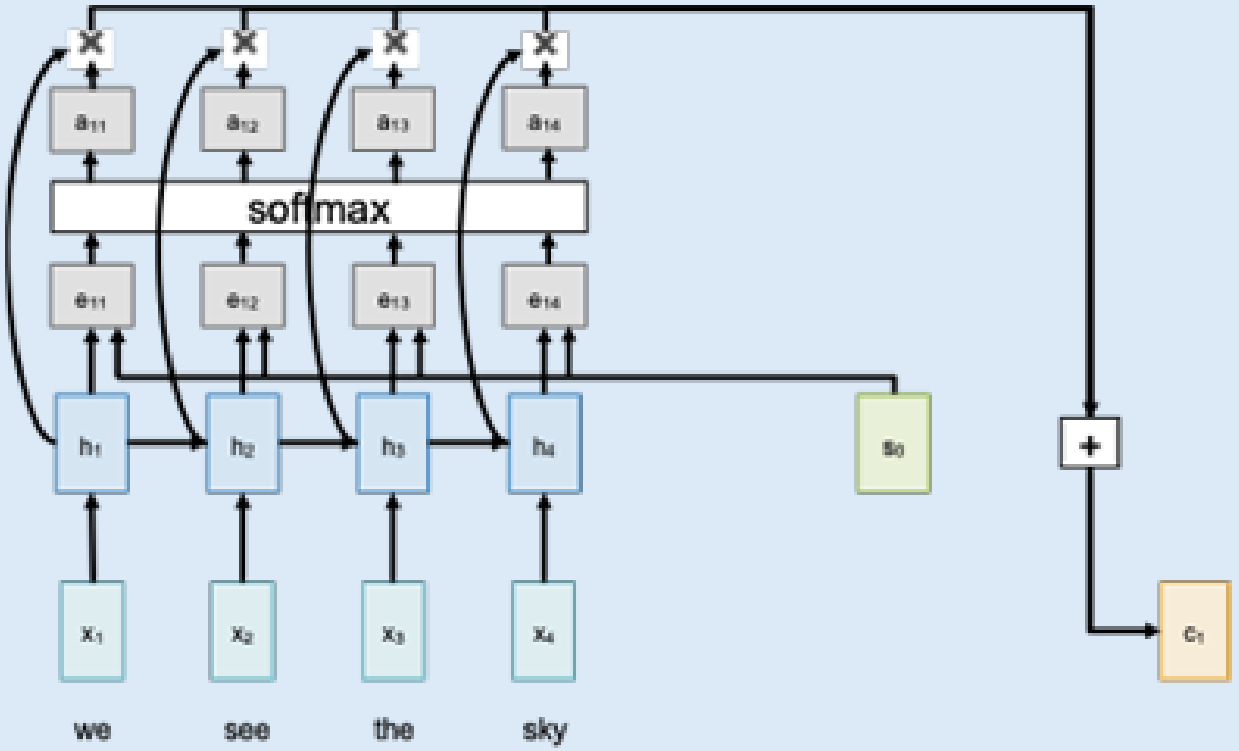


Computation:
Similarities: e $[N_X]$ $e_i = f_{att}(\mathbf{q}, \mathbf{X}_i)$
Attention weights: a = softmax(e) $[N_X]$
Output vector: **y** $= \sum_i a_i \mathbf{X}_i$ $[D_X]$

Let's generalize this!

Attention Layer

Inputs:
Query vector: q $[D_X]$
Data vectors: X $[N_X \times D_X]$

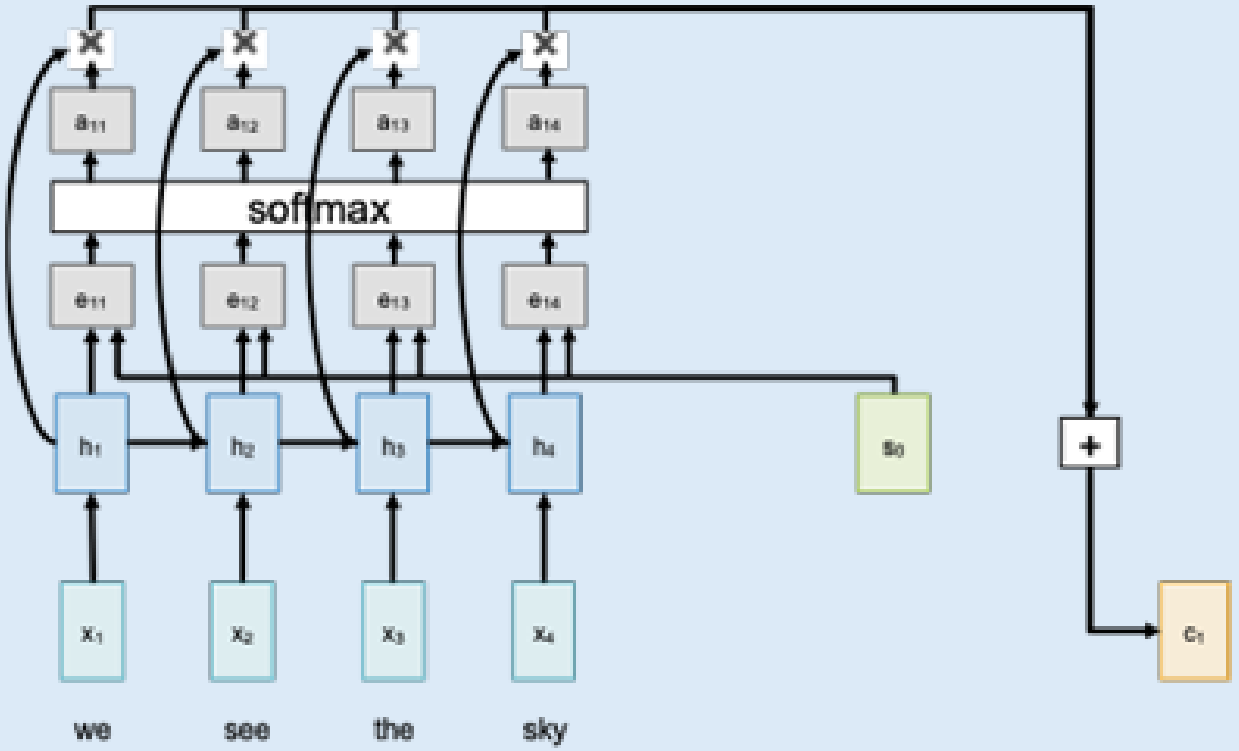


Computation:
Similarities: e $[N_X]$ $e_i = q \cdot X_i$
Attention weights: a = softmax(e) $[N_X]$
Output vector: y = $\sum_i a_i X_i$ $[D_X]$

Changes
- Use dot product for similarity
- Requires dimensions to match

Attention Layer

Inputs:
Query vector: **q** $[D_X]$
Data vectors: **X** $[N_X \times D_X]$



Computation:
Similarities: e $[N_X]$ $e_i = \mathbf{q} \cdot \mathbf{X}_i / \sqrt{D_X}$
Attention weights: a = softmax(e) $[N_X]$
Output vector: **y** $= \sum_i a_i \mathbf{X}_i$ $[D_X]$

Changes
- Use **scaled** dot product for similarity

Attention Layer

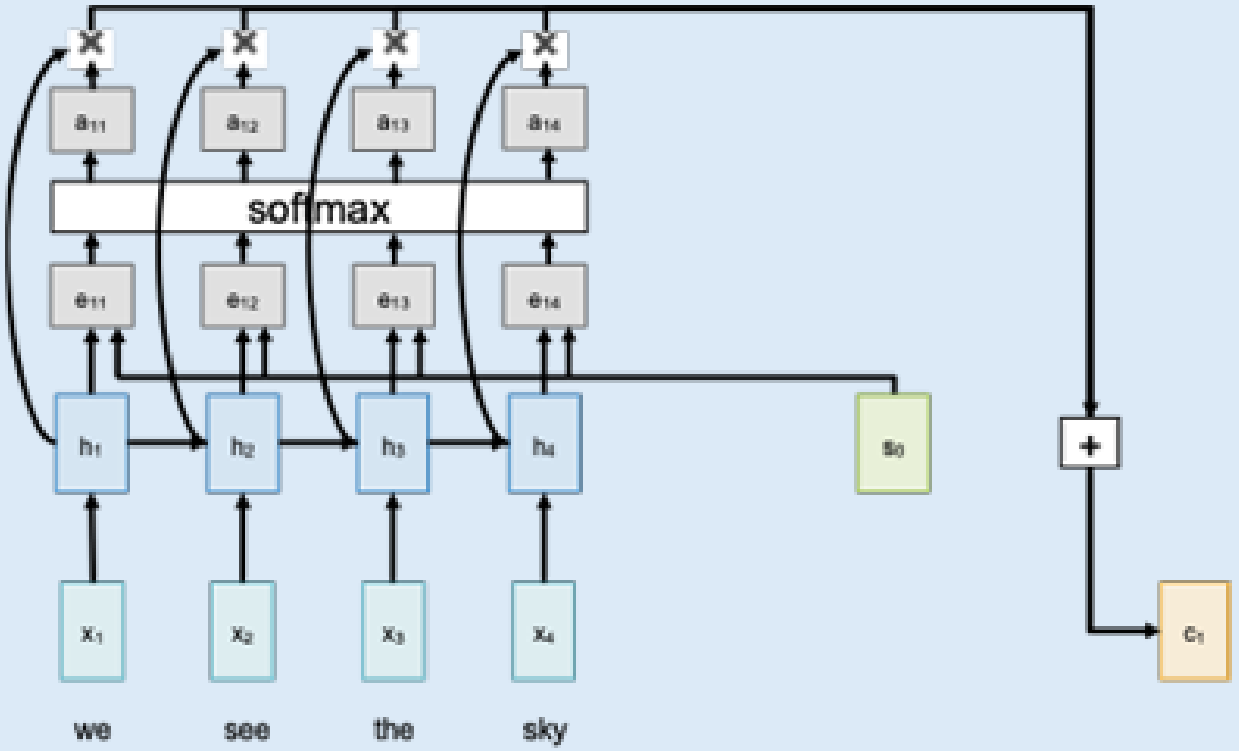
Inputs:
Query vector: **q** $[D_X]$
Data vectors: **X** $[N_X \times D_X]$

Large similarities will cause softmax to saturate and give vanishing gradients
Recall $a \cdot b = |a||b| \cos(\text{angle})$
Suppose that a and b are constant vectors of dimension D

Then $||a|| = \sqrt{\sum_i a^2} = |a|\sqrt{D}$

Computation:

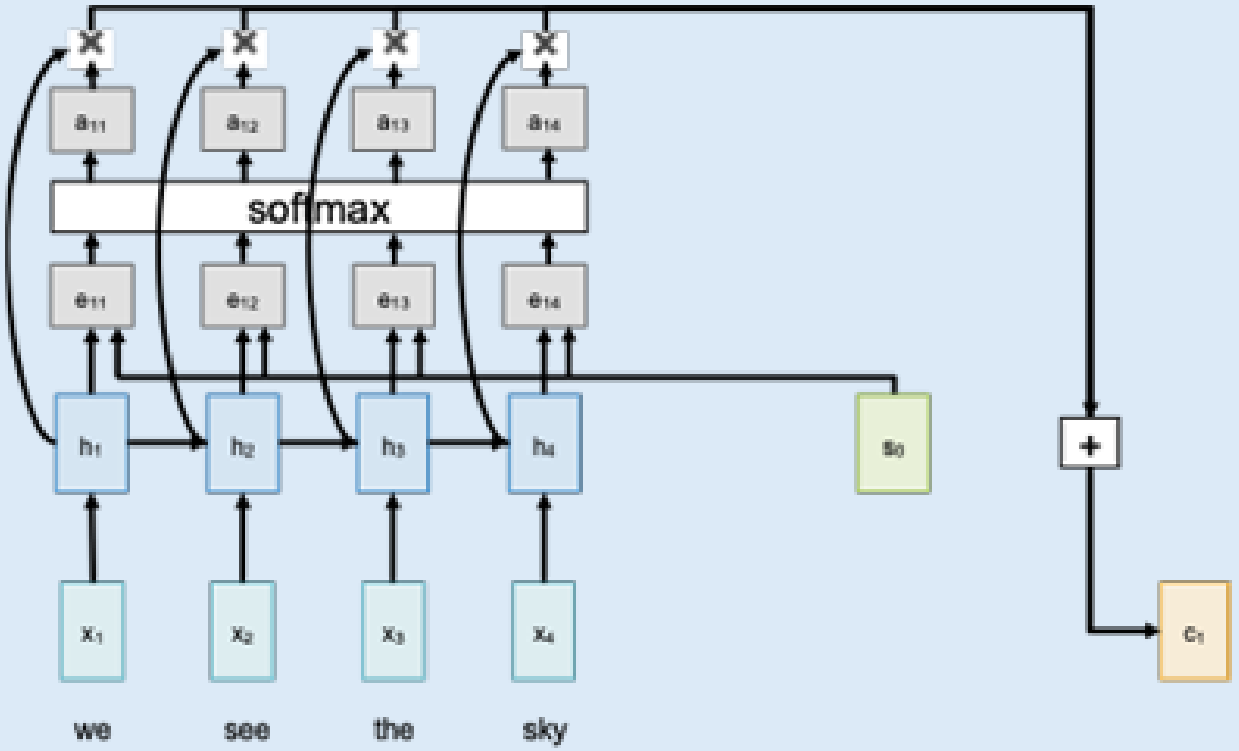
Similarities: e $[N_X]$ $e_i = \mathbf{q} \cdot \mathbf{X}_i / \sqrt{D_X}$
Attention weights: a = softmax(e) $[N_X]$
Output vector: **y** $= \sum_i a_i \mathbf{X}_i$ $[D_X]$



Changes
- Use **scaled** dot product for similarity

Attention Layer

Inputs:
Query vector: **Q** $[N_Q \times D_X]$
Data vectors: **X** $[N_X \times D_X]$



Computation:

Similarities: $E = \mathbf{Q}\mathbf{X}^T / \sqrt{D_X}$ $[N_Q \times N_X]$

$E_{ij} = \mathbf{Q}_i \cdot \mathbf{X}_j / \sqrt{D_X}$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N_Q \times N_X]$

Output vector: $\mathbf{Y} = A\mathbf{X}$ $[N_Q \times D_X]$

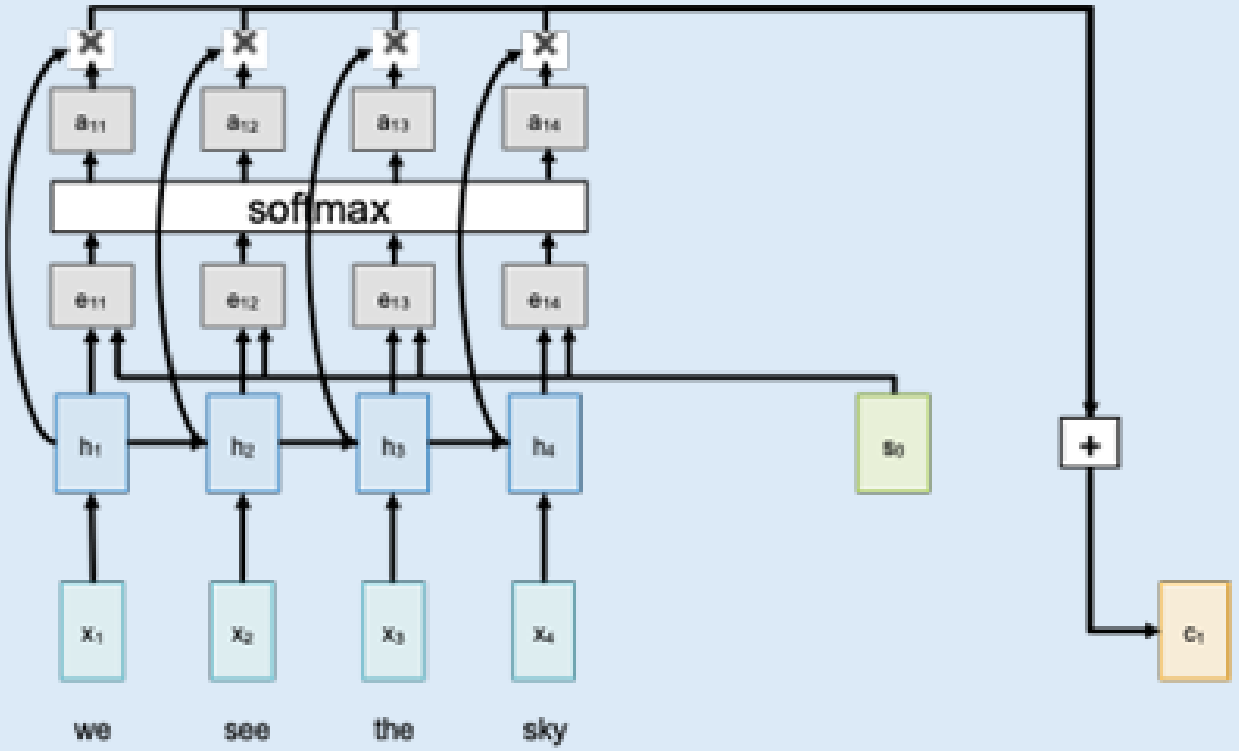
$\mathbf{Y}_i = \sum_j A_{ij} \mathbf{X}_j$

Changes

- Use scaled dot product for similarity
- Multiple **query** vectors

Attention Layer

Inputs:
Query vector: Q $[N_Q \times D_Q]$
Data vectors: X $[N_X \times D_X]$
Key matrix: W_K $[D_X \times D_Q]$
Value matrix: W_V $[D_X \times D_V]$



Computation:
Keys: $K = XW_K$ $[N_X \times D_Q]$
Values: $V = XW_V$ $[N_X \times D_V]$
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N_Q \times N_X]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: A = softmax(E, dim=1) $[N_Q \times N_X]$
Output vector: Y = AV $[N_Q \times D_V]$

$$Y_i = \sum_j A_{ij} V_j$$

Changes
- Use scaled dot product for similarity
- Multiple **query** vectors
- Separate **key** and **value**
- No longer requires $D_Q = D_X$

Attention Layer

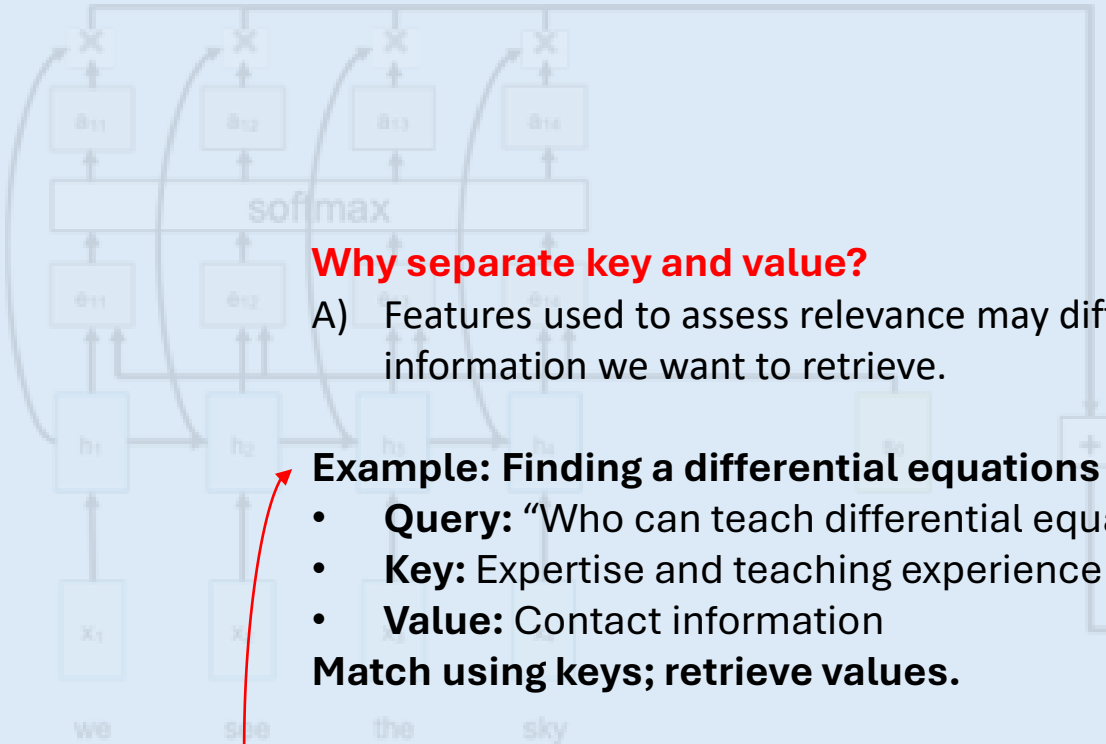
Inputs:
Query vector: Q $[N_Q \times D_Q]$
Data vectors: X $[N_X \times D_X]$
Key matrix: W_K $[D_X \times D_Q]$
Value matrix: W_V $[D_X \times D_V]$

Computation:
Keys: $K = XW_K$ $[N_X \times D_Q]$
Values: $V = XW_V$ $[N_X \times D_V]$
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N_Q \times N_X]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N_Q \times N_X]$
Output vector: $Y = AV$ $[N_Q \times D_V]$

$$Y_i = \sum_j A_{ij} V_j$$



Why separate key and value?

A) Features used to assess relevance may differ from the information we want to retrieve.

Example: Finding a differential equations instructor

- **Query:** "Who can teach differential equations?"
- **Key:** Expertise and teaching experience
- **Value:** Contact information

Match using keys; retrieve values.

Changes

- Use scaled dot product for similarity
- Multiple **query** vectors
- Separate **key** and **value**
- No longer requires $D_Q = D_X$

Attention Layer

Inputs:

Query vector: Q $[N_Q \times D_Q]$

Data vectors: X $[N_X \times D_X]$

Key matrix: W_K $[D_X \times D_Q]$

Value matrix: W_V $[D_X \times D_V]$

Computation:

Keys: $K = XW_K$ $[N_X \times D_Q]$

Values: $V = XW_V$ $[N_X \times D_V]$

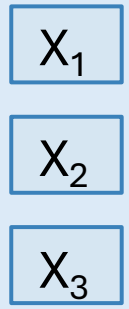
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N_Q \times N_X]$

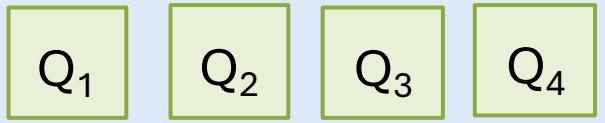
$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N_Q \times N_X]$

Output vector: $Y = AV$ $[N_Q \times D_V]$

$$Y_i = \sum_j A_{ij} V_j$$





Attention Layer

Inputs:

Query vector: Q $[N_Q \times D_Q]$

Data vectors: X $[N_X \times D_X]$

Key matrix: W_K $[D_X \times D_Q]$

Value matrix: W_V $[D_X \times D_V]$

Computation:

Keys: $K = XW_K$ $[N_X \times D_Q]$

Values: $V = XW_V$ $[N_X \times D_V]$

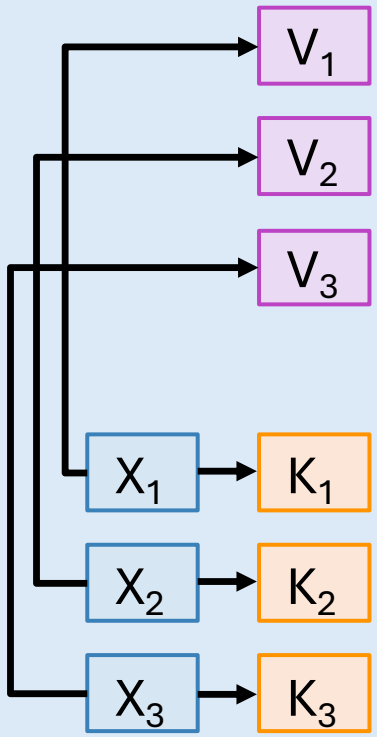
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N_Q \times N_X]$

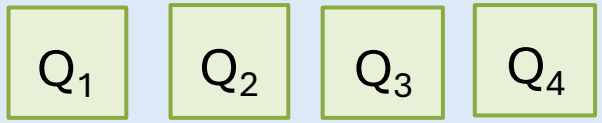
$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N_Q \times N_X]$

Output vector: $Y = AV$ $[N_Q \times D_V]$

$$Y_i = \sum_j A_{ij} V_j$$





Attention Layer

Inputs:

Query vector: Q $[N_Q \times D_Q]$

Data vectors: X $[N_X \times D_X]$

Key matrix: W_K $[D_X \times D_Q]$

Value matrix: W_V $[D_X \times D_V]$

Computation:

Keys: $K = XW_K$ $[N_X \times D_Q]$

Values: $V = XW_V$ $[N_X \times D_V]$

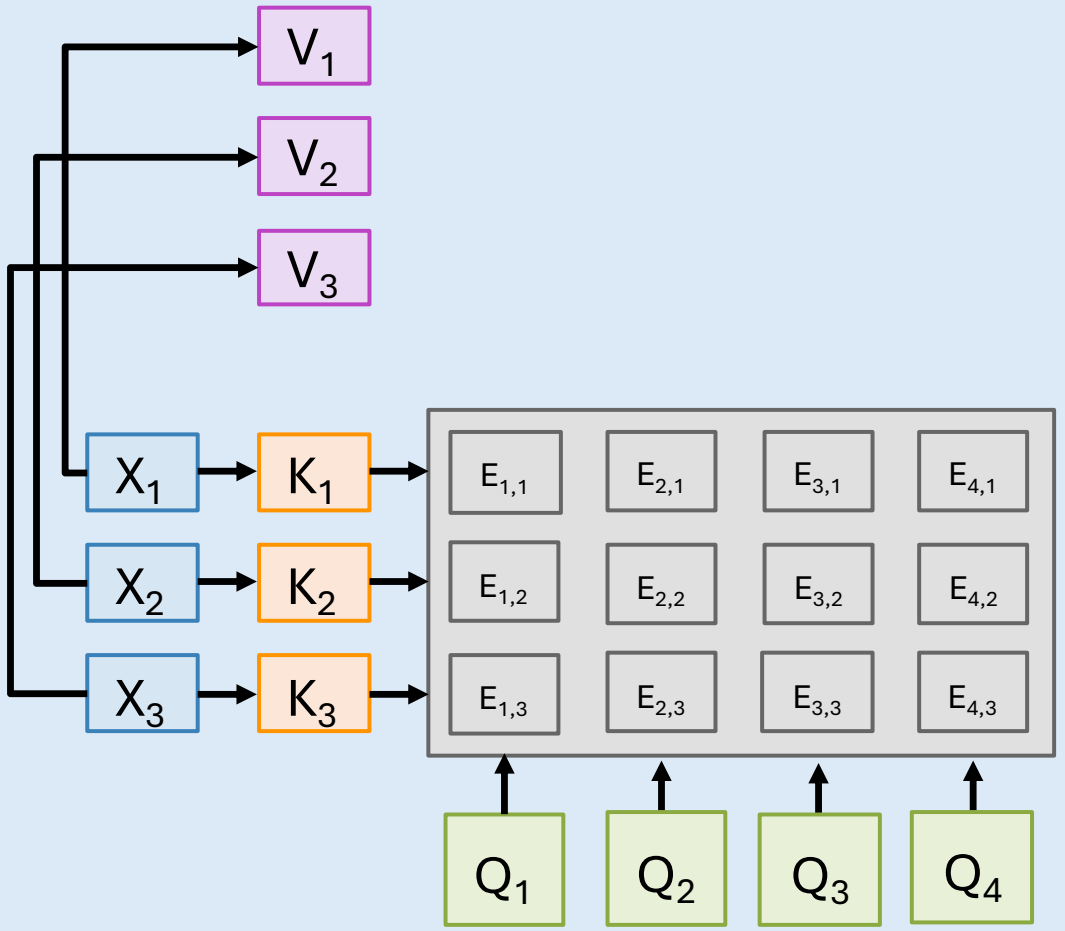
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N_Q \times N_X]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N_Q \times N_X]$

Output vector: $Y = AV$ $[N_Q \times D_V]$

$$Y_i = \sum_j A_{ij} V_j$$



Attention Layer

Inputs:
Query vector: Q $[N_Q \times D_Q]$
Data vectors: X $[N_X \times D_X]$
Key matrix: W_K $[D_X \times D_Q]$
Value matrix: W_V $[D_X \times D_V]$

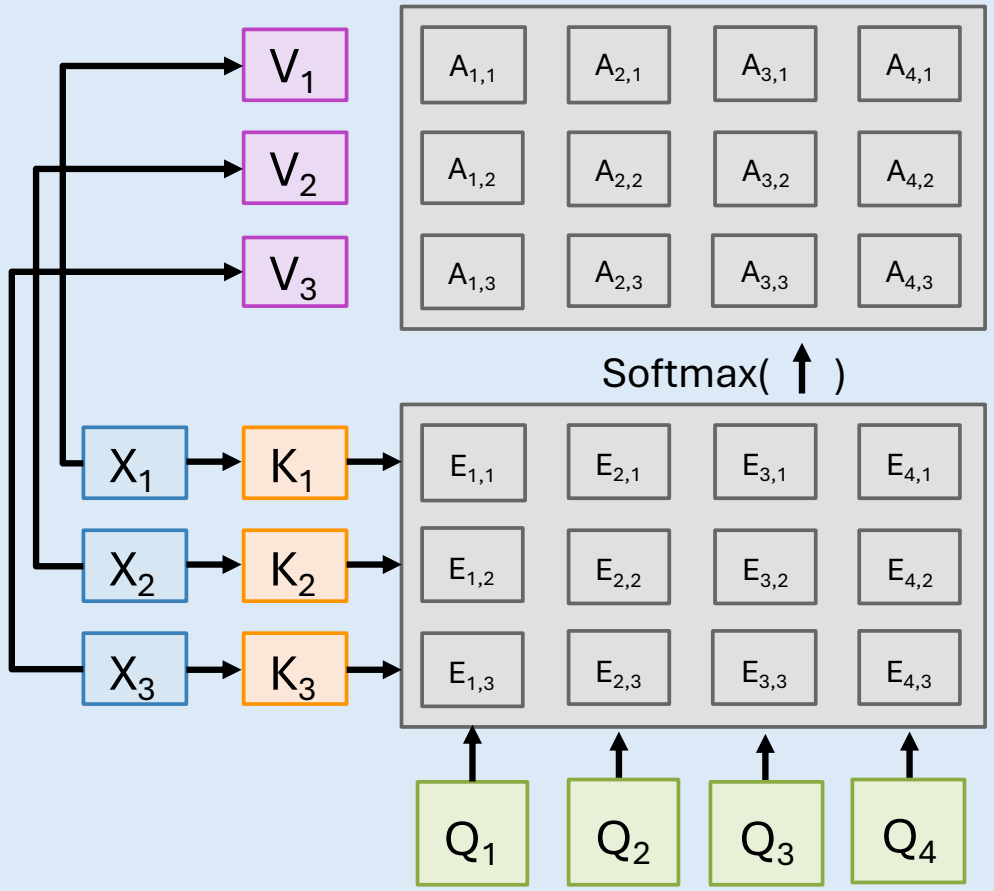
Computation:
Keys: $K = XW_K$ $[N_X \times D_Q]$
Values: $V = XW_V$ $[N_X \times D_V]$
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N_Q \times N_X]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: A = softmax(E, dim=1) $[N_Q \times N_X]$
Output vector: Y = AV $[N_Q \times D_V]$

$$Y_i = \sum_j A_{ij} V_j$$

Softmax normalizes each column: each **query** predicts a distribution over the **keys**



Attention Layer

Inputs:
Query vector: Q $[N_Q \times D_Q]$
Data vectors: X $[N_X \times D_X]$
Key matrix: W_K $[D_X \times D_Q]$
Value matrix: W_V $[D_X \times D_V]$

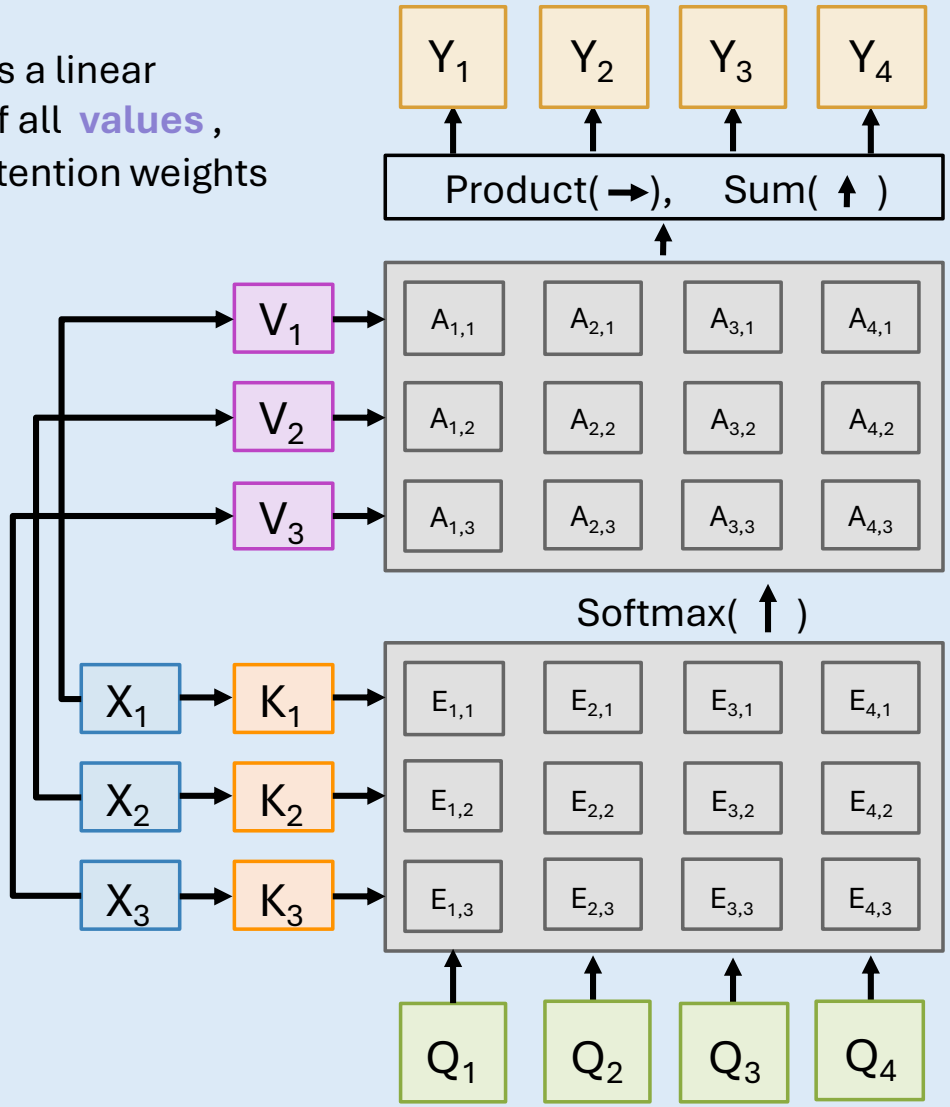
Computation:
Keys: $K = XW_K$ $[N_X \times D_Q]$
Values: $V = XW_V$ $[N_X \times D_V]$
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N_Q \times N_X]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: A = softmax(E, dim=1) $[N_Q \times N_X]$
Output vector: $Y = AV$ $[N_Q \times D_V]$

$$Y_i = \sum_j A_{ij} V_j$$

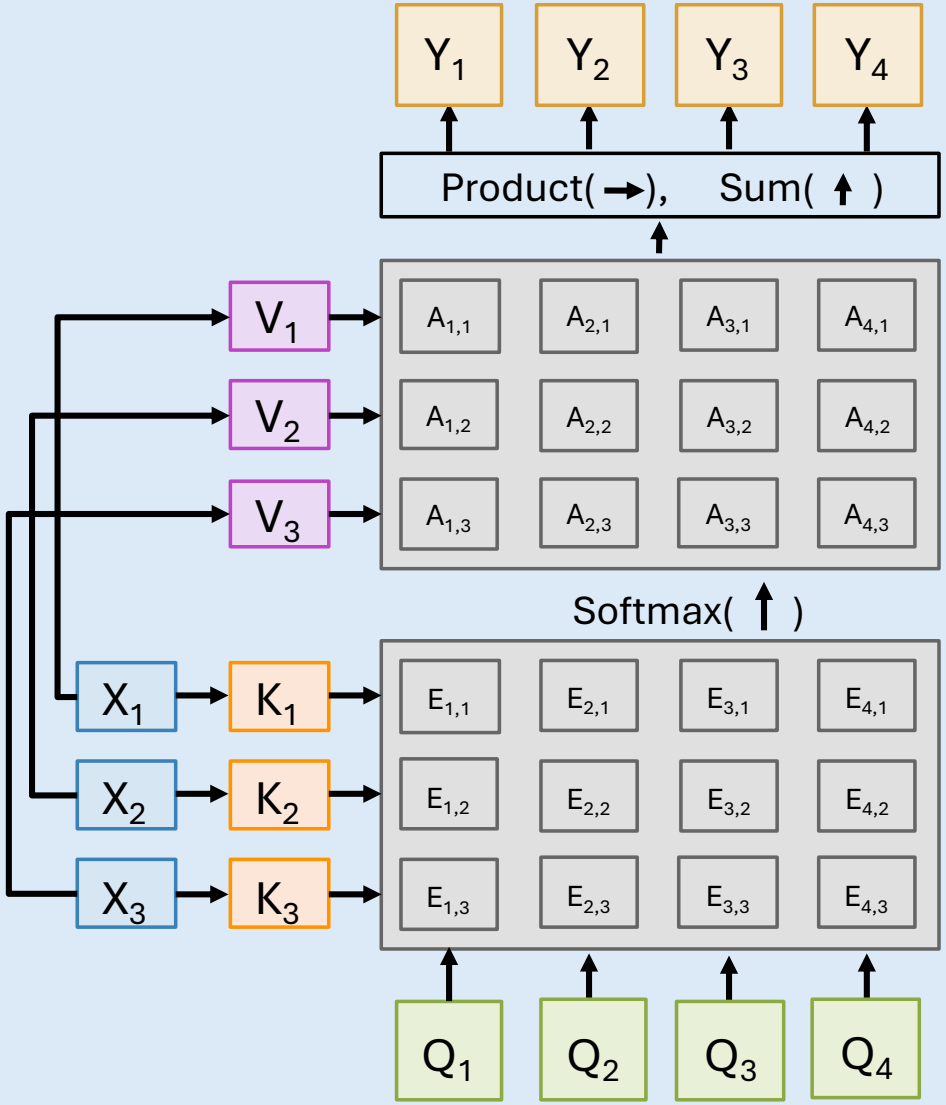
Each **output** is a linear combination of all **values**, weighted by attention weights



Cross-Attention Layer

Inputs:
Query vector: Q $[N_Q \times D_Q]$
Data vectors: X $[N_X \times D_X]$
Key matrix: W_K $[D_X \times D_Q]$
Value matrix: W_V $[D_X \times D_V]$

Each **query** produces one **output**, which is a mix of information in the **data** vectors



Computation:
Keys: $K = XW_K$ $[N_X \times D_Q]$
Values: $V = XW_V$ $[N_X \times D_V]$
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N_Q \times N_X]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: A = softmax(E, dim=1) $[N_Q \times N_X]$
Output vector: $Y = AV$ $[N_Q \times D_V]$

$$Y_i = \sum_j A_{ij} V_j$$

Self-Attention Layer

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Each **input** produces one **output**, which is a mix of information from all **inputs**

Shapes get a little simpler:

- N input vectors, each D_{in}
- Almost always $D_Q = D_V = D_{out}$

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

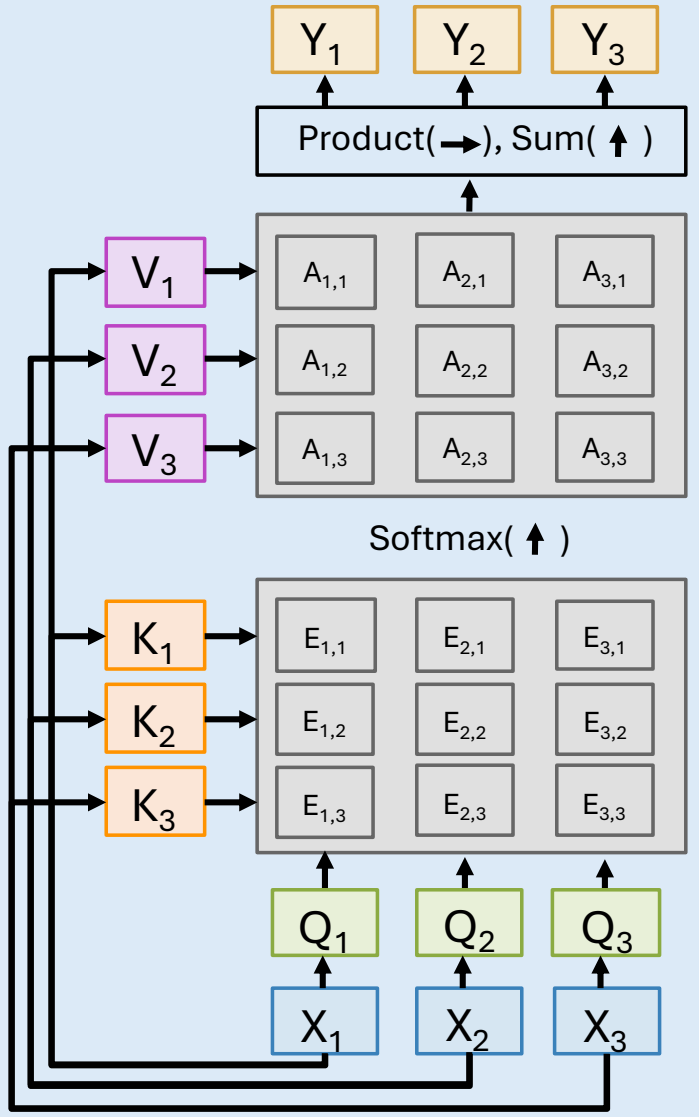
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$



Self-Attention Layer

Inputs:

Input vectors: X [N x D_{in}]

Key matrix: W_K [D_{in} x D_{out}]

Value matrix: W_V [D_{in} x D_{out}]

Query matrix: W_Q [D_{in} x D_{out}]

Each **input** produces one **output**, which is a mix of information from all **inputs**

Computation:

Queries: $Q = XW_Q$ [N x D_{out}]

Keys: $K = XW_K$ [N x D_{out}]

Values: $V = XW_V$ [N x D_{out}]

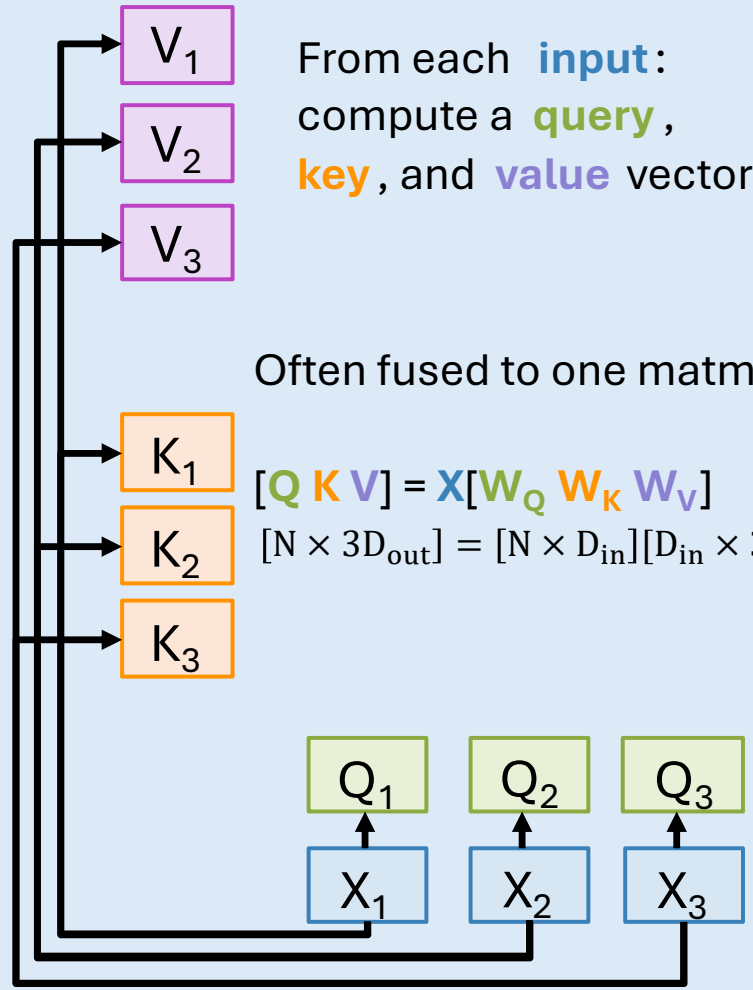
Similarities: $E = QK^T / \sqrt{D_Q}$ [N x N]

$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$

Attention weights: A = softmax(E, dim=1) [N x N]

Output vector: $Y = AV$ [N x D_{out}]

$Y_i = \sum_j A_{ij} V_j$



From each **input**: compute a **query**, **key**, and **value** vector

Often fused to one matmul:

$[Q\ K\ V] = X[W_Q\ W_K\ W_V]$

$[N \times 3D_{out}] = [N \times D_{in}][D_{in} \times 3D_{out}]$

Self-Attention Layer

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Each **input** produces one **output**, which is a mix of information from all **inputs**

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

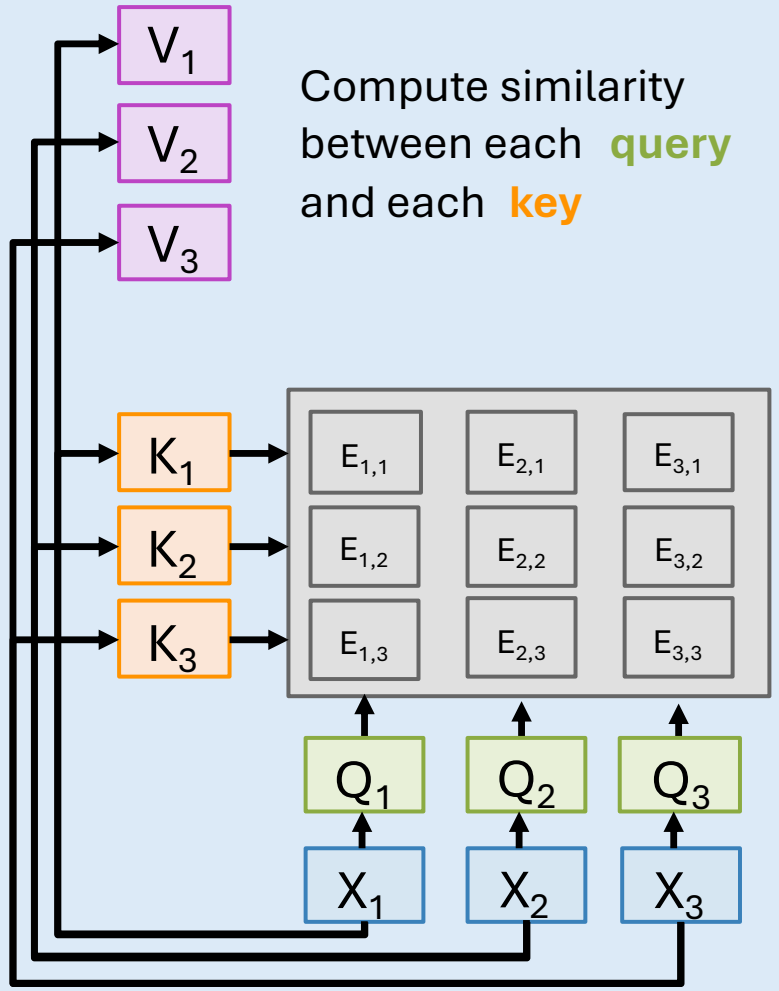
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$



Compute similarity between each **query** and each **key**

Self-Attention Layer

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Each **input** produces one **output**, which is a mix of information from all **inputs**

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

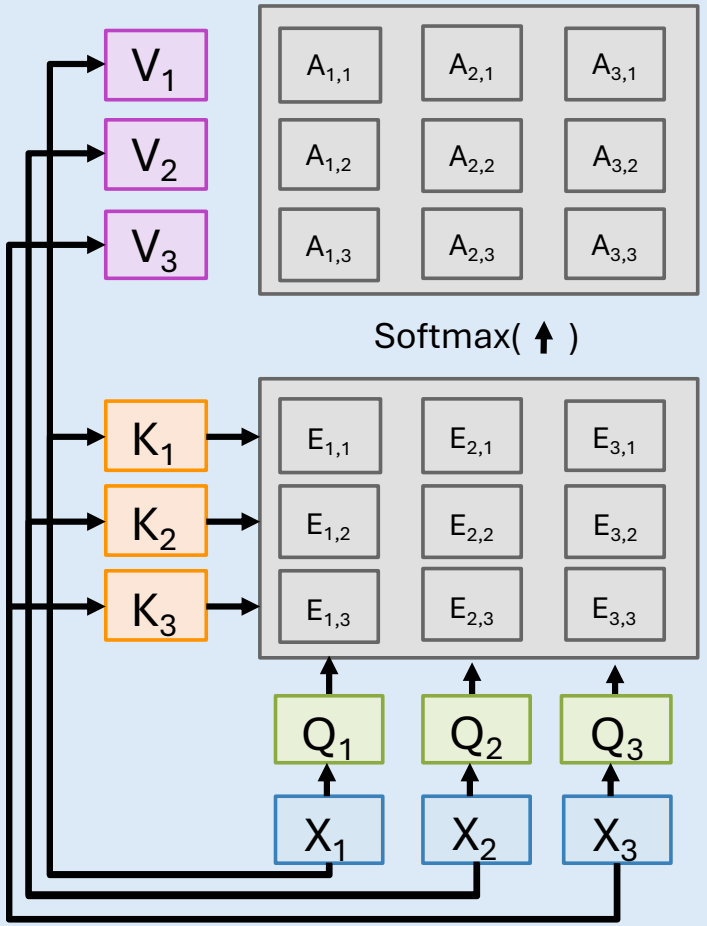
$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$

Normalize over each column: each **query** computes a distribution over **keys**



Self-Attention Layer

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Each **input** produces one **output**, which is a mix of information from all **inputs**

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

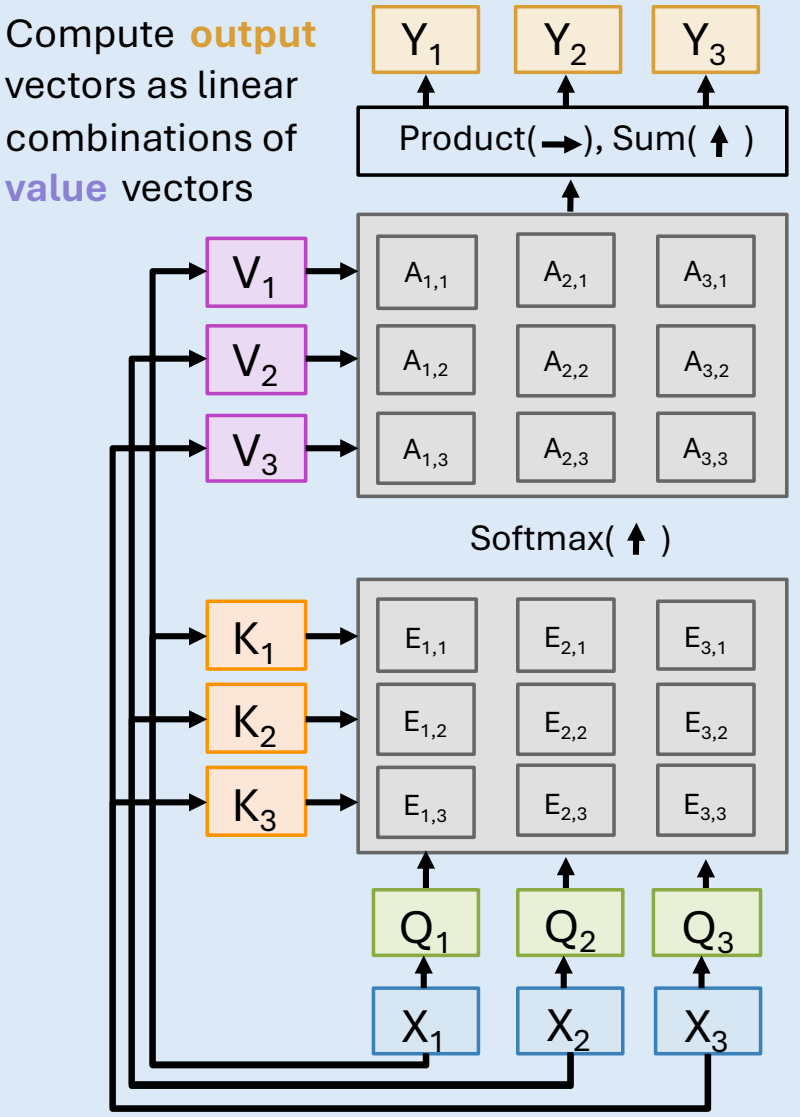
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$



Self-Attention Layer

Consider permuting **inputs**:

Inputs:

Input vectors: X [N x D_{in}]

Key matrix: W_K [D_{in} x D_{out}]

Value matrix: W_V [D_{in} x D_{out}]

Query matrix: W_Q [D_{in} x D_{out}]

Computation:

Queries: $Q = XW_Q$ [N x D_{out}]

Keys: $K = XW_K$ [N x D_{out}]

Values: $V = XW_V$ [N x D_{out}]

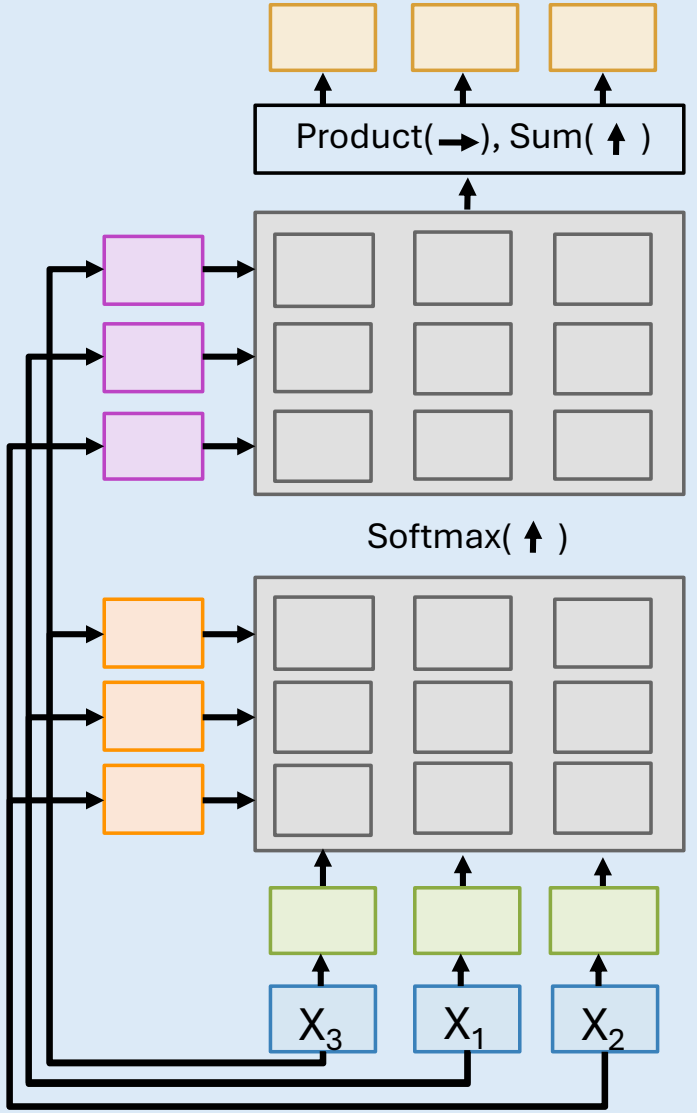
Similarities: $E = QK^T / \sqrt{D_Q}$ [N x N]

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: A = softmax(E, dim=1) [N x N]

Output vector: $Y = AV$ [N x D_{out}]

$$Y_i = \sum_j A_{ij} V_j$$



Self-Attention Layer

Inputs:
Input vectors: X [N x D_{in}]
Key matrix: W_K [D_{in} x D_{out}]
Value matrix: W_V [D_{in} x D_{out}]
Query matrix: W_Q [D_{in} x D_{out}]

Computation:
Queries: $Q = XW_Q$ [N x D_{out}]
Keys: $K = XW_K$ [N x D_{out}]
Values: $V = XW_V$ [N x D_{out}]
Similarities: $E = QK^T / \sqrt{D_Q}$ [N x N]

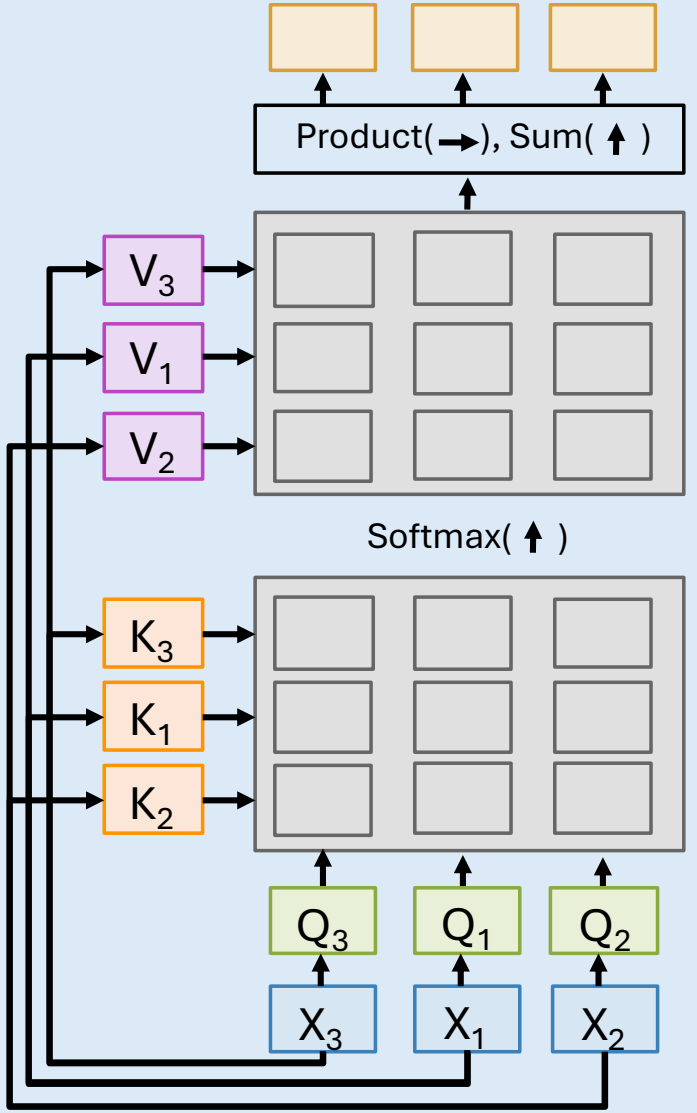
$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: A = softmax(E, dim=1) [N x N]
Output vector: $Y = AV$ [N x D_{out}]

$$Y_i = \sum_j A_{ij} V_j$$

Consider permuting **inputs**:

Queries, **keys**, and **values** will be the same but permuted



Self-Attention Layer

Inputs:
Input vectors: X $[N \times D_{in}]$
Key matrix: W_K $[D_{in} \times D_{out}]$
Value matrix: W_V $[D_{in} \times D_{out}]$
Query matrix: W_Q $[D_{in} \times D_{out}]$

Computation:
Queries: $Q = XW_Q$ $[N \times D_{out}]$
Keys: $K = XW_K$ $[N \times D_{out}]$
Values: $V = XW_V$ $[N \times D_{out}]$
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

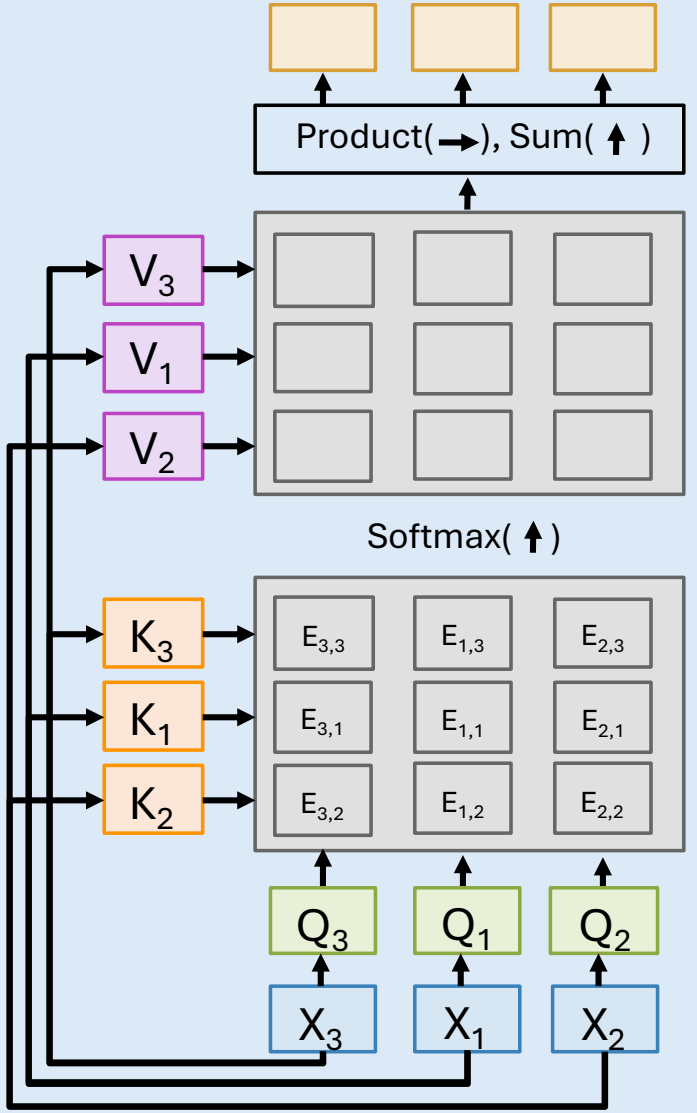
Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$
Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$

Consider permuting **inputs**:

Queries, **keys**, and **values** will be the same but permuted

Similarities are the same but permuted



Self-Attention Layer

Inputs:
Input vectors: X $[N \times D_{in}]$
Key matrix: W_K $[D_{in} \times D_{out}]$
Value matrix: W_V $[D_{in} \times D_{out}]$
Query matrix: W_Q $[D_{in} \times D_{out}]$

Computation:
Queries: $Q = XW_Q$ $[N \times D_{out}]$
Keys: $K = XW_K$ $[N \times D_{out}]$
Values: $V = XW_V$ $[N \times D_{out}]$
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$
Output vector: $Y = AV$ $[N \times D_{out}]$

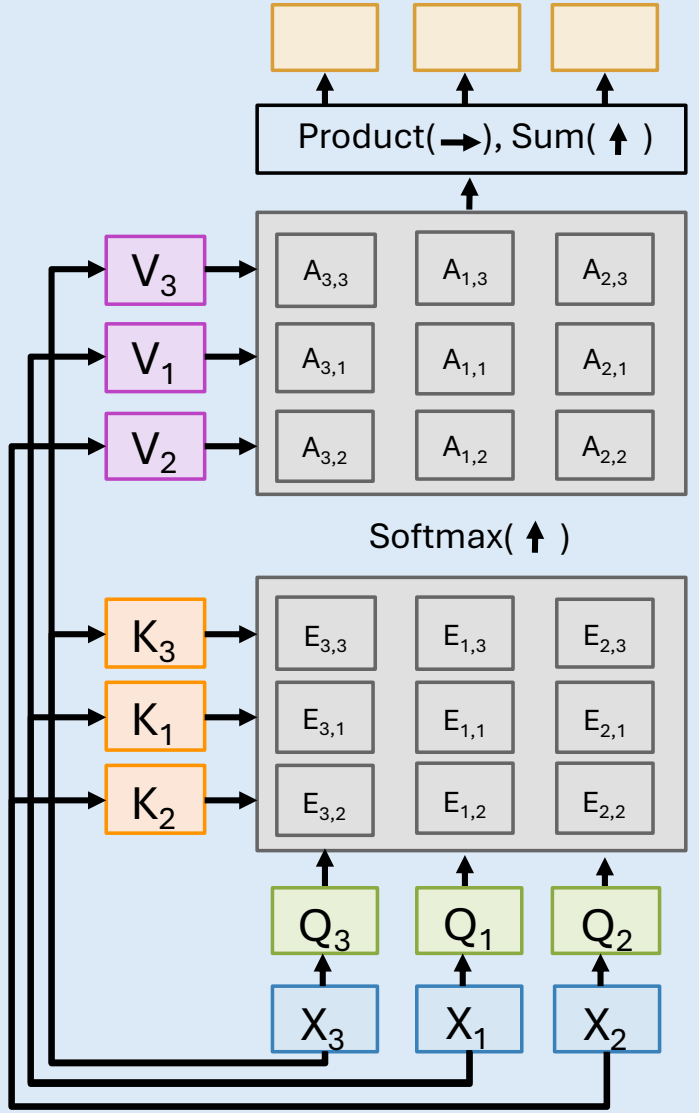
$$Y_i = \sum_j A_{ij} V_j$$

Consider permuting **inputs**:

Queries, **keys**, and **values** will be the same but permuted

Similarities are the same but permuted

Attention weights are the same but permuted



Self-Attention Layer

Inputs:
Input vectors: X $[N \times D_{in}]$
Key matrix: W_K $[D_{in} \times D_{out}]$
Value matrix: W_V $[D_{in} \times D_{out}]$
Query matrix: W_Q $[D_{in} \times D_{out}]$

Computation:
Queries: $Q = XW_Q$ $[N \times D_{out}]$
Keys: $K = XW_K$ $[N \times D_{out}]$
Values: $V = XW_V$ $[N \times D_{out}]$
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$
Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$

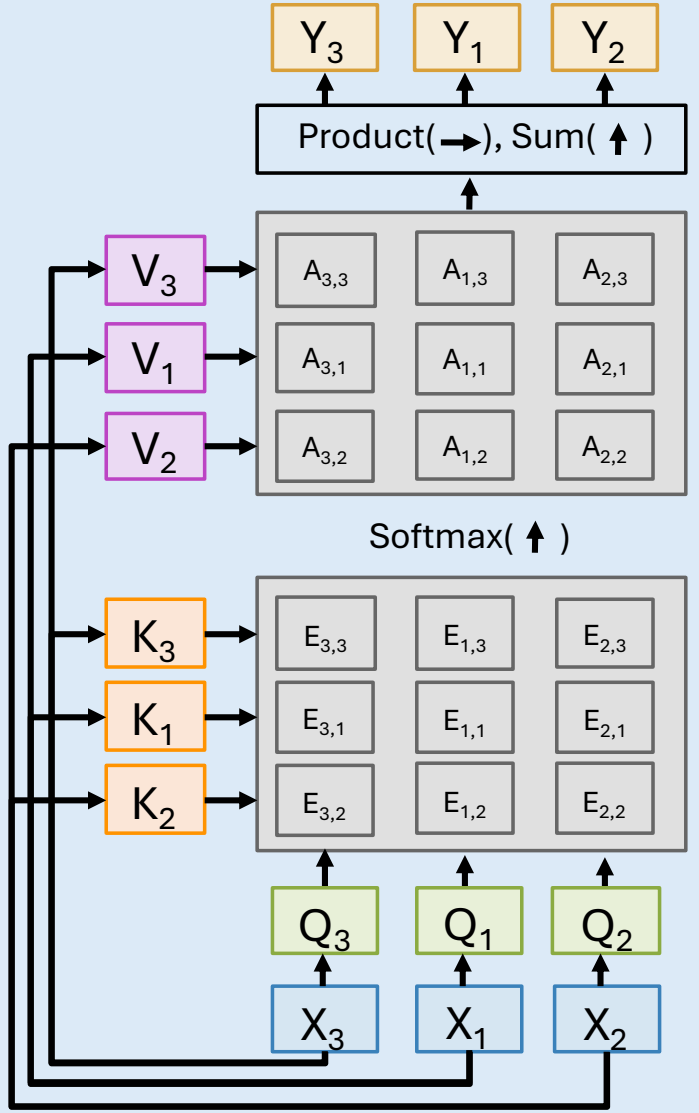
Consider permuting **inputs**:

Queries, **keys**, and **values** will be the same but permuted

Similarities are the same but permuted

Attention weights are the same but permuted

Outputs are the same but permuted



Self-Attention Layer

Inputs:
Input vectors: X $[N \times D_{in}]$
Key matrix: W_K $[D_{in} \times D_{out}]$
Value matrix: W_V $[D_{in} \times D_{out}]$
Query matrix: W_Q $[D_{in} \times D_{out}]$

Self-Attention is permutation equivariant:
$F(\sigma(X)) = \sigma(F(X))$

Computation:
Queries: $Q = XW_Q$ $[N \times D_{out}]$
Keys: $K = XW_K$ $[N \times D_{out}]$
Values: $V = XW_V$ $[N \times D_{out}]$

This means that Self-Attention works on *sets of vectors*

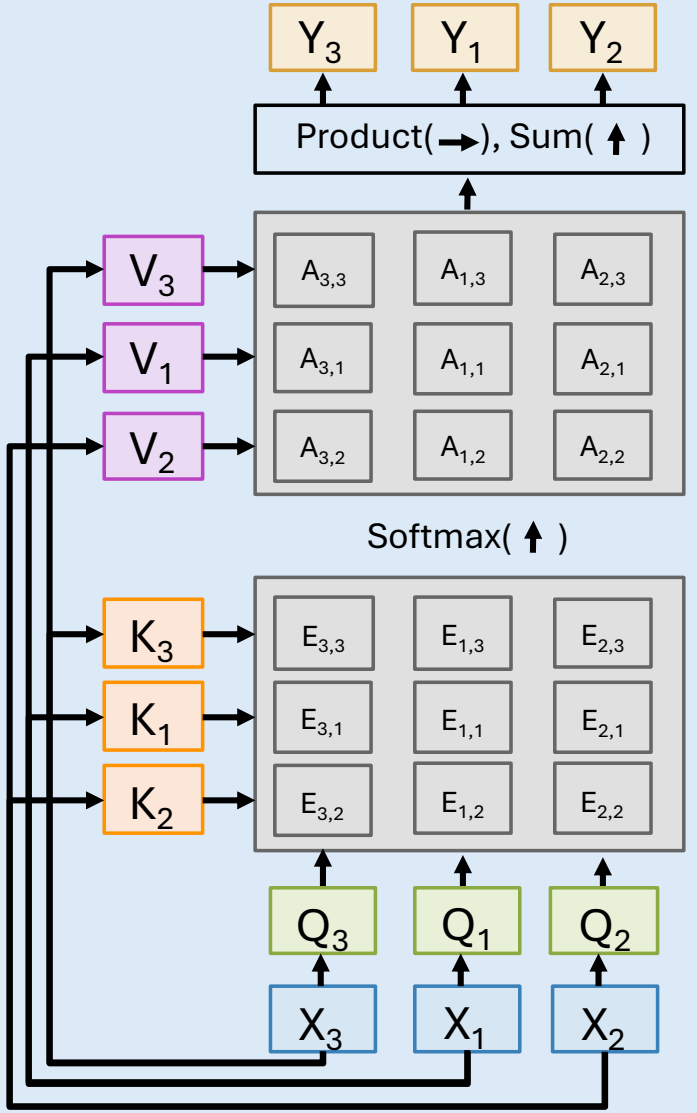
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$



Self-Attention Layer

Inputs:
Input vectors: X [N x D_{in}]
Key matrix: W_K [D_{in} x D_{out}]
Value matrix: W_V [D_{in} x D_{out}]
Query matrix: W_Q [D_{in} x D_{out}]

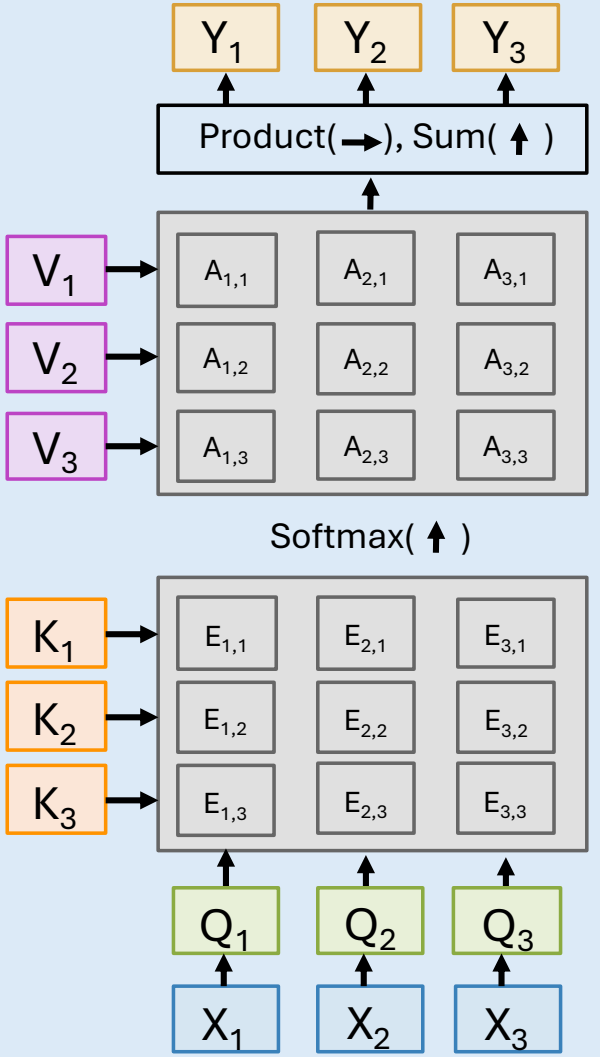
Problem : Self-Attention does not know the order of the sequence

Computation:
Queries: $Q = XW_Q$ [N x D_{out}]
Keys: $K = XW_K$ [N x D_{out}]
Values: $V = XW_V$ [N x D_{out}]
Similarities: $E = QK^T / \sqrt{D_Q}$ [N x N]

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: A = softmax(E, dim=1) [N x N]
Output vector: $Y = AV$ [N x D_{out}]

$$Y_i = \sum_j A_{ij} V_j$$



Self-Attention Layer

Inputs:
Input vectors: X [N x D_{in}]
Key matrix: W_K [D_{in} x D_{out}]
Value matrix: W_V [D_{in} x D_{out}]
Query matrix: W_Q [D_{in} x D_{out}]

Computation:
Queries: $Q = XW_Q$ [N x D_{out}]
Keys: $K = XW_K$ [N x D_{out}]
Values: $V = XW_V$ [N x D_{out}]
Similarities: $E = QK^T / \sqrt{D_Q}$ [N x N]

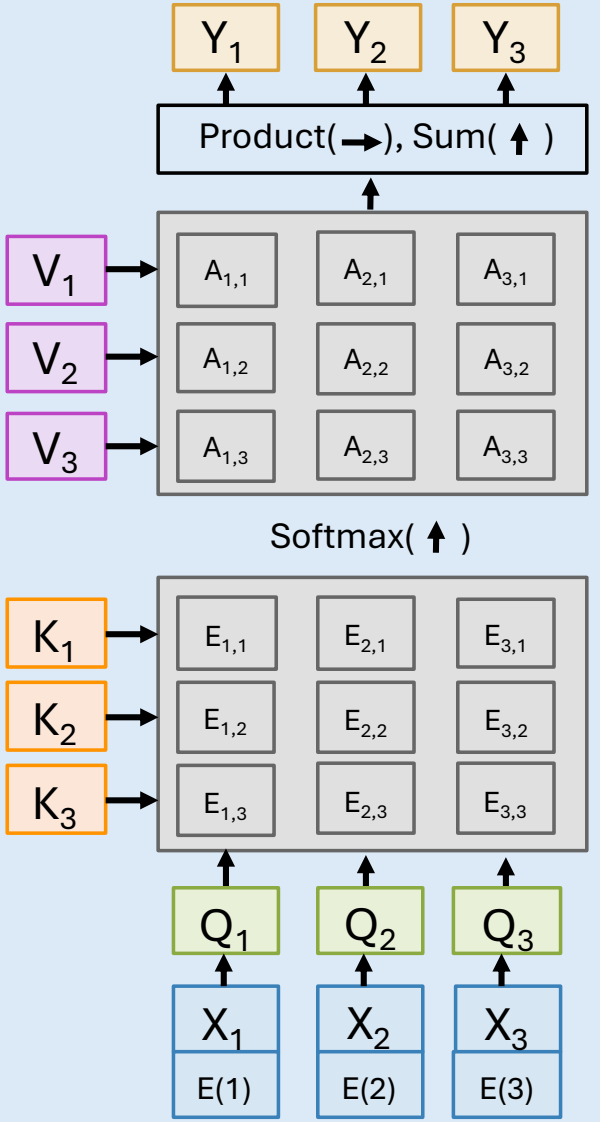
$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: A = softmax(E, dim=1) [N x N]
Output vector: $Y = AV$ [N x D_{out}]

$$Y_i = \sum_j A_{ij} V_j$$

Problem : Self-Attention does not know the order of the sequence

Solution : Add positional encoding to each input; this is a vector that is a fixed function of the index



Self-Attention Layer

Inputs:

Input vectors: X [N x D_{in}]

Key matrix: W_K [D_{in} x D_{out}]

Value matrix: W_V [D_{in} x D_{out}]

Query matrix: W_Q [D_{in} x D_{out}]

Computation:

Queries: $Q = XW_Q$ [N x D_{out}]

Keys: $K = XW_K$ [N x D_{out}]

Values: $V = XW_V$ [N x D_{out}]

Similarities: $E = QK^T / \sqrt{D_Q}$ [N x N]

$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$

Attention weights: A = softmax(E, dim=1) [N x N]

Output vector: $Y = AV$ [N x D_{out}]

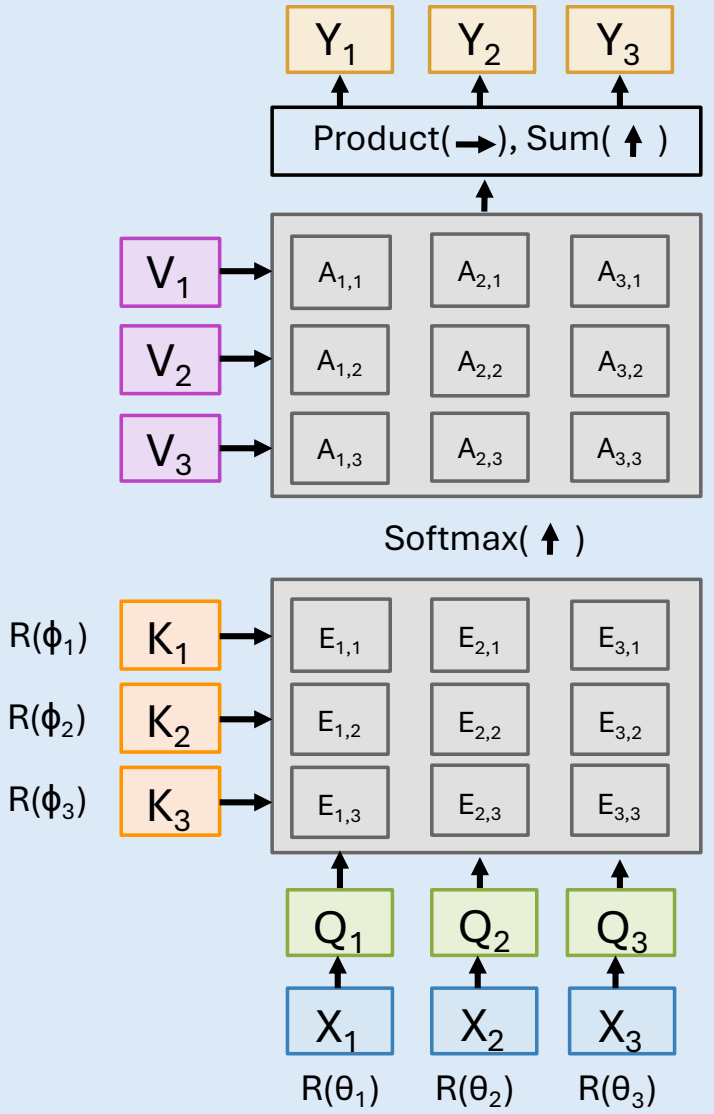
$Y_i = \sum_j A_{ij} V_j$

Problem : Self-Attention does not know the order of the sequence

Solution : Map positions to angles, and *rotate* queries and keys to encode relative positions (RoPE)

$$(R(\theta_i)q_i)^T(R(\varphi_j)k_j) = q_i{}^T R(\varphi_j - \theta_i)k_j$$

Su et al, "RoFormer: Enhanced Transformer with Rotary Position Embedding", arXiv 2021



Masked Self-Attention Layer

Don't let vectors "look ahead" in the sequence

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Override similarities with -inf; this controls which inputs each vector is allowed to look at.

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

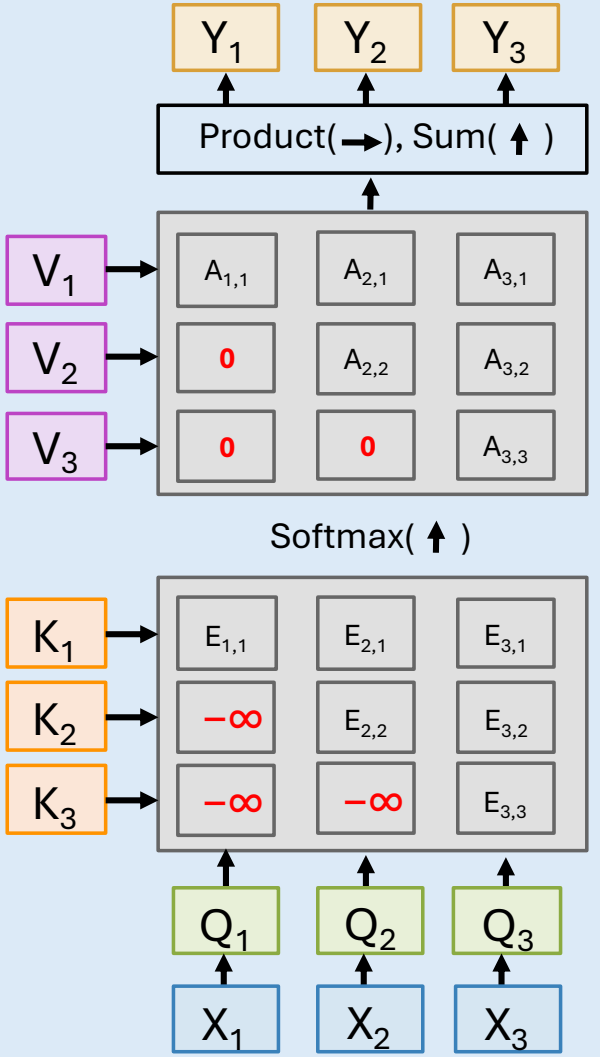
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$



Masked Self-Attention Layer

Don't let vectors "look ahead" in the sequence

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$

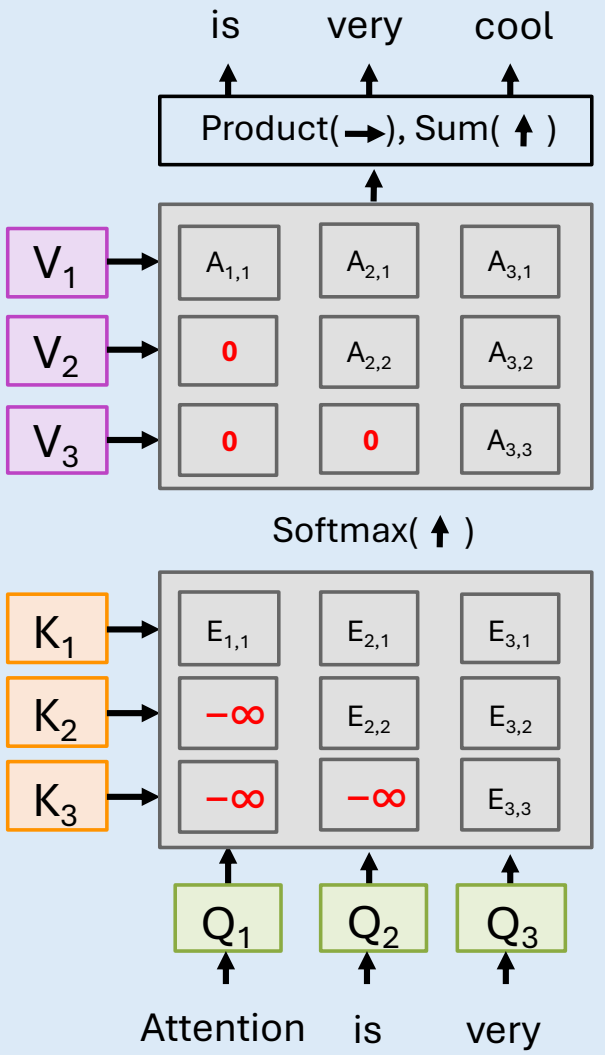
Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$Y_i = \sum_j A_{ij} V_j$

Override similarities with -inf; this controls which inputs each vector is allowed to look at.

Used for language modeling where you want to predict the next word



Multihead Self-Attention Layer

Run H copies of Self-Attention in parallel

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

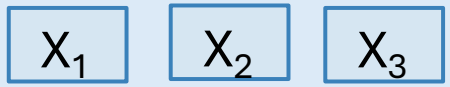
Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$



Multihead Self-Attention Layer

Run H copies of Self-Attention in parallel

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

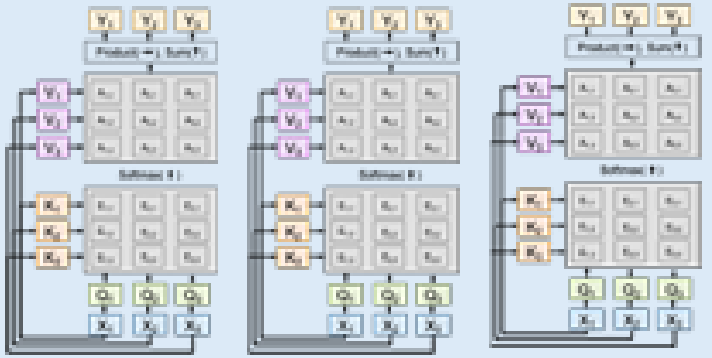
$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$

H = 3 independent self-attention layers (called heads), each with their own weights



X_1 X_2 X_3

Multihead Self-Attention Layer

Run H copies of Self-Attention in parallel

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

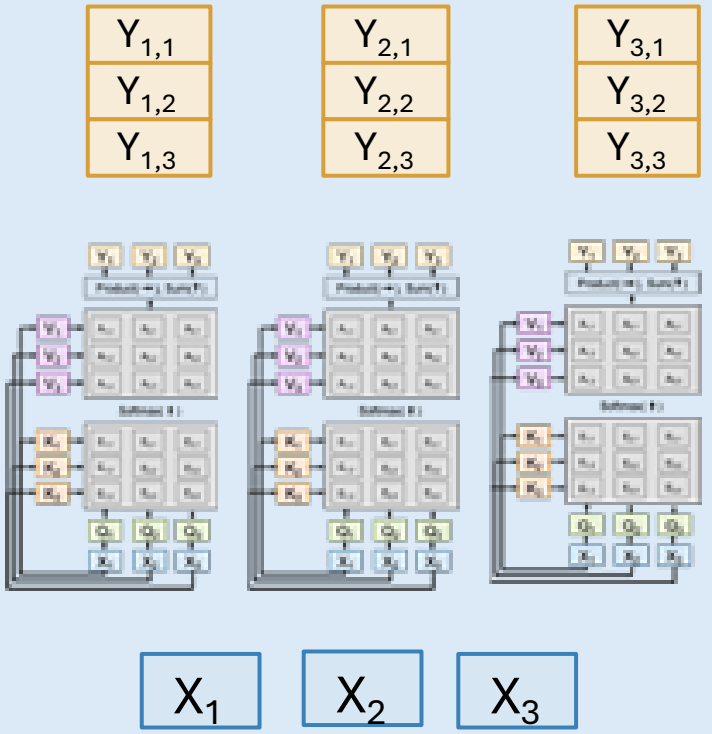
Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$

Stack up the H independent outputs for each input X

H = 3 independent self-attention layers (called heads), each with their own weights



Multihead Self-Attention Layer

Run H copies of Self-Attention in parallel

Inputs:

Input vectors: X $[N \times D_{in}]$

Key matrix: W_K $[D_{in} \times D_{out}]$

Value matrix: W_V $[D_{in} \times D_{out}]$

Query matrix: W_Q $[D_{in} \times D_{out}]$

Computation:

Queries: $Q = XW_Q$ $[N \times D_{out}]$

Keys: $K = XW_K$ $[N \times D_{out}]$

Values: $V = XW_V$ $[N \times D_{out}]$

Similarities: $E = QK^T / \sqrt{D_Q}$ $[N \times N]$

$$E_{ij} = Q_i \cdot K_j / \sqrt{D_Q}$$

Attention weights: $A = \text{softmax}(E, \text{dim}=1)$ $[N \times N]$

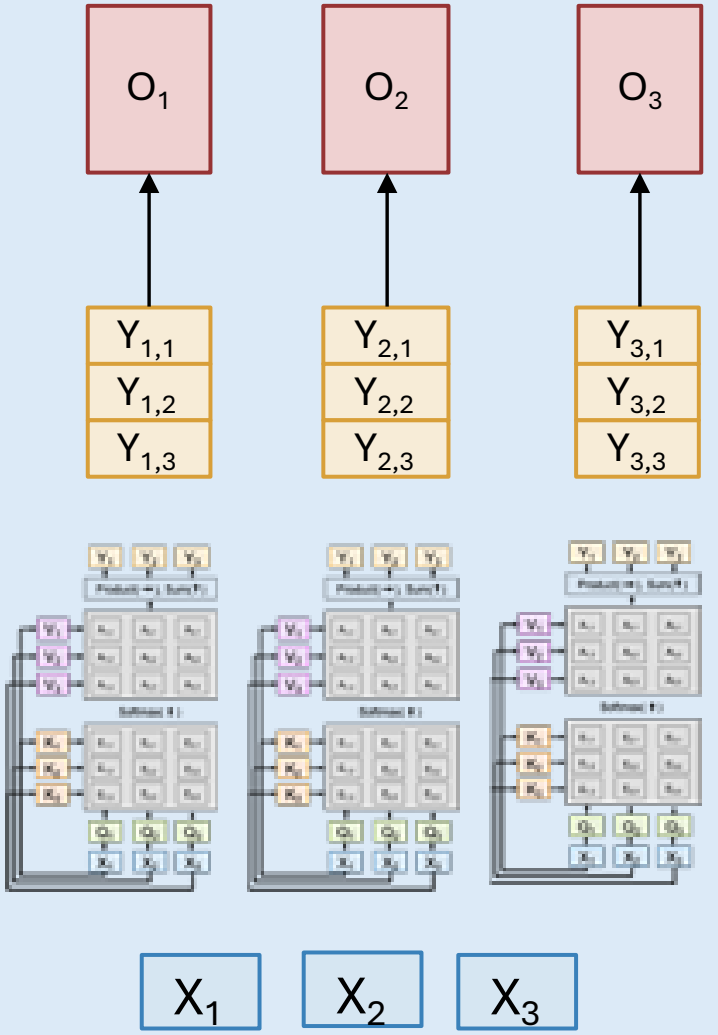
Output vector: $Y = AV$ $[N \times D_{out}]$

$$Y_i = \sum_j A_{ij} V_j$$

Output projection fuses data from each head

Stack up the H independent outputs for each input X

H = 3 independent self-attention layers (called heads), each with their own weights



Multihead Self-Attention Layer

Run H copies of Self-Attention in parallel

Inputs:

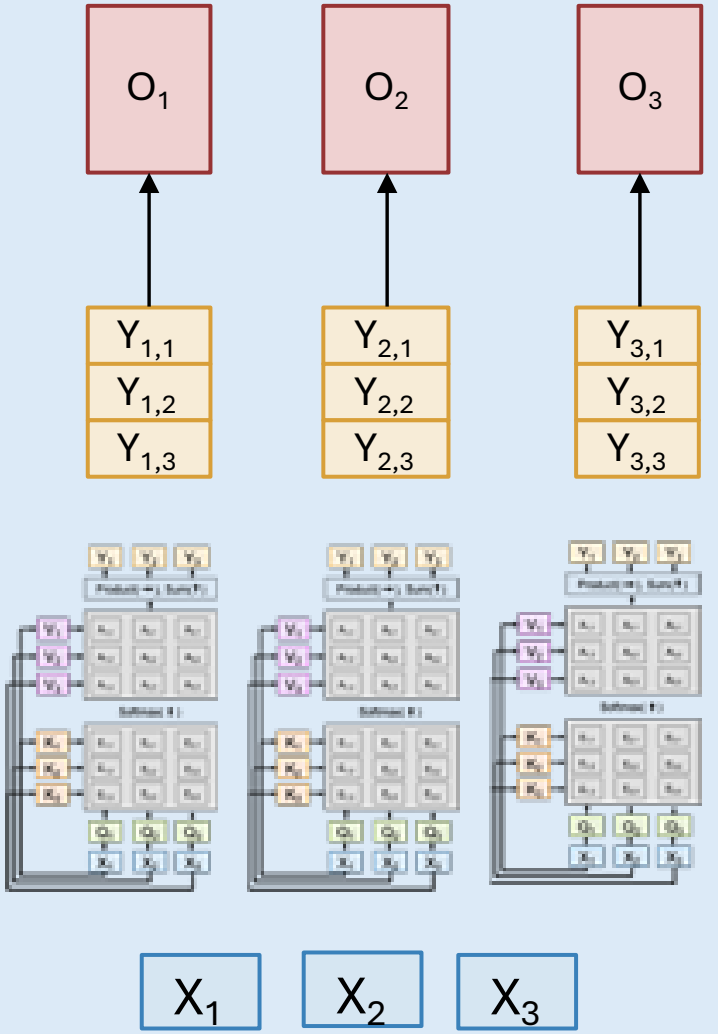
Input vectors : X [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Each of the H parallel layers use a qkv dim of D_H = "head dim"

Usually D_H = D / H, so inputs and outputs have the same dimension

Computation:

Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
Attention weights: A = softmax(E, dim=2) [H x N x N]
Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]



Multihead Self-Attention Layer

Run H copies of Self-Attention in parallel

Inputs:

Input vectors : X [N x D]

Key matrix: W_K [D x HD_H]

Value matrix: W_V [D x HD_H]

Query matrix: W_Q [D x HD_H]

Output matrix: W_O [HD_H x D]

In practice, compute all H heads **in parallel** using batched matrix multiply operations.

Used everywhere in practice.

Computation:

Queries: $Q = XW_Q$ [H x N x D_H]

Keys: $K = XW_K$ [H x N x D_H]

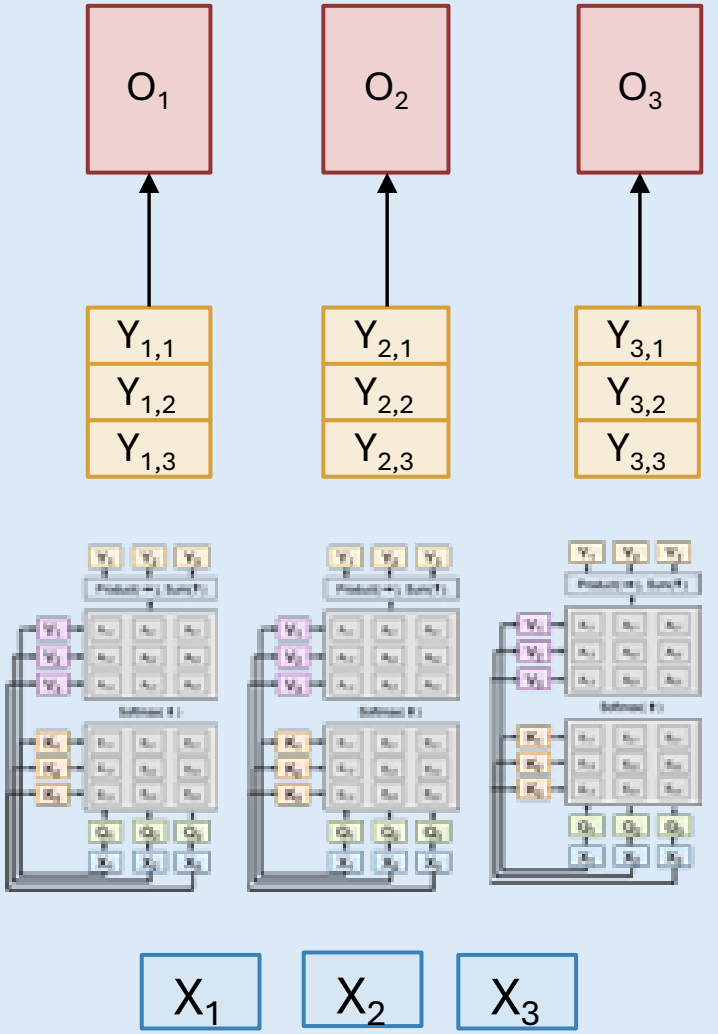
Values: $V = XW_V$ [H x N x D_H]

Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]

Attention weights: A = softmax(E, dim=2) [H x N x N]

Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]

Outputs: $O = YW_O$ [N x D]



Self-Attention is Four Matrix Multiplies!

Inputs:

Input vectors : X [N x D]

Key matrix: W_K [D x HD_H]

Value matrix: W_V [D x HD_H]

Query matrix: W_Q [D x HD_H]

Output matrix: W_O [HD_H x D]

Computation:

Queries: $Q = XW_Q$ [H x N x D_H]

Keys: $K = XW_K$ [H x N x D_H]

Values: $V = XW_V$ [H x N x D_H]

Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]

Attention weights: A = softmax(E, dim=2) [H x N x N]

Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]

Outputs: $O = YW_O$ [N x D]

Self-Attention is Four Matrix Multiplies!

Inputs:
Input vectors : **X** [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
Attention weights: A = softmax(E, dim=2) [H x N x N]
Head outputs: Y = AV [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
   Split and reshape to get **Q**, **K**, **V** each of shape [H x N x D_H]

Self-Attention is Four Matrix Multiplies!

Inputs:
Input vectors : X [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
Attention weights: A = softmax(E, dim=2) [H x N x N]
Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
   Split and reshape to get Q, K, V each of shape [H x N x D_H]
2. QK Similarity
   [H x N x D_H] [H x N x D_H] => [H x N x N]

Self-Attention is Four Matrix Multiplies!

Inputs:
Input vectors : X [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
Attention weights: A = softmax(E, dim=2) [H x N x N]
Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
   Split and reshape to get Q, K, V each of shape [H x N x D_H]
2. QK Similarity
   [H x N x D_H] [H x N x D_H] => [H x N x N]
3. V-Weighting
   [H x N x N] [H x N x D_H] => [H x N x D_H]
   Reshape to [N x HD_H]

Self-Attention is Four Matrix Multiplies!

Inputs:
Input vectors : X [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
Attention weights: A = softmax(E, dim=2) [H x N x N]
Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
   Split and reshape to get Q, K, V each of shape [H x N x D_H]
2. QK Similarity
   [H x N x D_H] [H x N x D_H] => [H x N x N]
3. V-Weighting
   [H x N x N] [H x N x D_H] => [H x N x D_H]
   Reshape to [N x HD_H]
4. Output Projection
   [N x HD_H] [HD_H x D] => [N x D]

Self-Attention is Four Matrix Multiplies!

Inputs:
Input vectors : X [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
Attention weights: A = softmax(E, dim=2) [H x N x N]
Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
   Split and reshape to get Q, K, V each of shape [H x N x D_H]
2. QK Similarity
   [H x N x D_H] [H x N x D_H] => [H x N x N]
3. V-Weighting
   [H x N x N] [H x N x D_H] => [H x N x D_H]
   Reshape to [N x HD_H]
4. Output Projection
   [N x HD_H] [HD_H x D] => [N x D]

Q: How much compute does this take as the number of vectors N increases?

Self-Attention is Four Matrix Multiplies!

Inputs:
Input vectors : X [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
Attention weights: A = softmax(E, dim=2) [H x N x N]
Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
   Split and reshape to get Q, K, V each of shape [H x N x D_H]
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   [H x N x D_H] [H x N x D_H] => [H x N x N]
3. V-Weighting
   [H x N x N] [H x N x D_H] => [H x N x D_H]
   Reshape to [N x HD_H]
4. Output Projection
   [N x HD_H] [HD_H x D] => [N x D]

Q: How much compute does this take as the number of vectors N increases?
A: $O(N^2)$

Self-Attention is Four Matrix Multiplies!

Inputs:
Input vectors : X [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
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Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
   Split and reshape to get Q, K, V each of shape [H x N x D_H]
2. QK Similarity
   [H x N x D_H] [H x N x D_H] => [H x N x N]
3. V-Weighting
   [H x N x N] [H x N x D_H] => [H x N x D_H]
   Reshape to [N x HD_H]
4. Output Projection
   [N x HD_H] [HD_H x D] => [N x D]

Q: How much memory does this take as the number of vectors N increases?

Self-Attention is Four Matrix Multiplies!

Inputs:
Input vectors : X [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
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Head outputs: $Y = AV$ [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
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   [H x N x N] [H x N x D_H] => [H x N x D_H]
   Reshape to [N x HD_H]
4. Output Projection
   [N x HD_H] [HD_H x D] => [N x D]

Q: How much memory does this take as the number of vectors N increases?
A: $O(N^2)$

Self-Attention is Four Matrix Multiplies!

If N=100K, H=64 then HxNxN attention weights take 1.192 TB! GPUs don't have that much memory...

Inputs:
Input vectors : X [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
Attention weights: A = softmax(E, dim=2) [H x N x N]
Head outputs: Y = AV [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
   Split and reshape to get Q, K, V each of shape [H x N x D_H]
2. QK Similarity
   [H x N x D_H] [H x N x D_H] => [H x N x N]
3. V-Weighting
   [H x N x N] [H x N x D_H] => [H x N x D_H]
   Reshape to [N x HD_H]
4. Output Projection
   [N x HD_H] [HD_H x D] => [N x D]

Q: How much memory does this take as the number of vectors N increases?
A: $O(N^2)$

Self-Attention is Four Matrix Multiplies!

Inputs:
Input vectors : **X** [N x D]
Key matrix: W_K [D x HD_H]
Value matrix: W_V [D x HD_H]
Query matrix: W_Q [D x HD_H]
Output matrix: W_O [HD_H x D]

Computation:
Queries: $Q = XW_Q$ [H x N x D_H]
Keys: $K = XW_K$ [H x N x D_H]
Values: $V = XW_V$ [H x N x D_H]
Similarities: $E = QK^T / \sqrt{D_Q}$ [H x N x N]
Attention weights: A = softmax(E, dim=2) [H x N x N]
Head outputs: Y = AV [H x N x D_H] => [N x HD_H]
Outputs: $O = YW_O$ [N x D]

Flash Attention algorithm computes 2+3 at the same time without storing the full attention matrix!

Makes large N possible

(Note) Processing attention block by block may seem slower, but it is often faster: moving data takes time on GPUs, and FlashAttention greatly reduces memory transfers.

1. QKV Projection
   [N x D] [D x $3HD_H$] => [N x $3HD_H$]
   Split and reshape to get Q, K, V each of shape [H x N x D_H]
2. QK Similarity
   [H x N x D_H] [H x N x D_H] => [H x N x N]
3. V-Weighting
   [H x N x N] [H x N x D_H] => [H x N x D_H]
   Reshape to [N x HD_H]
4. Output Projection
   [N x HD_H] [HD_H x D] => [N x D]

Q: How much memory does this take as the number of vectors N increases?
A: O(N) with Flash Attention

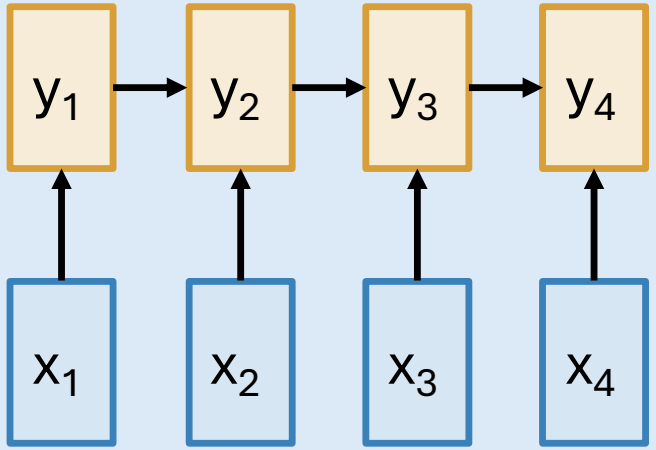
Dao et al, "FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness", 2022

Three Ways of Processing Sequences



Three Ways of Processing Sequences

Recurrent Neural Network



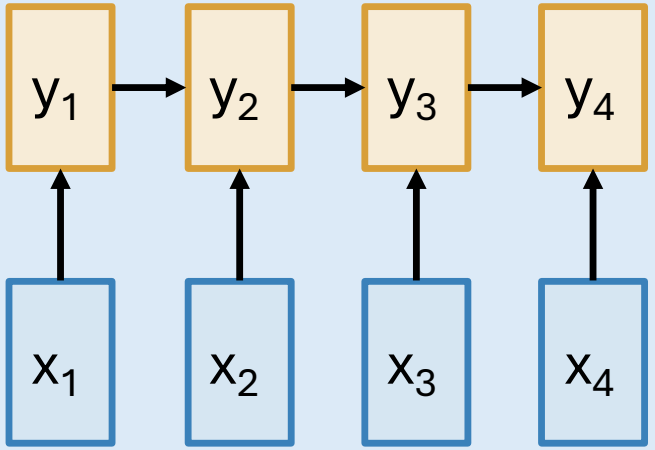
Works on **1D ordered sequences**

(+) Theoretically good at long sequences: O(N) compute and memory for a sequence of length N

(-) Not parallelizable. Need to compute hidden states sequentially

Three Ways of Processing Sequences

Recurrent Neural Network

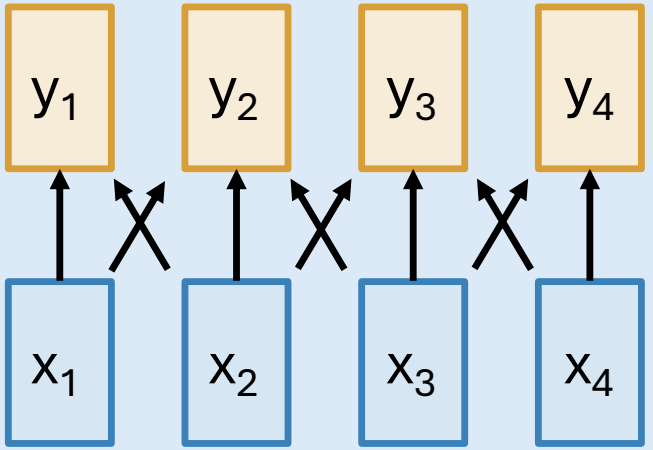


Works on **1D ordered sequences**

(+) Theoretically good at long sequences: O(N) compute and memory for a sequence of length N

(-) Not parallelizable. Need to compute hidden states sequentially

Convolution



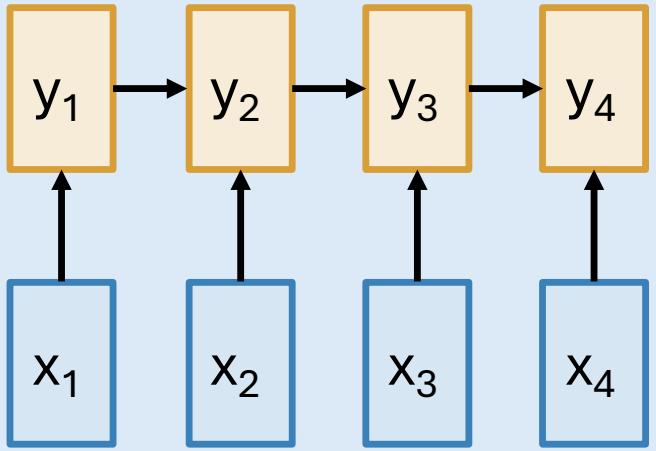
Works on **N-dimensional grids**

(-) Bad for long sequences: need to stack many layers to build up large receptive fields

(+) Parallelizable, outputs can be computed in parallel

Three Ways of Processing Sequences

Recurrent Neural Network

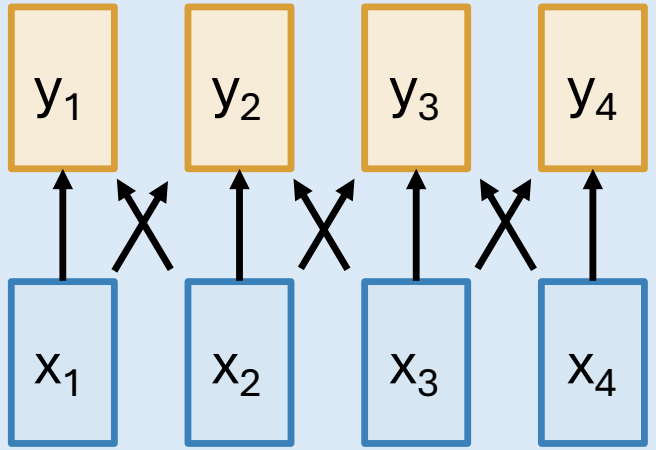


Works on **1D ordered sequences**

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Convolution

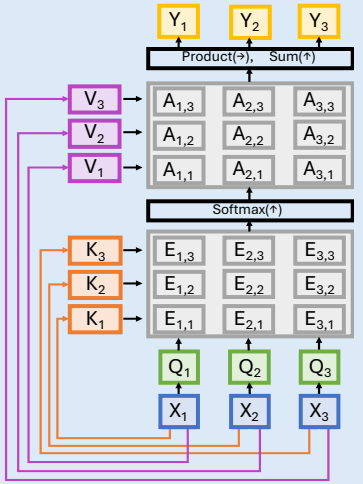


Works on **N-dimensional grids**

(-) Bad for long sequences: need to stack many layers to build up large receptive fields

(+) Parallelizable, outputs can be computed in parallel

Self-Attention



Works on **sets of vectors**

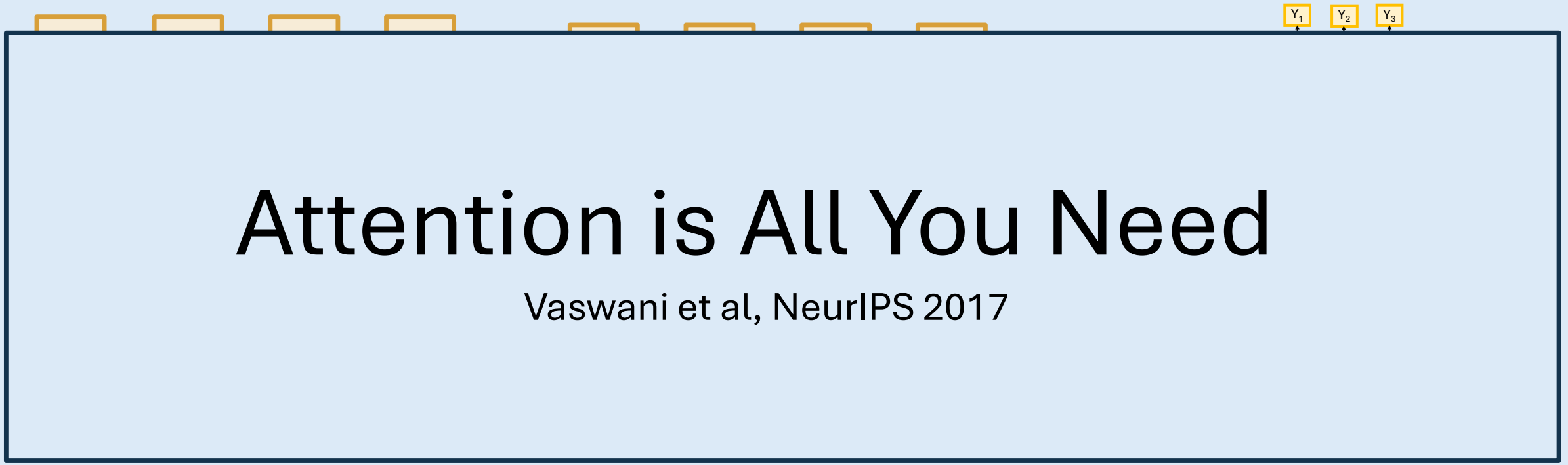
(+) Great for long sequences; each output depends directly on all inputs

(+) Highly parallel, it's just 4 matmuls

(−) Expensive: $O(N^2)$ compute, O(N) memory for sequence of length N

Three Ways of Processing Sequences

Recurrent Neural Network | Convolution | Self-Attention



Attention is All You Need

Vaswani et al, NeurIPS 2017

memory for a sequence of length N

(-) Not parallelizable. Need to compute hidden states sequentially

receptive fields

(+) Parallelizable, outputs can be computed in parallel

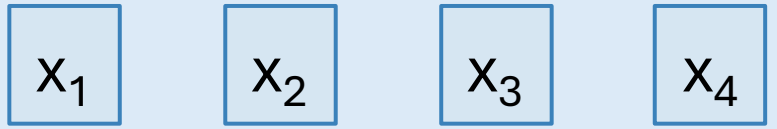
(+) Highly parallel, it's just 4 matmuls

(−) Expensive: $O(N^2)$ compute, $O(N)$ memory for sequence of length N

The Transformer

Transformer Block

Input: Set of vectors x



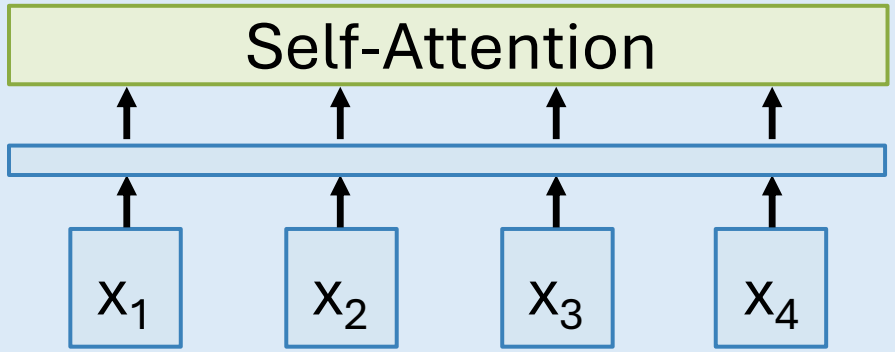
Vaswani et al, “Attention is all you need,” NeurIPS 2017

The Transformer

Transformer Block

Input: Set of vectors x

All vectors interact through (multiheaded) Self-Attention



Vaswani et al, “Attention is all you need,” NeurIPS 2017

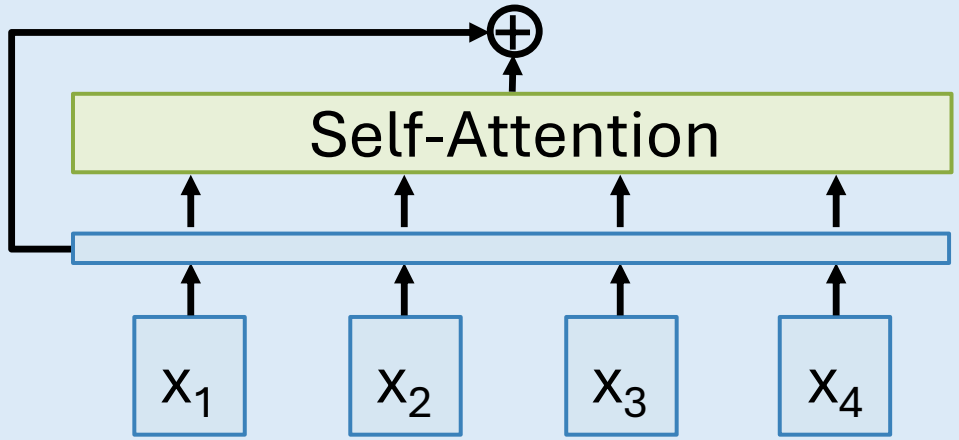
The Transformer

Transformer Block

Input: Set of vectors x

Residual connection

All vectors interact through (multiheaded) Self-Attention



Vaswani et al, "Attention is all you need," NeurIPS 2017

The Transformer

Transformer Block

Input: Set of vectors x

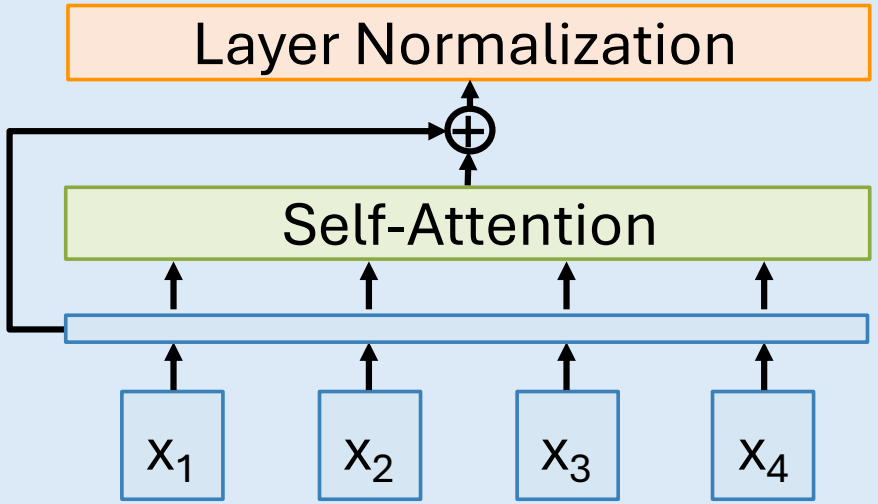
Recall **Layer Normalization** :
Given $h_1, \ldots, h_N$ (Shape: D)
scale: γ (Shape: D)
shift: β (Shape: D)
$\mu_i = (\sum_j h_{i,j})/D$ (scalar)
$\sigma_i = (\sum_j (h_{i,j} - \mu_i)^2/D)^{1/2}$ (scalar)
$z_i = (h_i - \mu_i) / \sigma_i$
$y_i = \gamma * z_i + \beta$

Ba et al, 2016

Layer normalization normalizes all vectors

Residual connection

All vectors interact through (multiheaded) Self-Attention



Vaswani et al, "Attention is all you need," NeurIPS 2017

The Transformer

Transformer Block

Input: Set of vectors x

Usually a two-layer MLP;
classic setup is
D => 4D => D

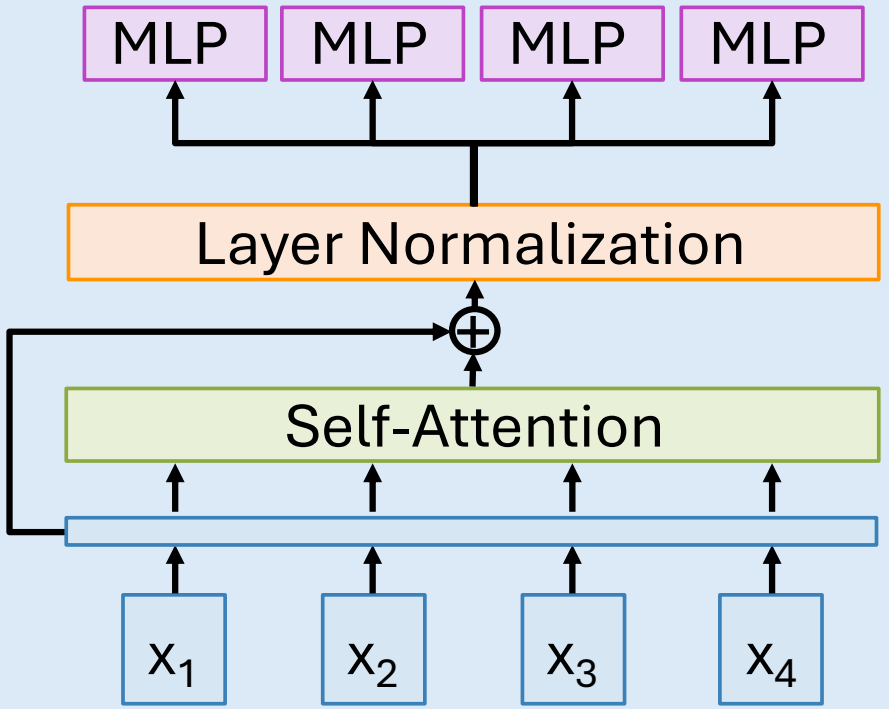
Also sometimes called FFN
(Feed-Forward Network)

MLP independently
on each vector

Layer normalization
normalizes all vectors

Residual connection

All vectors interact through
(multiheaded) Self-Attention



Vaswani et al, "Attention is all you need," NeurIPS 2017

The Transformer

Transformer Block

Input: Set of vectors x

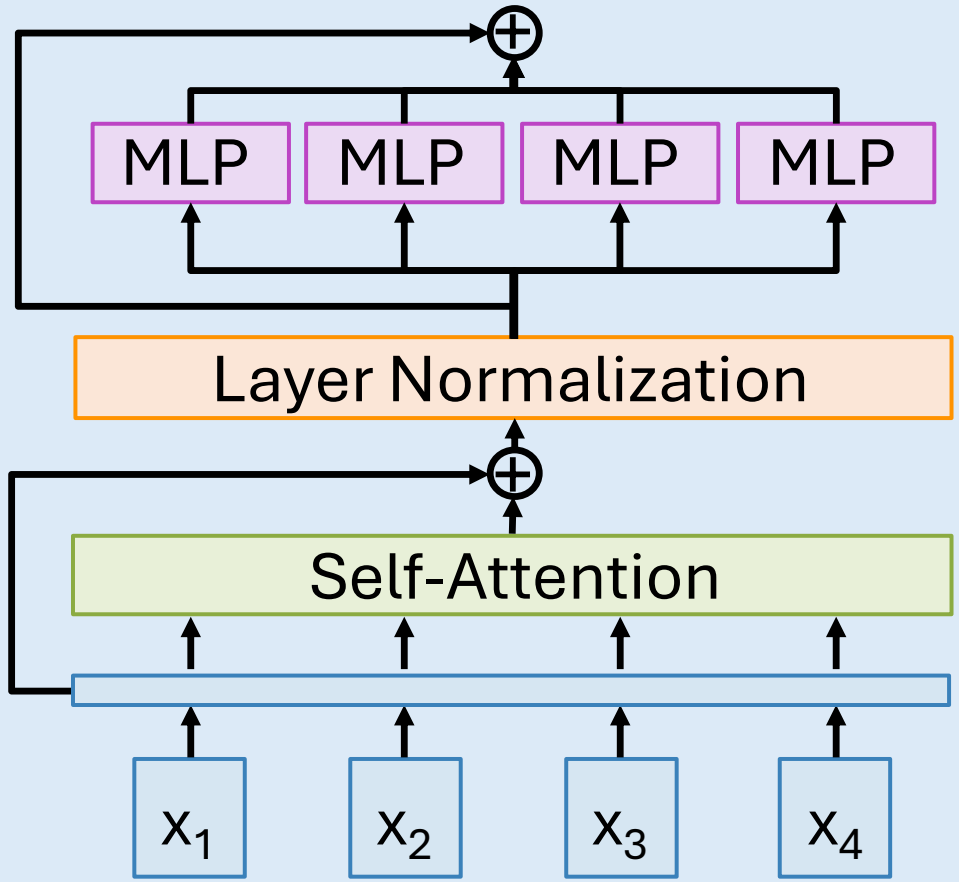
Residual connection

MLP independently on each vector

Layer normalization normalizes all vectors

Residual connection

All vectors interact through (multiheaded) Self-Attention



Vaswani et al, "Attention is all you need," NeurIPS 2017

The Transformer

Transformer Block

Input: Set of vectors x

Another Layer Norm

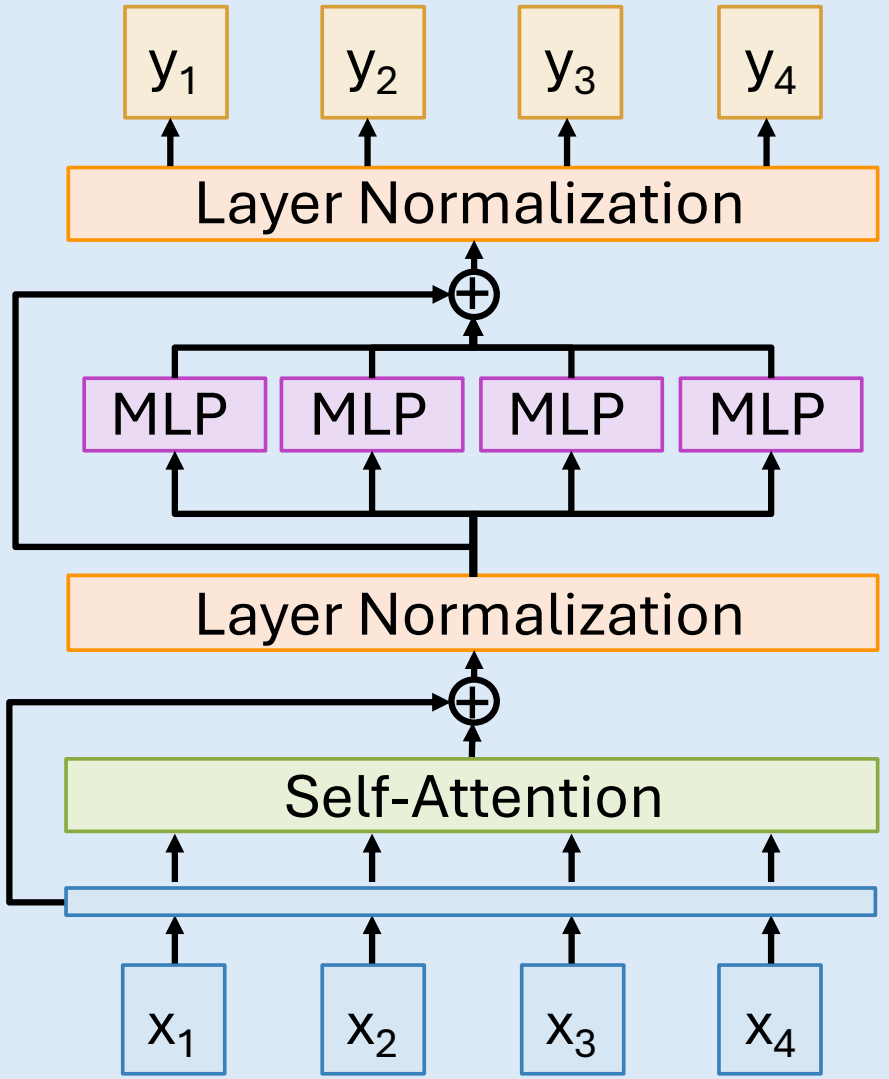
Residual connection

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Vaswani et al, "Attention is all you need," NeurIPS 2017

The Transformer

Transformer Block

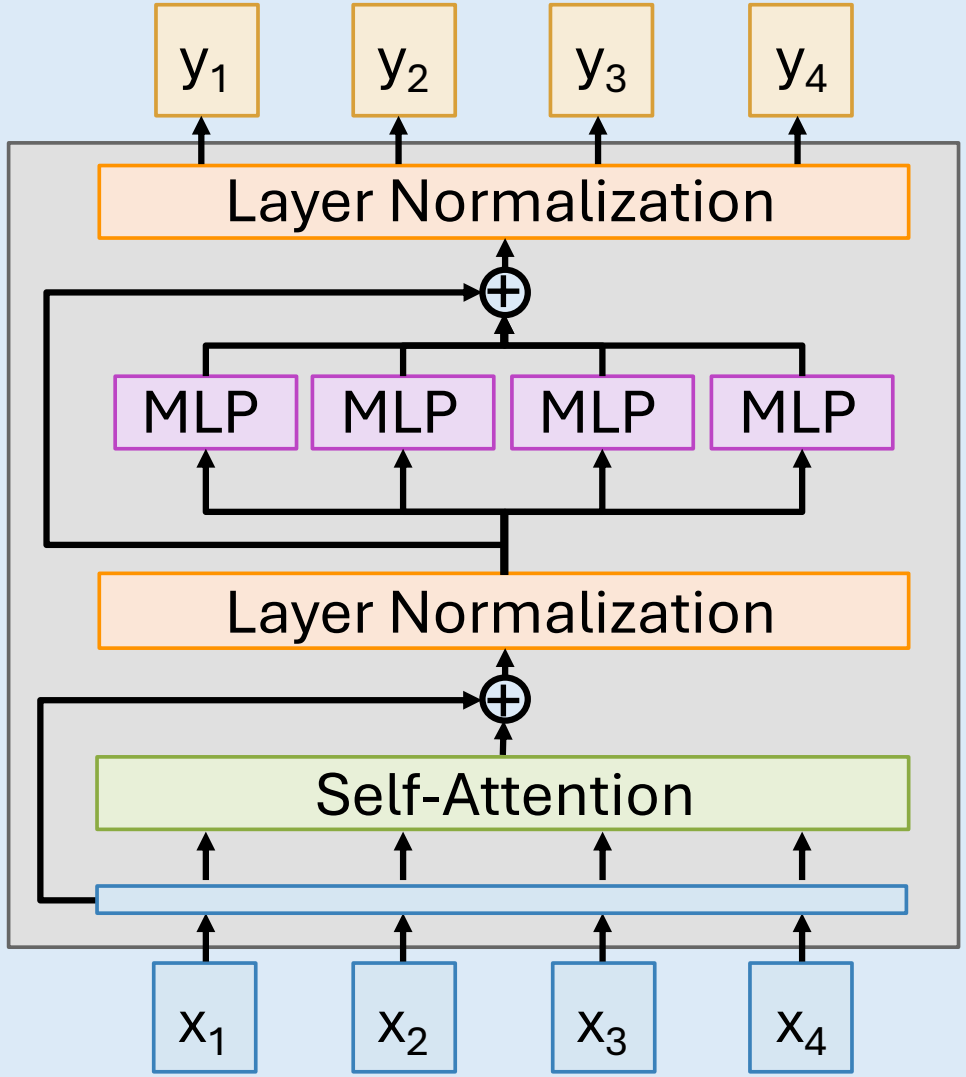
Input: Set of vectors x
Output: Set of vectors y

Self-Attention is the only interaction between vectors

LayerNorm and MLP work on each vector independently

Highly scalable and parallelizable, most of the compute is just 6 matmuls:

4 from Self-Attention
2 from MLP



Vaswani et al, "Attention is all you need," NeurIPS 2017

The Transformer

Transformer Block

Input: Set of vectors x
Output: Set of vectors y

Self-Attention is the only interaction between vectors

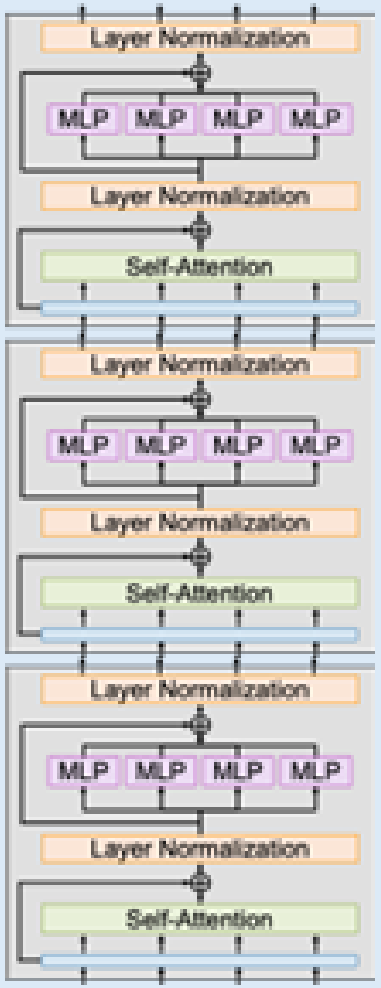
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A **Transformer** is just a stack of identical Transformer blocks!

They have not changed much since 2017… but have gotten a lot bigger



Vaswani et al, “Attention is all you need,” NeurIPS 2017

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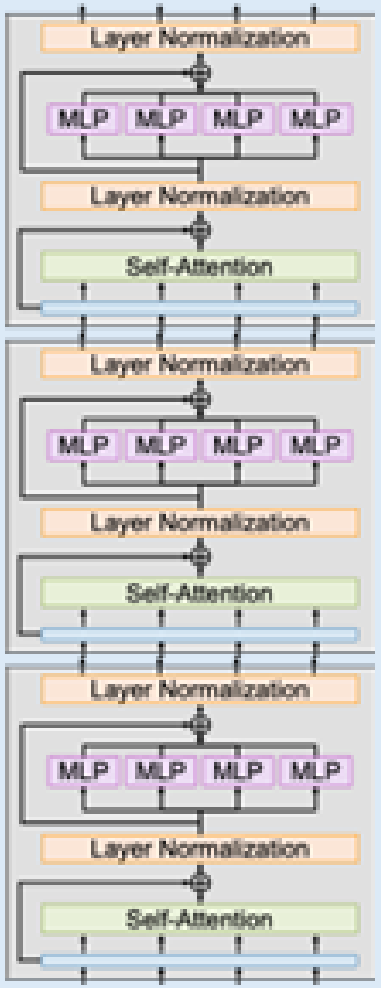
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12 blocks, D=1024, H=16, N=512
213M params



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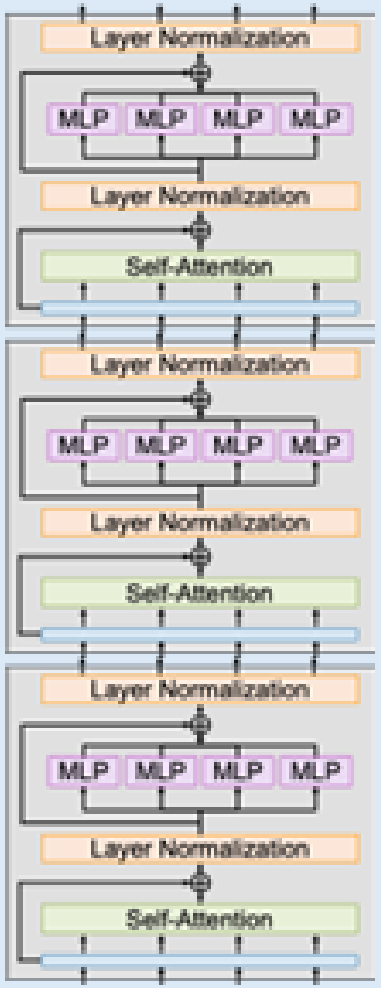
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12 blocks, D=1024, H=16, N=512
213M params

GPT-2: [Radford et al, 2019]
48 blocks, D=1600, H=25, N=1024
1.5B params



Vaswani et al, "Attention is all you need," NeurIPS 2017

The Transformer

Transformer Block

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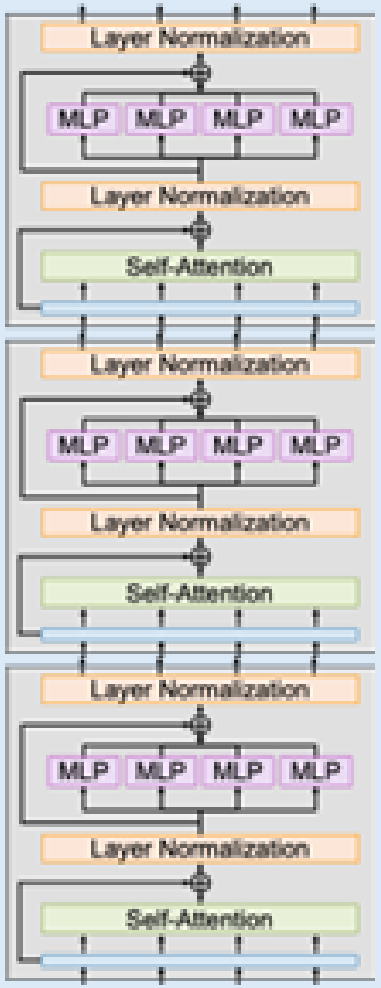
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213M params

GPT-2: [Radford et al, 2019]
48 blocks, D=1600, H=25, N=1024
1.5B params

GPT-3: [Brown et al, 2020]
96 blocks, D=12288, H=96, N=2048
175B params

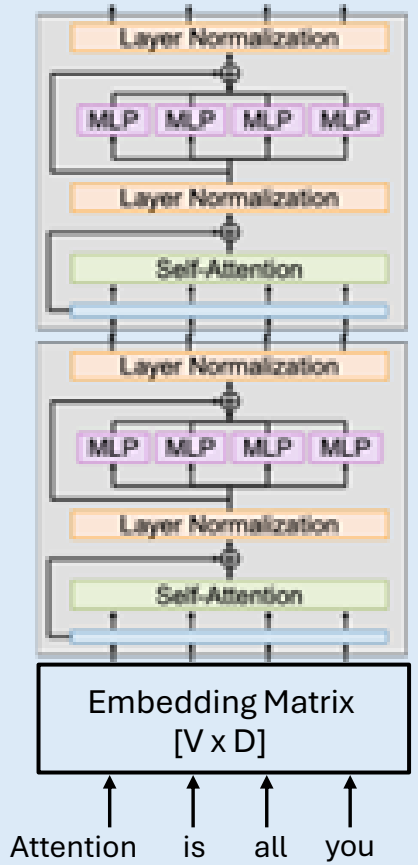


Vaswani et al, “Attention is all you need,” NeurIPS 2017

Transformers for Language Modeling

Learn an embedding matrix at the start of the model to convert words into vectors.

Given vocab size V and model dimension D, it's a lookup table of shape [V x D]

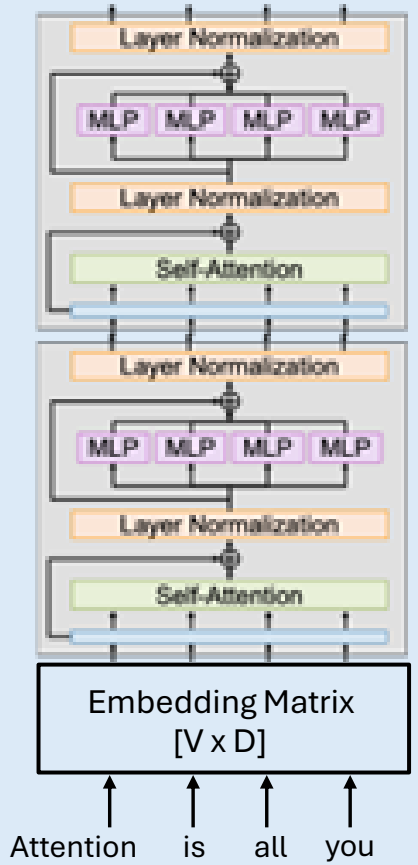


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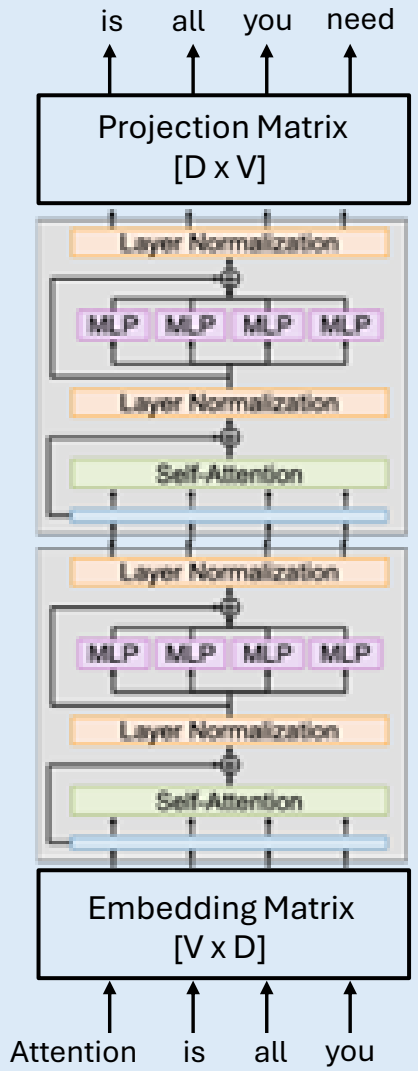
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Transformers for Language Modeling

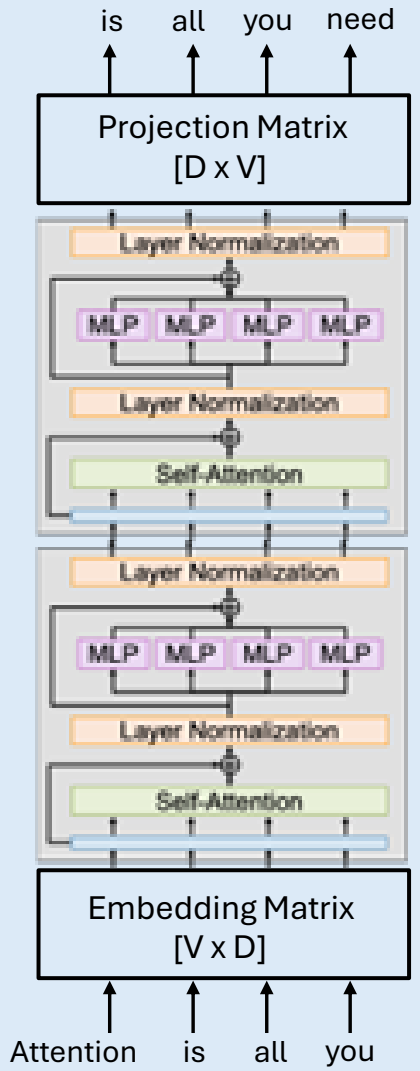
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Train to predict next token using softmax + cross-entropy loss



Vision Transformers (ViT)



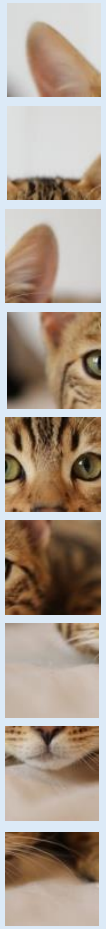
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Dosovitskiy et al, “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale”, ICLR 2021

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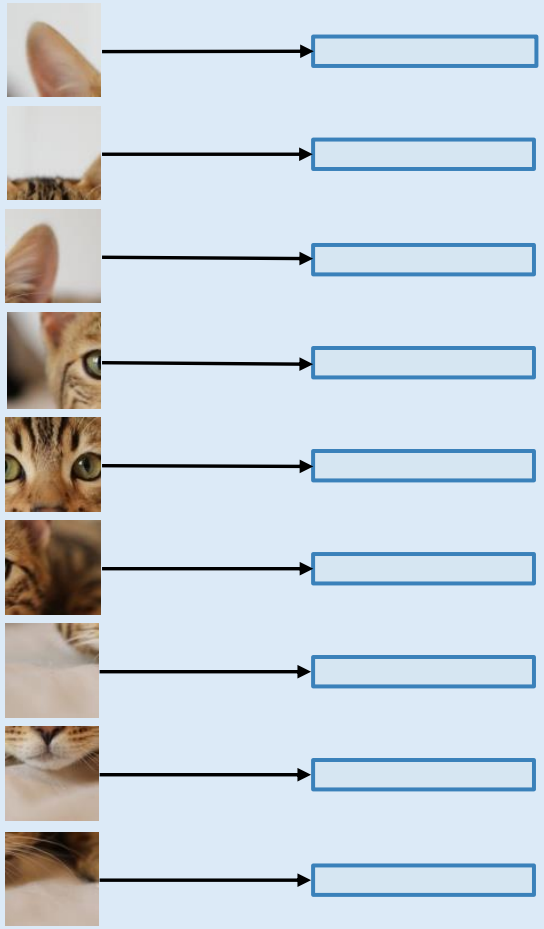
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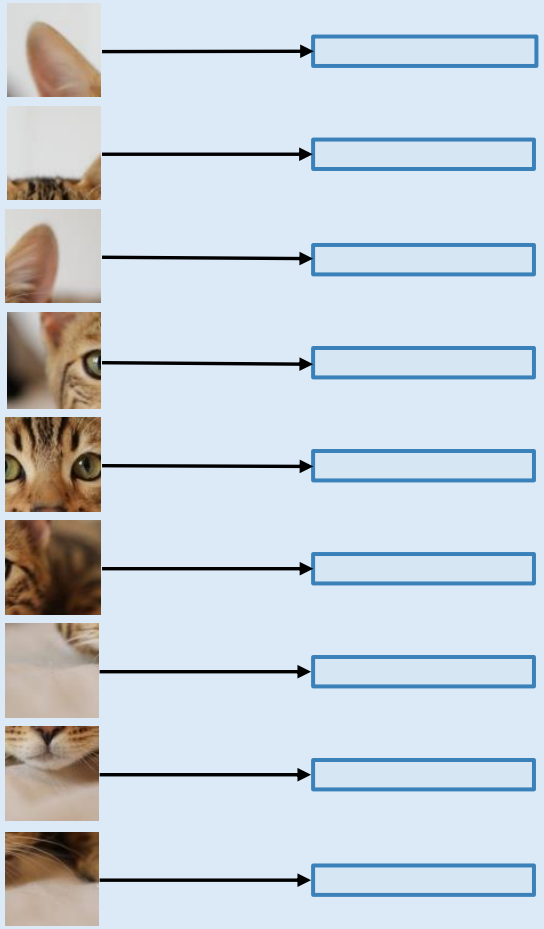
Flatten and apply a linear transform 768 => D

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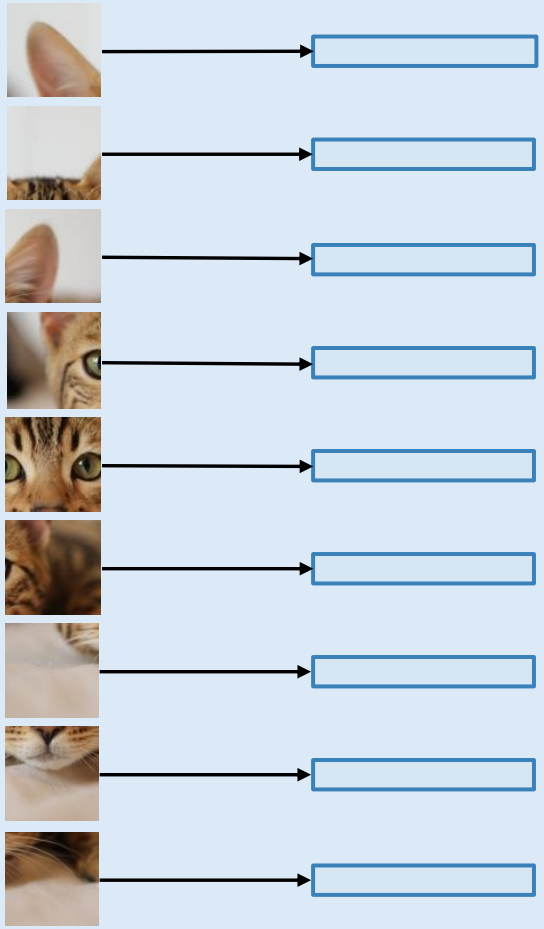
Q: Any other way to describe this operation?

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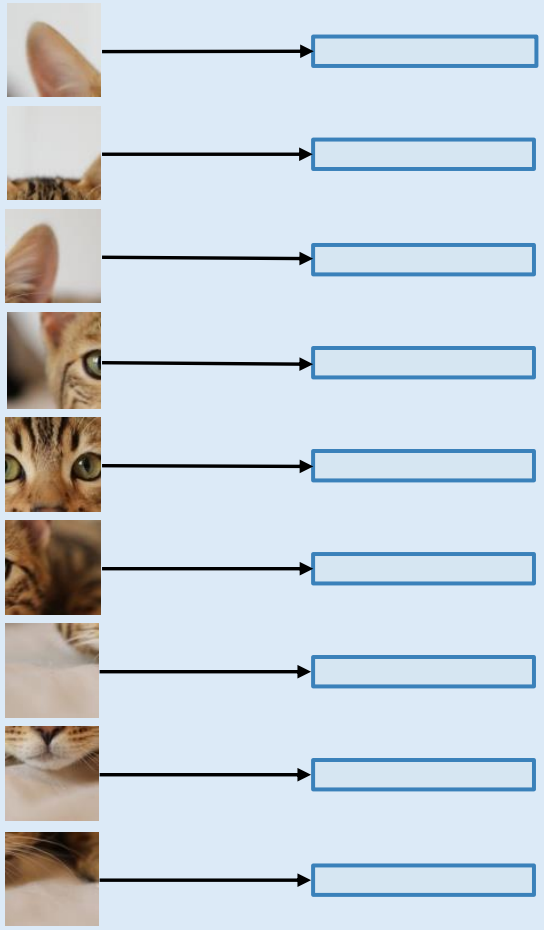
A: 16x16 conv with stride 16, 3 input channels, D output channels

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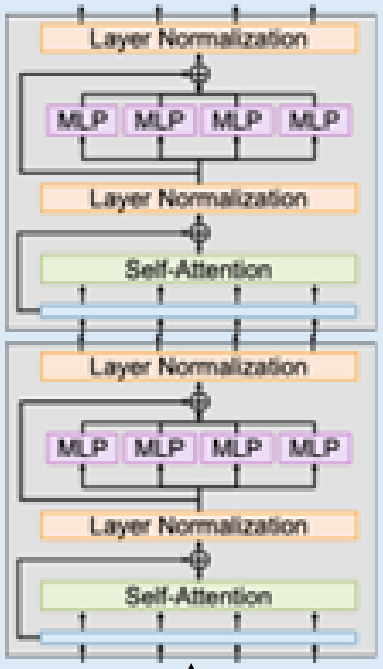


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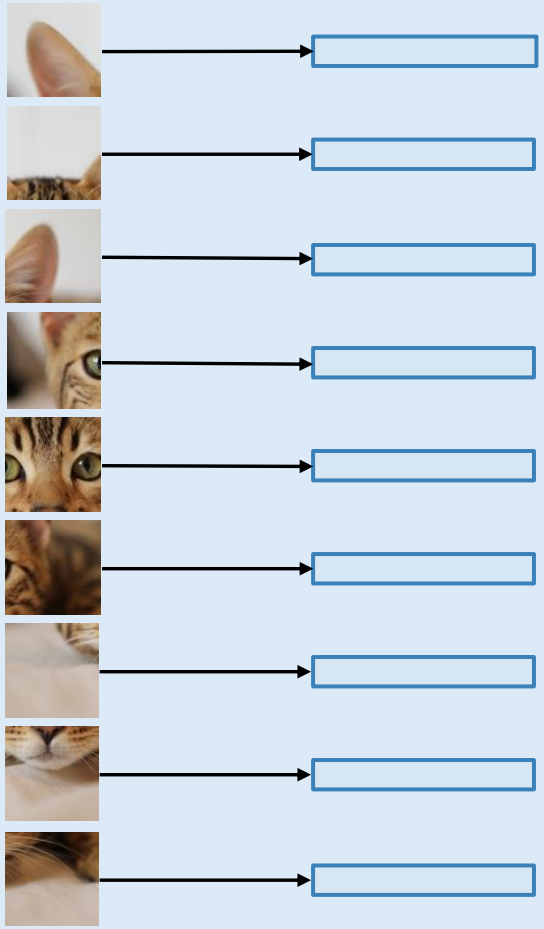
D-dim vector per patch are the input vectors to the Transformer

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Vision Transformers (ViT)

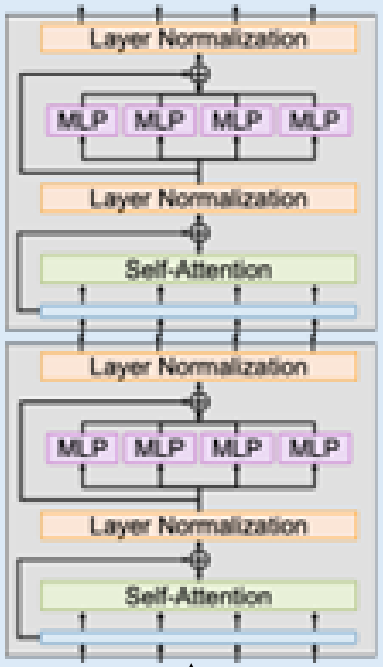


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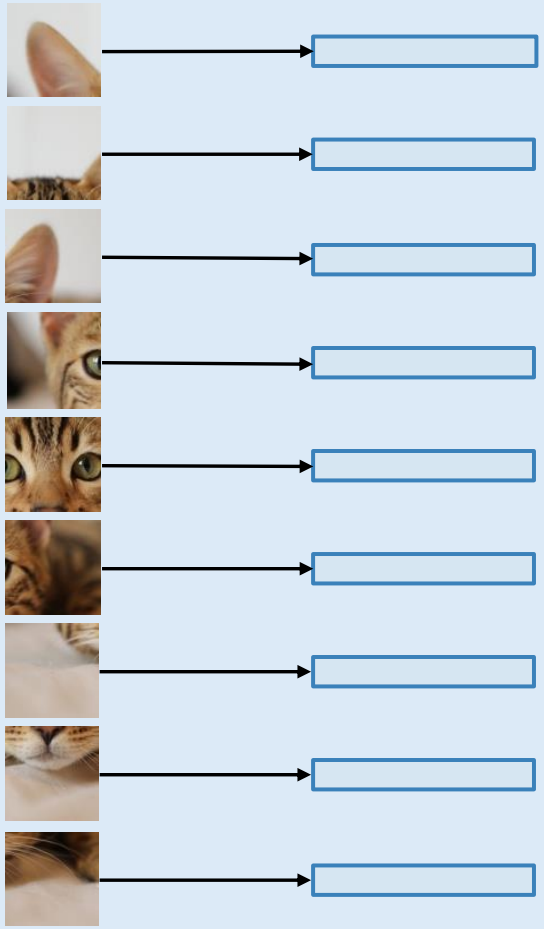
Use positional encoding to tell the transformer the 2D position of each patch

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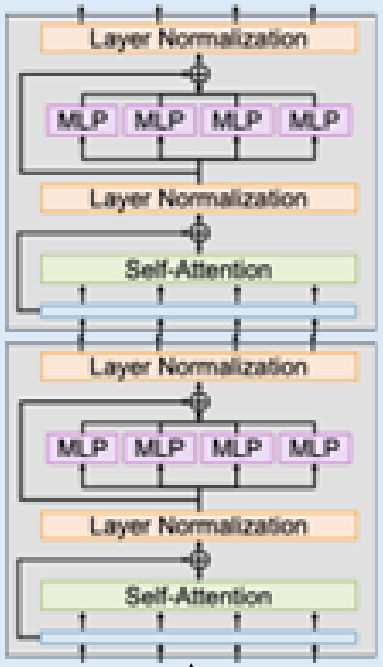


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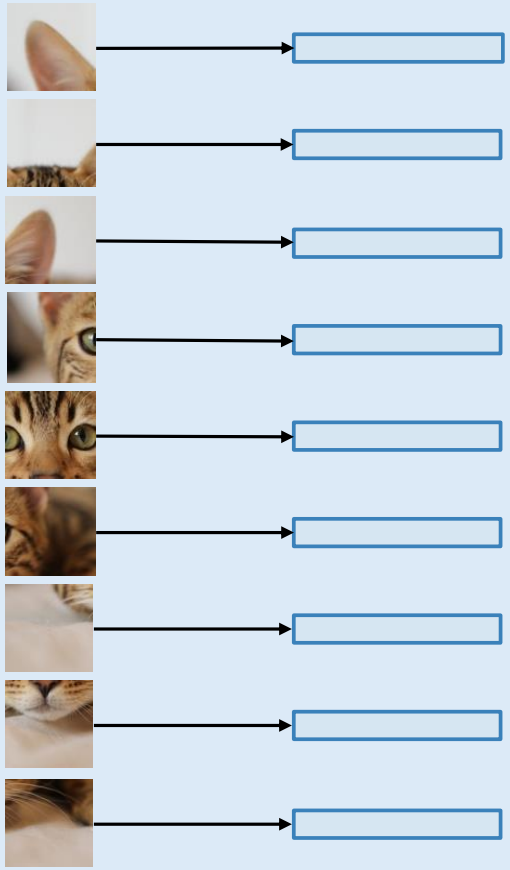
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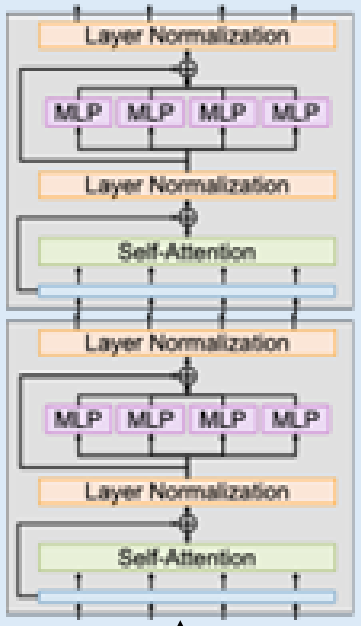


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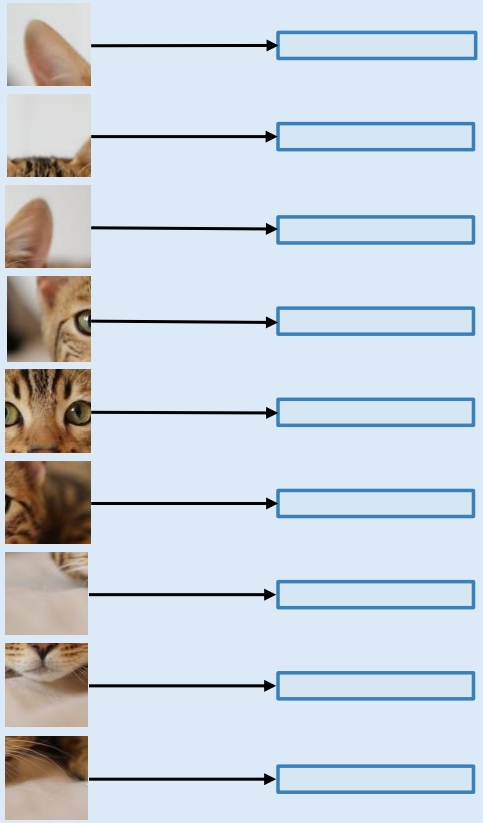
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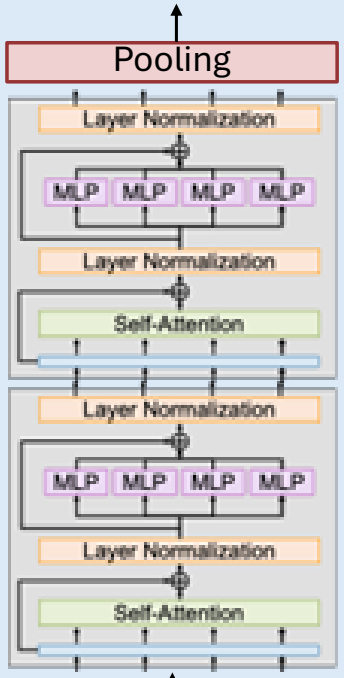


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Flatten and apply a linear transform 768 => D



Average pool NxD vectors to 1xD, apply a linear layer D=>C to predict class scores

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Transformer gives an output vector per patch

Don't use any masking; each image patch can look at all other image patches

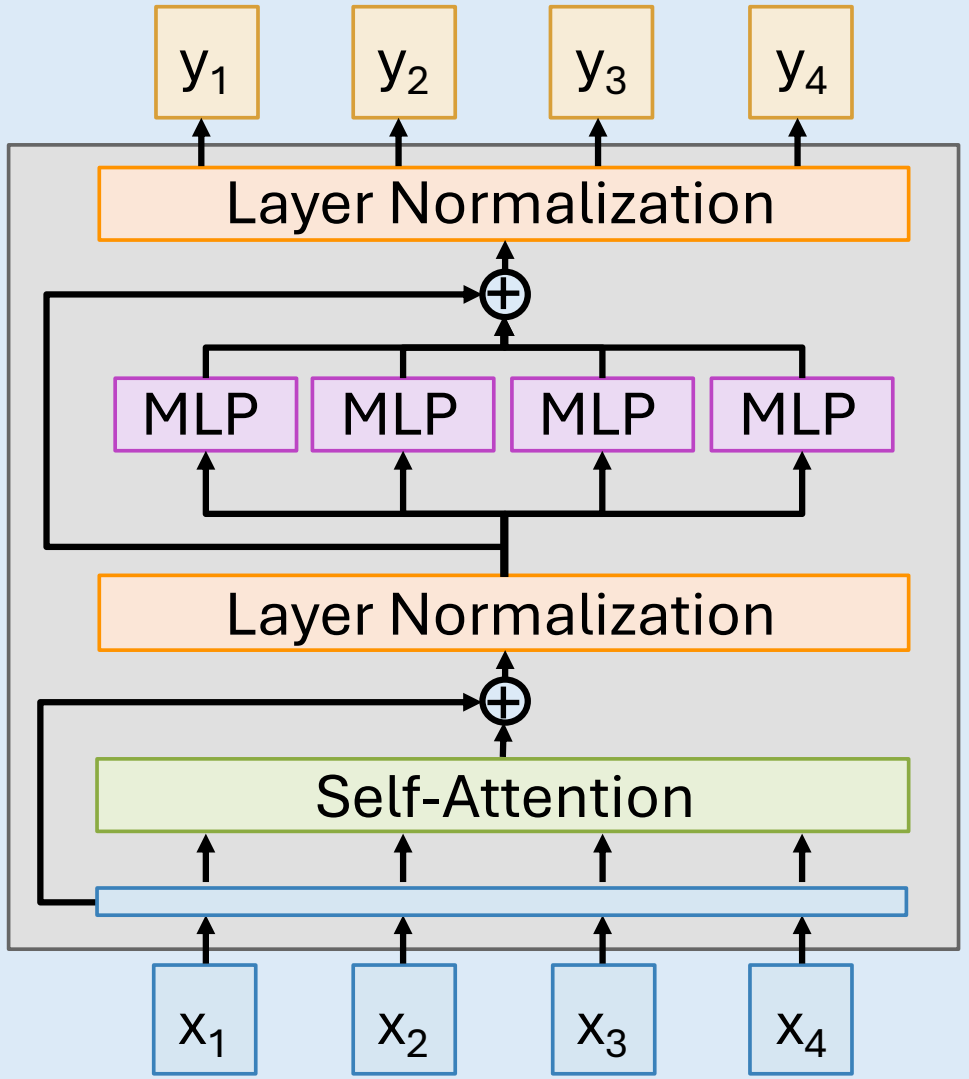
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Tweaking Transformers

The Transformer architecture has not changed much since 2017.

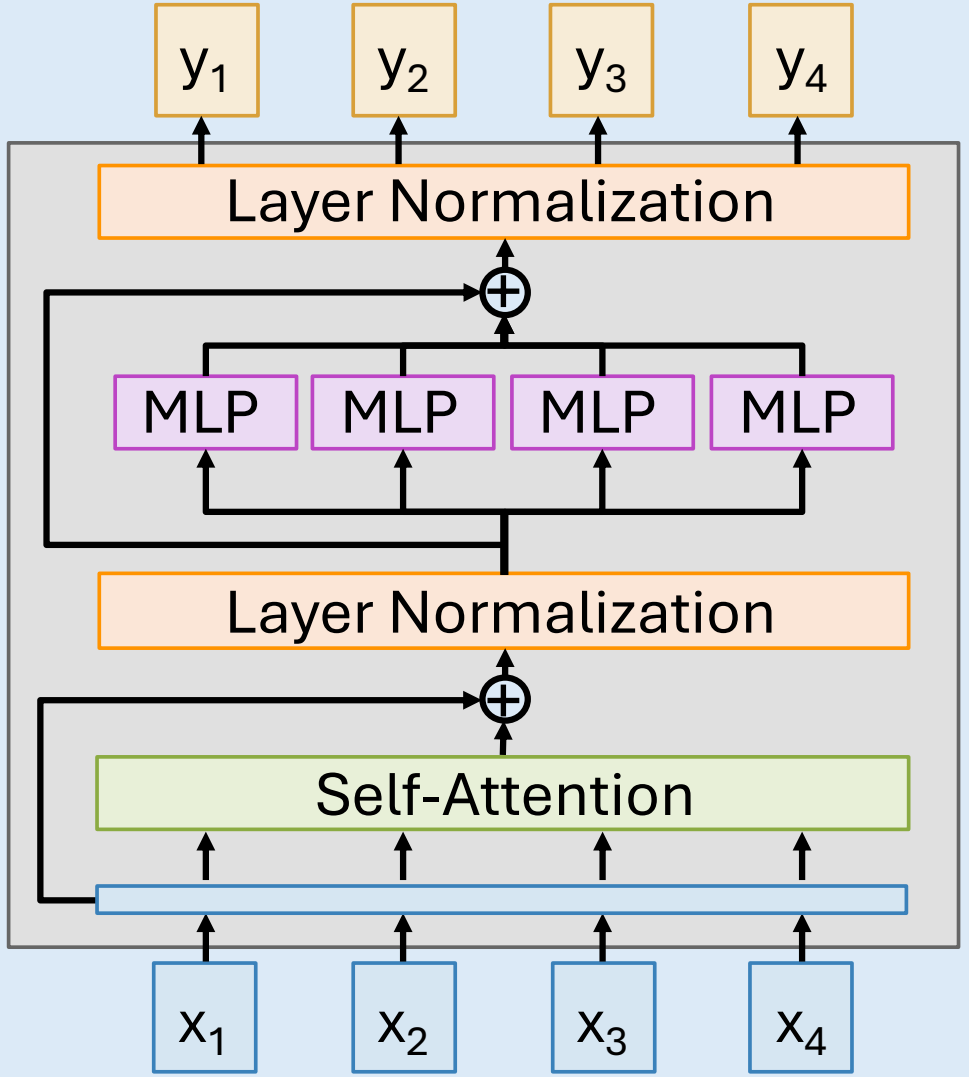
But a few changes have become common:



Pre-Norm Transformer

Layer normalization is outside the residual connections

Kind of weird, the model can't actually learn the identify function



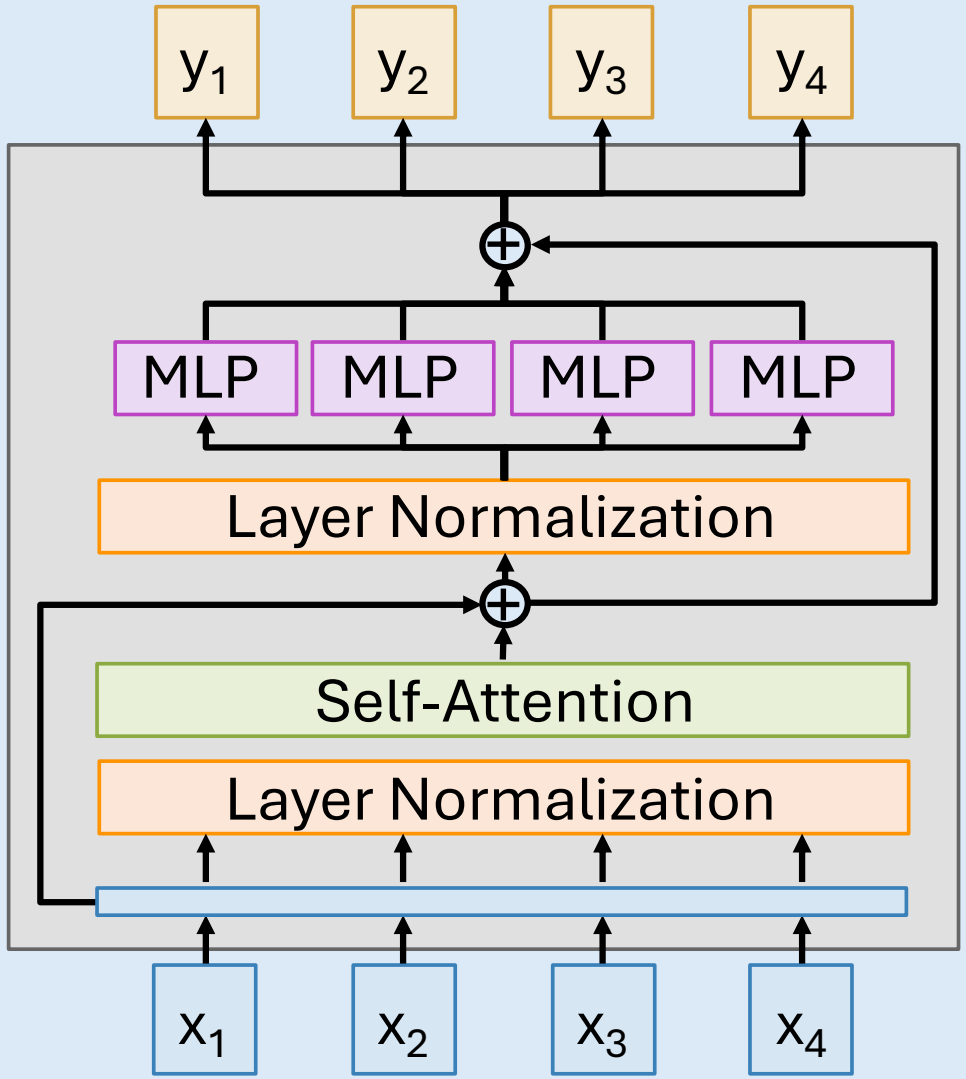
Baevski & Auli, "Adaptive Input Representations for Neural Language Modeling", arXiv 2018

Pre-Norm Transformer

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Solution : Move layer normalization before the Self-Attention and MLP, inside the residual connections. Training is more stable.



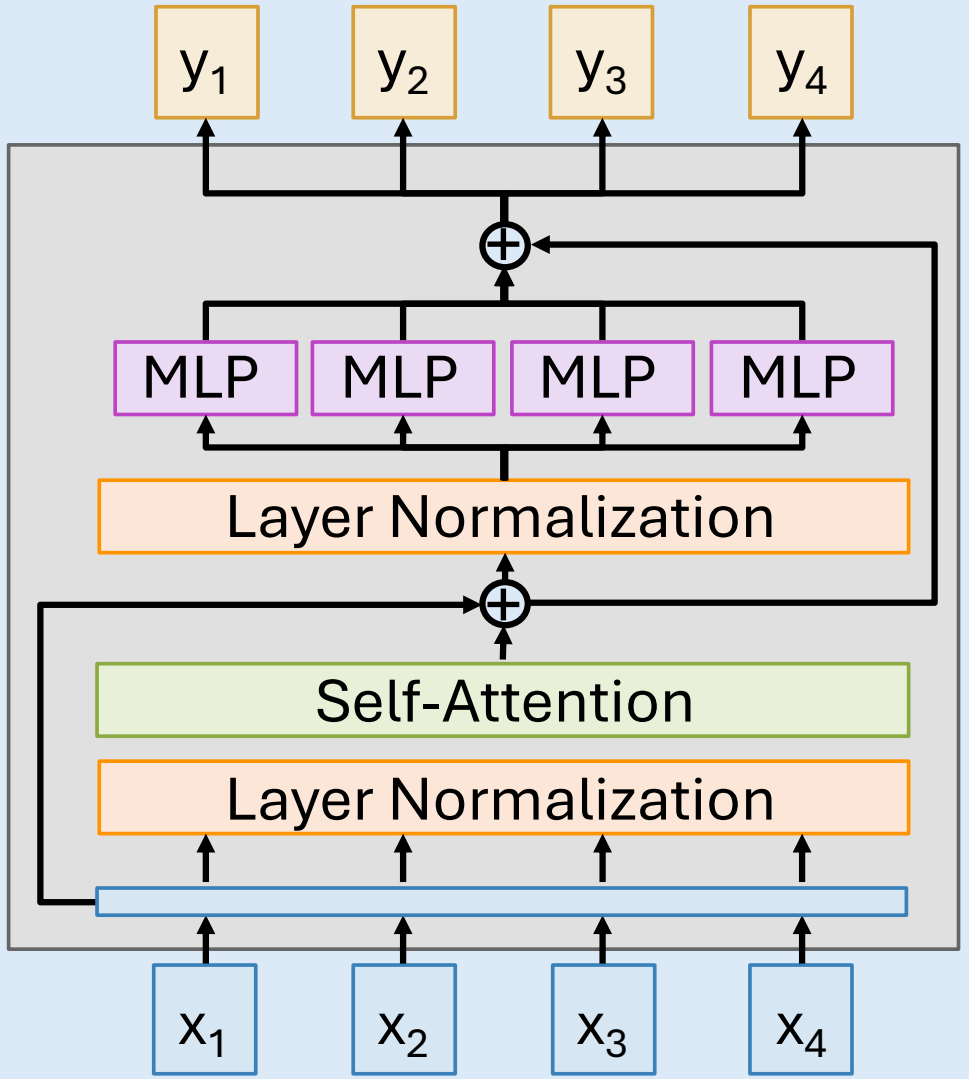
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QK-Norm

Normalize queries and keys before computing attention similarities

Queries: $Q = \text{normalize}(XW_Q)$ $[H \times N \times D_H]$
Keys: $K = \text{normalize}(XW_K)$ $[H \times N \times D_H]$
Values: $V = XW_V$ $[H \times N \times D_H]$
Similarities: $E = QK^T / \sqrt{D_Q}$ $[H \times N \times N]$

Prevents gradient spikes, stabilizes training



Zhang and Sennrich, "Root Mean Square Layer Normalization", NeurIPS 2019

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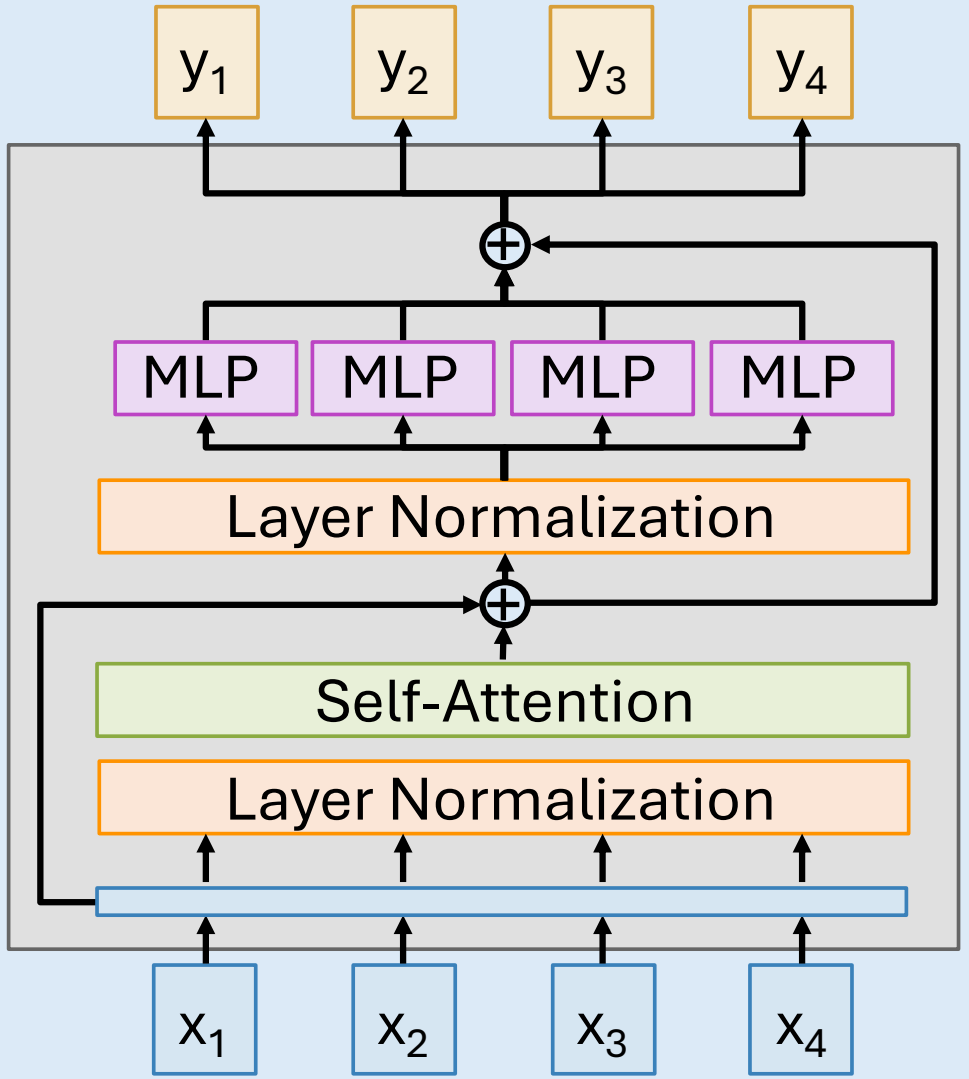
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Normalize with RMSNorm:

$$y_i = \frac{x_i}{\text{RMS}(x)} \cdot \gamma_i$$

$$\text{RMS}(x) = \sqrt{\varepsilon + \frac{1}{N}\sum_{i=1}^{N} x_i^2}$$

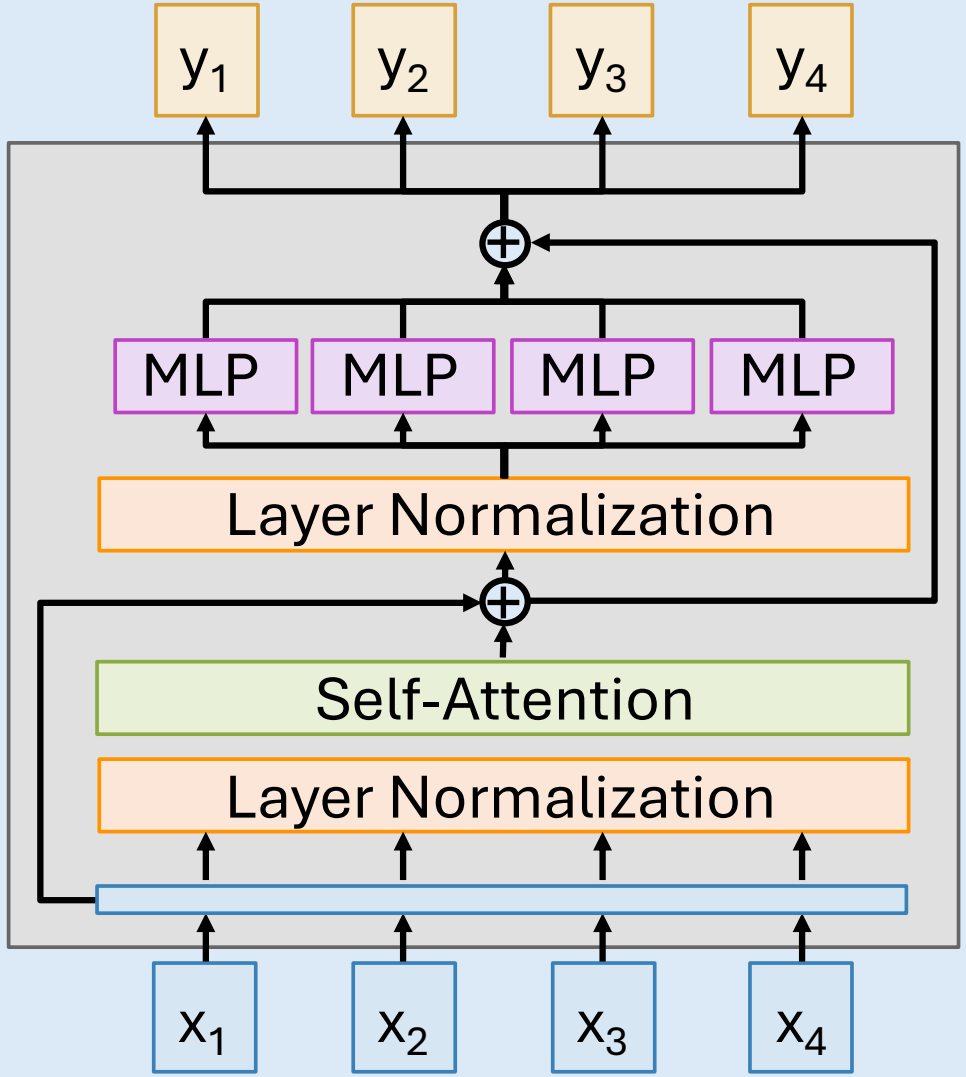
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SwiGLU MLP

Classic MLP :

Input: X [N x D]
Weights: W_1 [D x 4D]
W_2 [4D x D]
Output: $Y = \sigma(XW_1)W_2$ [N x D]



Shazeer, "GLU Variants Improve Transformers", 2020

SwiGLU MLP

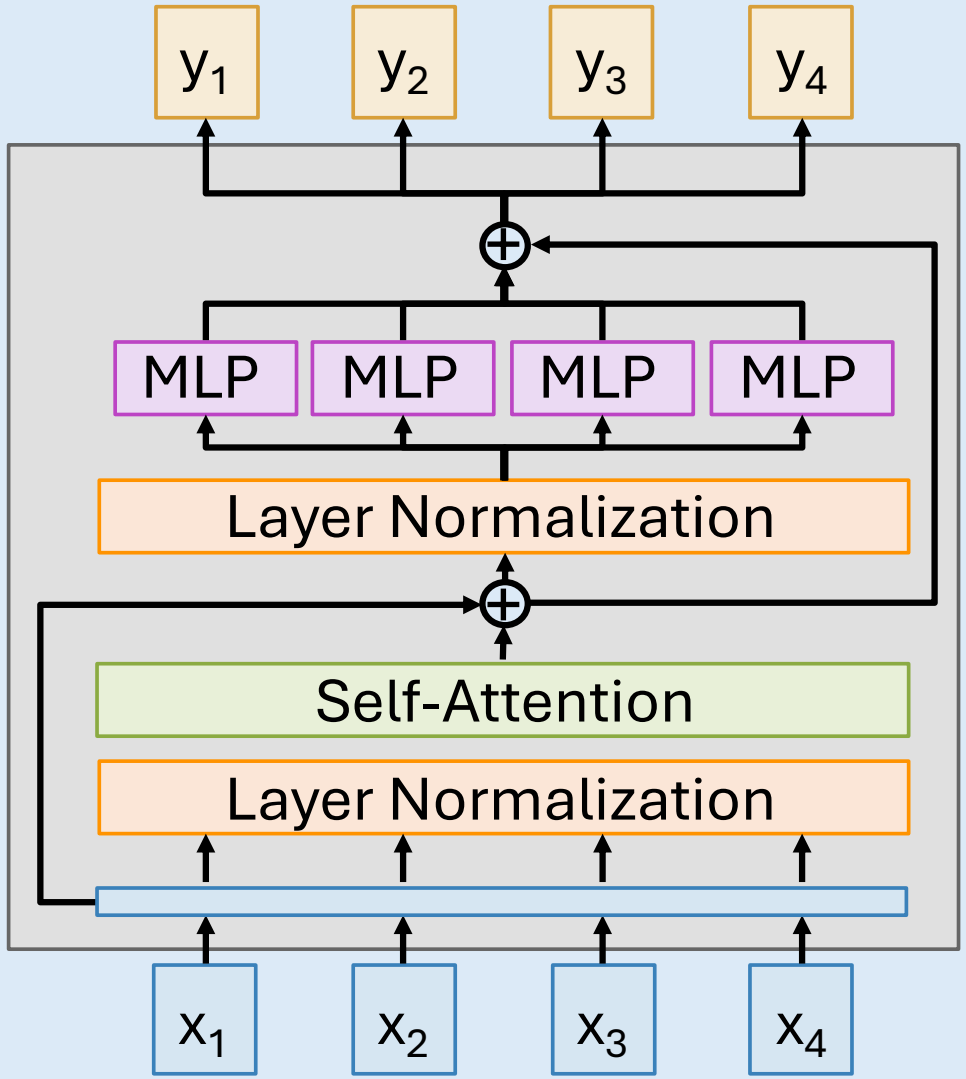
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SwiGLU MLP :

Input: X [N x D]
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Output:
$$Y = (\mathrm{SiLU}(XW_1) \odot XW_2)W_3$$

Setting H = 8D/3 keeps same total params



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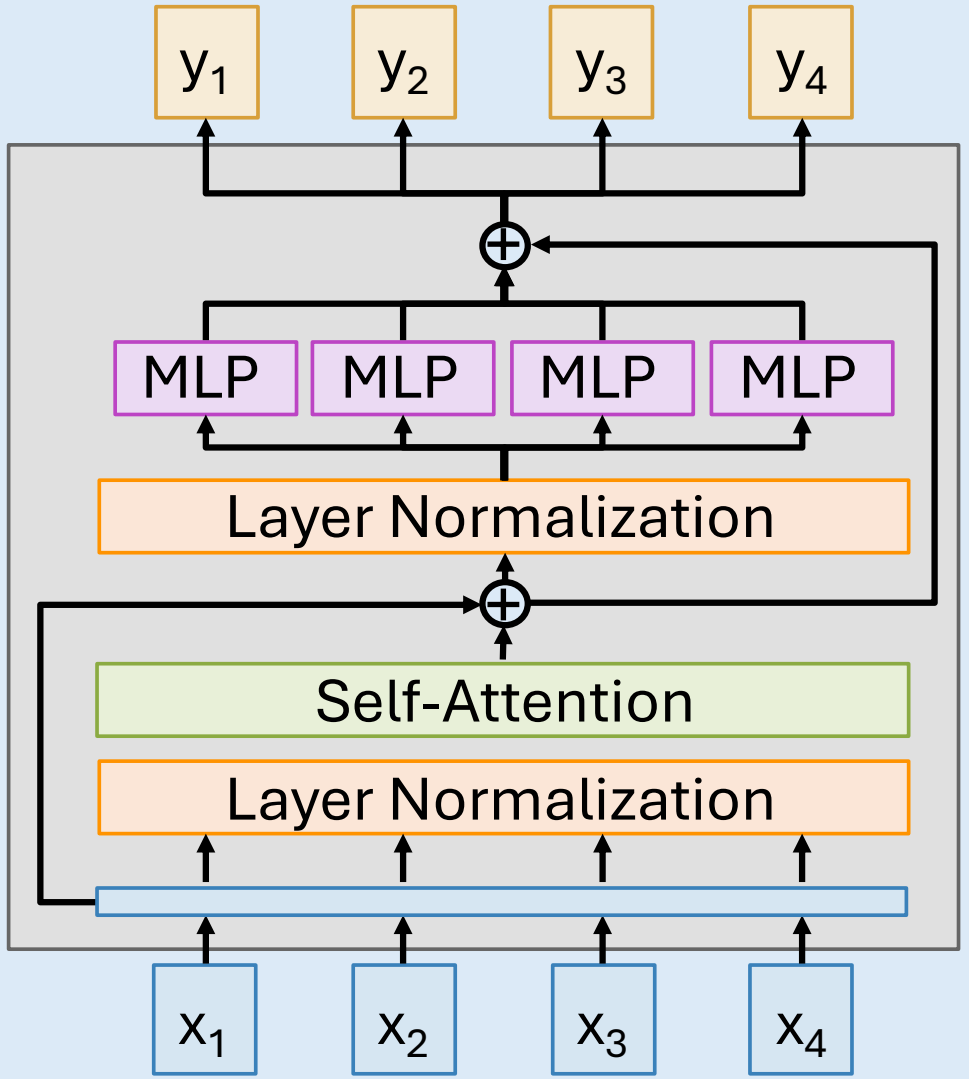
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We offer no explanation as to why these architectures seem to work; we attribute their success, as all else, to divine benevolence.

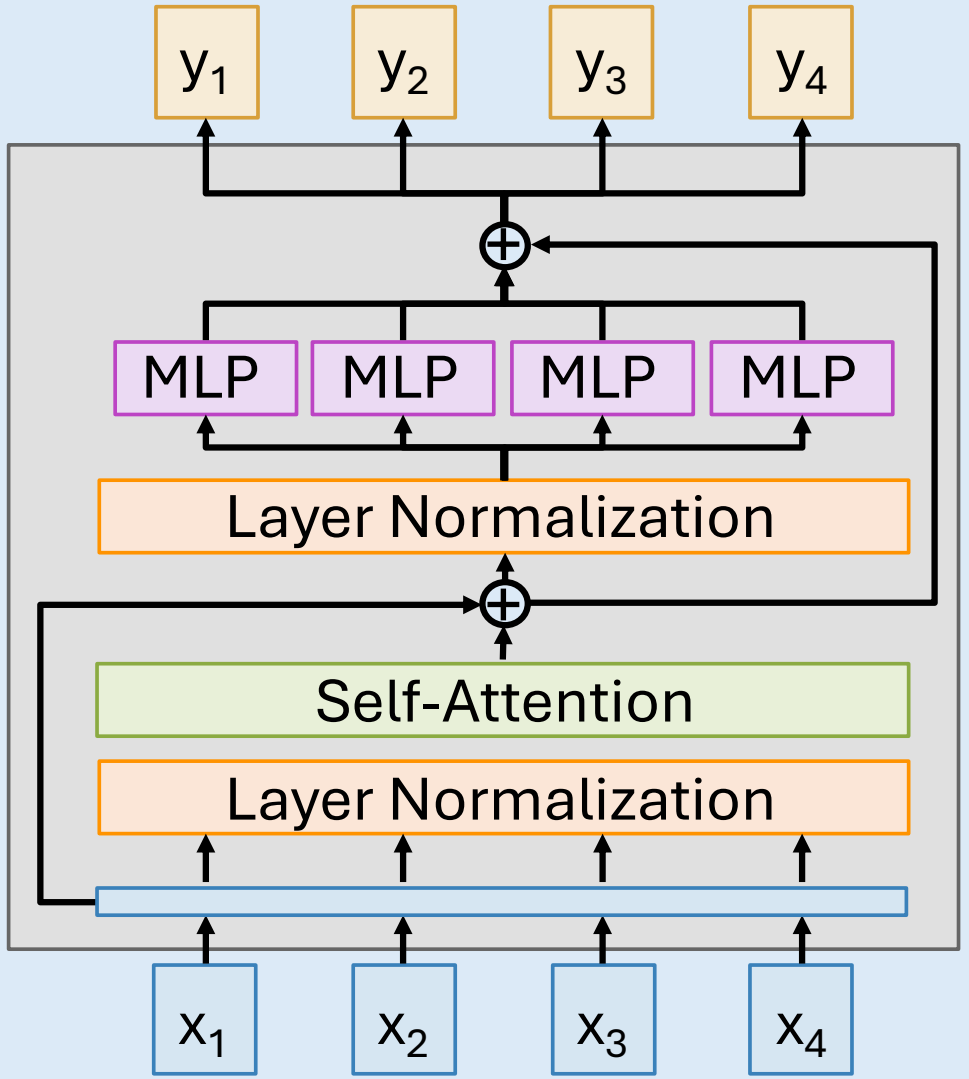


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Mixture of Experts (MoE)

Learn E separate sets of MLP weights in each block; each MLP is an *expert*

W_1: [D x 4D] => [E x D x 4D]
W_2: [4D x D] => [E x 4D x D]



Shazeer et al, “Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer”, 2017

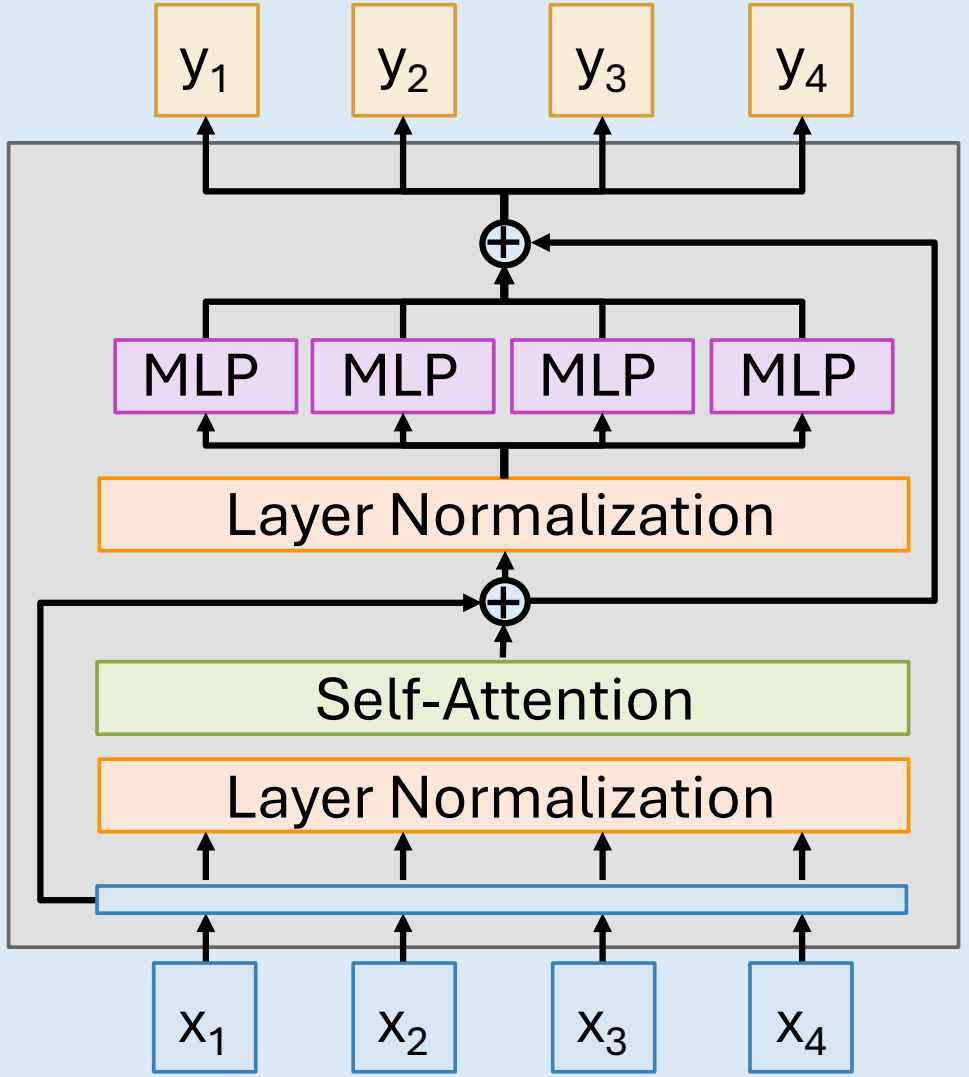
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Each token gets *routed* to A < E of the experts. These are the *active experts* .

Increases params by E,
But only increases compute by A



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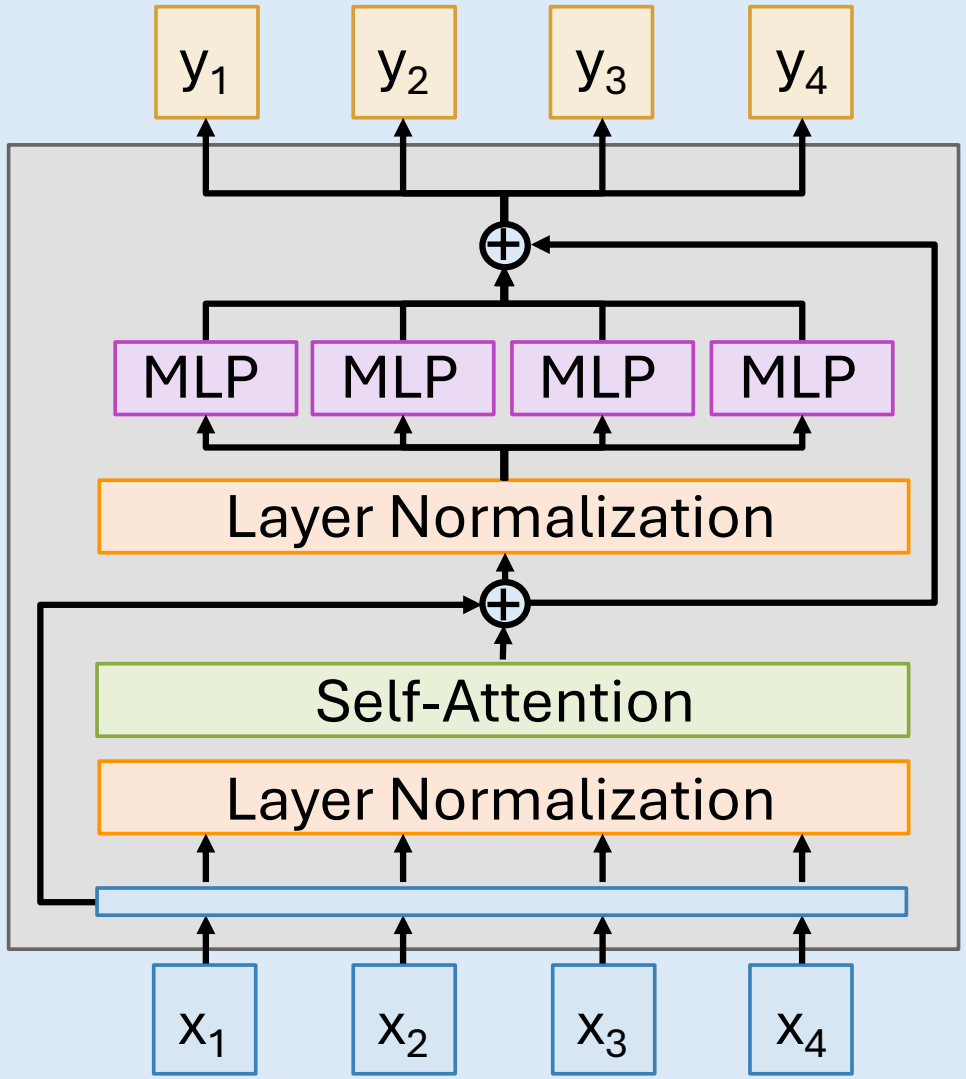
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Example: Gemma4 26B-A4B (4/2/2026)

- 1 “shared expert” processes all tokens
- 128 “routed experts”; each token selects 8/128 to process it
- 26B total params, but each token only “activates” 4B params

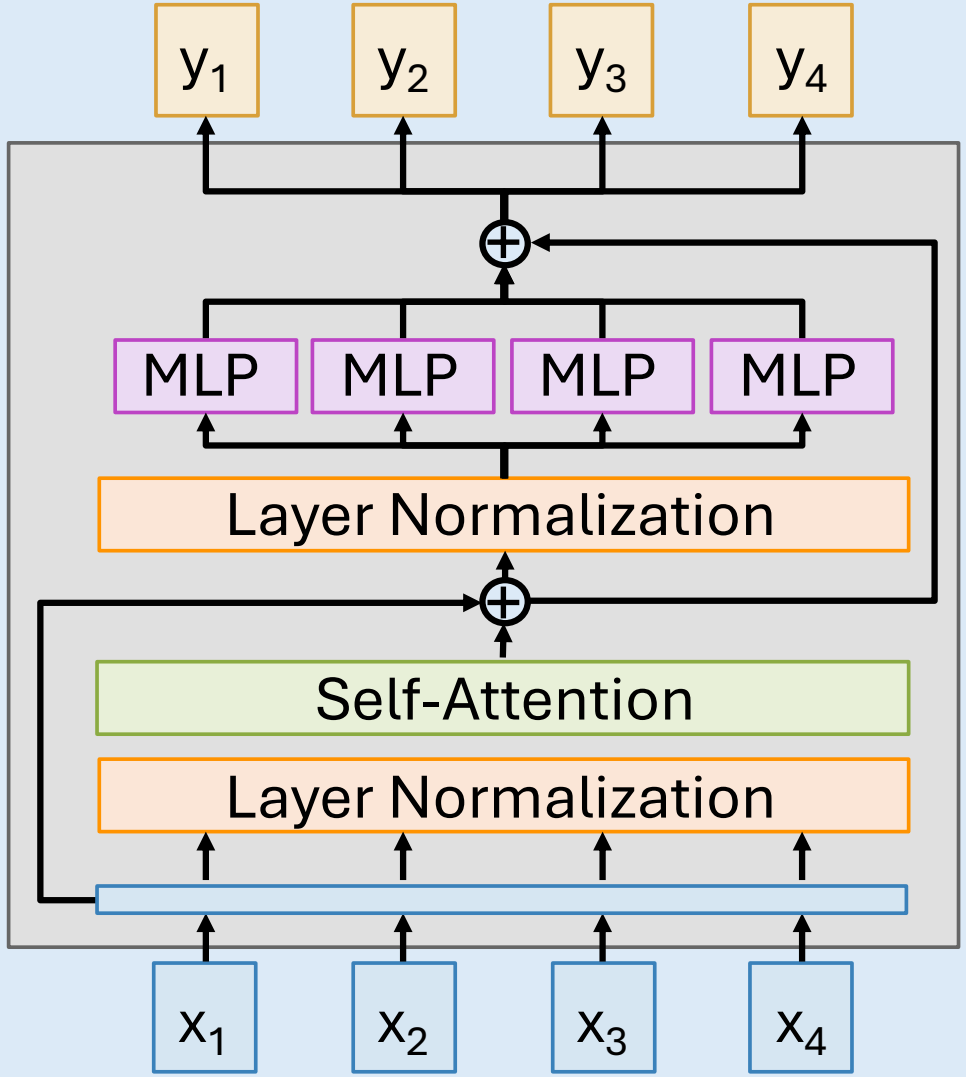


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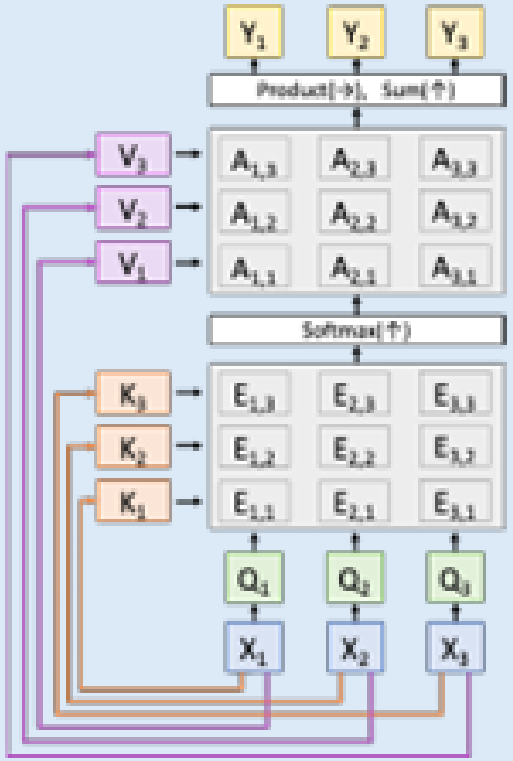
But a few changes have become common:

- **Pre-Norm:** Move normalization inside residual
- **QK-Norm** : Normalize keys and queries
- **SwiGLU** : Different MLP architecture
- **Mixture of Experts (MoE)** : Learn E different MLPs, use A < E of them per token. Massively increase params, modest increase to compute cost.



Summary: Attention + Transformers

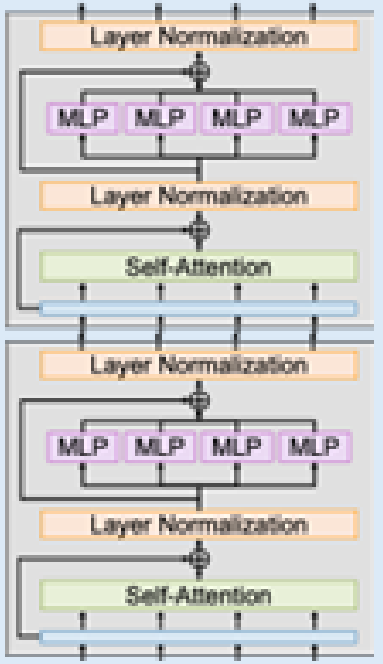
Attention : A new primitive that operates on sets of vectors



Transformers are the backbone of all large AI models today!

Used for language, vision, speech, …

Transformer : A neural network architecture that uses attention everywhere



References

- Stanford CS231n

- Mathematical Foundations of Deep Neural Networks at Seoul National University.

